

Late to Recessions: Stocks and the Business Cycle

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Abstract

I find that returns are predictably negative for several months after the onset of recessions, becoming high only thereafter. I identify business cycle turning points by estimating a state-space model using macroeconomic data. Conditioning on the business cycle further reveals that returns exhibit momentum in recessions, whereas in expansions, they display the mild reversals expected from discount rate changes. A strategy exploiting this pattern produces positive alphas. Using analyst forecast data, I show my findings are consistent with investors' slow reaction to recessions. When expected returns are negative, analysts are too optimistic and their downward expectations revisions are exceptionally high.

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1 Introduction

A large and expanding literature explores the relation between stock returns and the business cycle. The traditional view is that stocks are rationally priced to immediately reflect investors' expectations about future economic activity and that expected excess returns on stocks are positive, vary over time, and display countercyclical behavior.¹ I argue that stock prices are less efficient at incorporating business cycle fluctuations than what the literature has previously acknowledged. Specifically, I show that stock prices, in an economically significant way, are late in reflecting information signaling the onset of recessions. This then has important consequences for the behavior of expected returns.²

I show that the state of the business cycle is far more informative about expected stock returns than previously recognized. I identify business cycle turning points by estimating a state-space model using real-time macroeconomic data. The central finding is that returns are predictably negative for several months following the onset of recessions before rising to persistent higher-than-average levels. Further conditioning on the business cycle reveals a strong state-dependence in autocorrelations. Returns exhibit substantial momentum in recessions, whereas in expansions, they display the mild reversals expected from discount rate changes. I propose a model of expected and realized returns that optimally accounts for these business cycle patterns. Using the model's expected return estimate, I then construct a trading strategy to analyze the implied return predictability and assess my findings' economic relevance. The strategy produces significant risk-adjusted returns: for the aggregate market, it earns an annualized alpha of 5.4%, increases Sharpe ratios by 30% over the fully invested buy-and-hold approach, and produces a significant monthly out-of-sample R^2 statistic of 1.7%.

The right panel of Figure 1 exemplifies the central finding. An investor who buys stocks in the first four months of a recession, as defined by the recession probability (see

¹References on the explanatory power of stock price changes for macro aggregates include: Barro (1990), Fama (1990), Stock and Watson (2003). References on countercyclical variations in excess returns and Sharpe ratios include: Fama and French (1989), Ferson and Harvey (1991), Lettau and Ludvigson (2010), Lustig and Verdelhan (2012), and Kozak and Santosh (2019).

²Consistent with the convention in the literature, I sometimes use the expression risk premium/discount rate to describe the expected excess return implied by the econometric model. However, it is important to note that agents' subjective ex-ante expectations of expected returns may differ quite substantially from the econometrician's forecast of the same object.

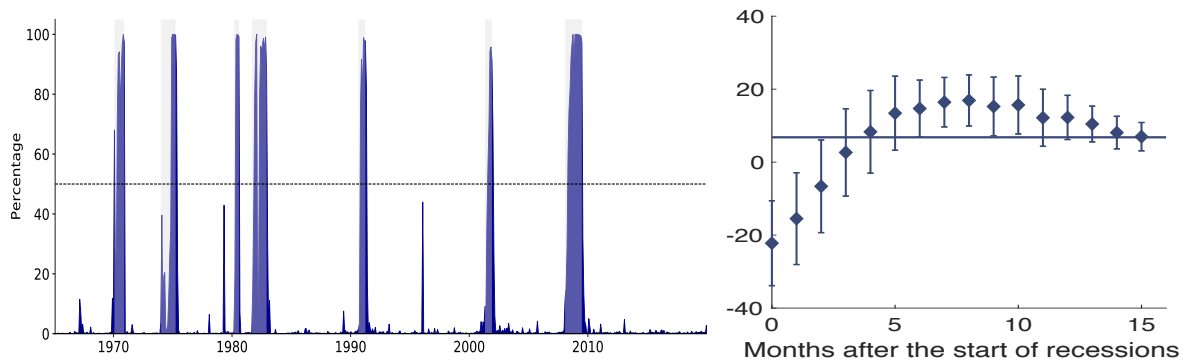


Fig. 1. **Probability for a recessionary regime and 6-month equity excess returns.** The left panel plots the real-time recession probability. Section 3.3 provides details for computing this probability. The light gray bars indicate the NBER recession dates. The right panel depicts the annualized 6-month realized excess market return during recessions. That is, the graph shows the excess returns for an investor who buys stocks m months after expecting to be in a recession and holds them for 6 months. I assume that an investor expects to be in a recession whenever the real-time recessionary probability first exceeds 50%. The vertical bars represent a 1 standard deviation error range and account for heteroskedasticity and autocorrelation in the residuals up to six lags.

the left panel of Figure 1), and holds them for six months earns an annualized average excess return of -8%.³ However, these negative returns are reversed in the following 12 months with persistently high excess market returns of around 13%. After a year, returns converge to the unconditional equity premium, depicted by the horizontal bar in the right panel of Figure 1. From a market participant perspective, this pattern suggests that the investor can profit by selling at the start of recessions and then increasing risk-taking several months later. From a risk-based perspective, the empirical pattern in Figure 1 presents a puzzle, because it is hard to explain negative average excess returns within any rational equilibrium model. Deepening the puzzle is the fact that these negative returns happen in precisely bad macroeconomic times, when risk-based theories generally suggest that compensation for risk should be positive and high. As I will argue, the pattern instead appears more consistent with the idea that investors are late to act on macroeconomic news that signals the start of recessions.

My proposed econometric model has two main ingredients. The first is a regime-switching model for expected and realized returns, where the regimes are linked to business cycle phases. I treat conditional expected returns as a latent variable that

³I assume that a recession occurs when the recession probability is above 50%. However, the 50% threshold rule is not essential. I obtain similar results with various threshold rules around 50%. The 6-month investment horizon is also not necessary for the pattern observed in the right panel of Figure 1. My investment strategy, described in Section 3, takes 1-month investment horizons and the continuous recession probability.

follows a first-order autoregressive process. Importantly, I allow for discount rate volatility to vary across the business cycle and for correlated shocks between the realized return and expected returns. The second ingredient identifies the economic regimes via an econometric model that uses macro variables to extract a diffusion index and a real-time recession probability forecast, as in [Hamilton \(1989\)](#), [Kim and Nelson \(1998\)](#), and [Chauvet \(1998\)](#).

To account for both model ingredients jointly, I draw from the growing literature on density forecast combination.⁴ Specifically, the model dynamically combines the predictive densities from two individual models that differ in the amount in which expected returns vary using time-varying weights. These dynamic model weights correspond to the probability of being in an expansion or a recession. My setup takes the form of a high-dimensional Markov-switching state-space model tailored towards monthly real-time data. A mixed-frequency approach and temporal aggregation enable me to sharpen inference about the regimes by incorporating the information contained in quarterly time-aggregated data and, at the same time, deal with variable reporting lags, early announcements, and missing observations of each particular macro series. Importantly, to avoid any look-ahead bias, I restrict the information set to those observations that were known and available to investors at each point in time and I only use data up to time t to forecast $t + 1$ returns.

The model utilizes the state-dependent serial correlation in returns to predict future returns. When expected returns are latent, realized lagged returns are noisy signals about discount rates. The model's estimation reveals that discount rates are persistent, fairly stable in expansions, and much more volatile during recessions. Moreover, return surprises and innovation in expected return are strongly negatively correlated, consistent with [Campbell \(1991\)](#), [Campbell and Ammer \(1993\)](#), and [Pástor and Stambaugh \(2009\)](#). All other things being equal, the persistence and variability of expected returns generate a positive autocorrelation in returns; thus, prior returns positively predict future returns. However, this positive effect is overwhelmed in expansions by the countervailing effect of

⁴For instance, [Geweke and Amisano \(2011\)](#) combine the predictive densities of alternative models using optimal linear weights. Similarly, [Del Negro, Hasegawa, and Schorfheide \(2016\)](#) assume a smooth autoregressive process for weight evolution to combine the predictive densities from two DSGE models.

return innovations that are highly negatively correlated with shocks to expected returns. In this state, returns exhibit mild negative autocorrelation, and, therefore, low past returns predict high future returns. When expected returns are much more volatile in recessions, the positive effect dominates so that returns are positively autocorrelated, and, consequently, the model updates forecasts of future returns in the same direction as recent realized returns. Notably, the weights used on lagged returns to compute the expected return measure are derived from optimal filtering techniques.

Using the model's estimate of expected returns, I analyze my findings' economic relevance via a trading strategy. Specifically, I treat the measure of expected returns as the conditional beta in a standard conditional CAPM framework and increase (decrease) risk-taking when the expected returns estimate is high (low). In line with Figure 1, the strategy generates a positive unconditional CAPM alpha by reducing risk-taking for several months following the onset of a recession. At such times, the estimates of expected returns are negative—not positive and high. Shortly after completely cashing out, the strategy returns to the market and increases risk-taking substantially. Put differently, the strategy produces profits in part by avoiding stock market crashes associated with recessions. The rest of the profits come from expansionary periods. During expansions, the strategy takes a contrarian position by exploiting mild mean reversion in returns. I show that although mean reversion is small in the data, it is crucial in a cumulative sense for forecasting future monthly returns. The strategy's substantial gains may be surprising given the prevailing consensus in the literature that stock return predictability is highly unstable (e.g., [Ferson, Sarkissian, and Simin \(2003a\)](#), [Goyal and Welch \(2008\)](#)) and concentrated in short-lived predictability pockets (e.g., [Timmermann \(2008\)](#), [Farmer, Schmidt, and Timmermann \(2018\)](#)).

Therefore, to evaluate the robustness of the strategy, I conduct a wide range of tests. The strategy overlaps with the time-series momentum factor of [Moskowitz, Ooi, and Pedersen \(2012\)](#) since, during recessions, the model of expected returns updates the forecasts of future returns in the same direction as recent realized returns. The strategy also coincides with the volatility-managed market factor of [Moreira and Muir \(2017\)](#), since the strategy also reduces risk-taking during these bad times. I show that my strategy is different from these two factors and produces positive alpha when controlling for

them. Specifically, I document that accounting for mild mean reversion in returns during expansions is important for return predictability. Moreover, one principal difference between the volatility-managed factor and my strategy is regarding when to increase risk-taking during recessions. The volatility-managed factor will gradually return to the market only when volatility decreases and not necessarily when the market rebounds.

I extend the empirical analysis of my strategy in the appendix. Significantly, I look at the cross-section of U.S. state portfolios and regional business cycles and show that the results are consistent. I also show that the strategy can be further improved by also timing volatility as in [Moreira and Muir \(2017\)](#). Moreover, the strategy produces fairly large benefits even when subject to tight constraints on leverage and short-selling and to realistic transaction costs. The strategy generates similar alphas across different subsample periods and for a wide range of pricing factors. I also verify that the strategy is profitable using utility-based performance measures, the manipulation-proof performance measures of [Goetzmann, Ingersoll, Spiegel, and Welch \(2007\)](#), or the regression-based timing measures of [Treyner and Mazuy \(1966\)](#) and [Henriksson and Merton \(1981\)](#). Notably, the strategy's outperformance vanishes if I turn off the state-dependent autocorrelation in returns. Given the strategy's profitability and robustness, I conclude that the model provides a good measure of the market expected returns over the monthly horizon.

I analyze these findings' economic implications in terms of the behavior of expected returns and examine the type of models we need to match these facts. I first explore standard models with rational expectations and countercyclical risk premia. Earning a positive unconditional alpha by reducing risk-taking during bad times is difficult to explain in terms of risk. In the context of these models, the alpha would be either negative or close to zero. As shown by [Lewellen and Nagel \(2006\)](#), if the conditional CAPM holds and the market volatility is constant, the unconditional alpha depends primarily on how the conditional risk exposure, β_t , covaries with the unobservable market risk premium, γ_t , that is $\alpha \approx \text{cov}(\beta_t, \gamma_t)$. Hence, a strategy that avoids taking risks during a recession's early stages will tend to earn a negative alpha because the market risk premium is precisely high during these times. If the risk premium and volatility move together, as implied by many asset-pricing theories, the volatility will tend to offset the impact of the risk premium and shrink the negative α towards zero.

Instead, the evidence appears to be more consistent with an initial underreaction to news. That is, news that signals the start of a recession is incorporated only slowly into prices, which induces the observed abnormal price drift during these times.⁵ To better understand this explanation, I explore a model in which investors use the present-value identity to value stocks but discount future cash flows using subjective expectations. If subjective beliefs about future cash flows are sluggish to react to new information, then expected returns are a function of a rational risk premia component and near-term forecastable changes in cash flows that are not yet incorporated into prices. Hence, overestimating future cash flows at the onset of the recession under the subjective measure offsets the positive increase in risk premia and imputes negative return predictability for the econometrician. Consistent with this sluggish subjective belief updating, I find that downward revisions in analyst expectations about cash flows are exceptionally high, and forecasts are too optimistic when my measure of expected returns is negative. These results are in line with recent papers that emphasize underreaction to aggregate news at short horizons. [Coibion and Gorodnichenko \(2015\)](#) find evidence of information stickiness in macro-forecasts from professional forecasters, while [Bouchaud, Krueger, Landier, and Thesmar \(2019\)](#) document economically significant expectation stickiness at the firm level from financial analysts' forecasts about future cash flows. Moreover, [Katz, Lustig, and Nielsen \(2017\)](#) show that local stock markets adjust sluggishly to changes in local inflation.

During expansions, I cannot determine whether the strategy's profits are a result of rational variation in expected returns or due to behavioral biases. Mean reversion in returns is consistent with rational discount rate movements as explained in [Cochrane \(2005\)](#). Alternatively, mild negative autocorrelation in returns could also reflect belief overreaction at longer horizons consistent with recent papers studying biases in beliefs (e.g., [Angeletos, Huo, and Sastry \(2021\)](#), [Bordalo, Gennaioli, Ma, and Shleifer \(2020\)](#), [De La O and Myers \(2021\)](#) and [Kohlhas and Walther \(2018\)](#)). See also [Angeletos et al. \(2021\)](#) for how a simple model with dispersed, noisy information and over-extrapolation

⁵Stock prices appear to drift after announcements or events related to earnings announcements (e.g., [Bernard and Thomas \(1989\)](#); [Mendenhall \(1991\)](#)), dividend initiations and omissions (e.g., [Michael, Thaler, and Womack \(1995\)](#)), and analysts' recommendations revisions (e.g., [Bjerring, Lakonishok, and Vermaelen \(1983\)](#)).

can explain initial underreaction and delayed overreaction of beliefs in response to shocks.

The paper is organized as follows. Section 2 presents the novel stylized facts on the business cycle variations in stock returns. Section 3 introduces the econometric framework. Section 4 presents the estimation results. Section 5 measures the economic relevance of my findings. Section 6 analyzes the time-series behavior of my expected returns estimates. Section 7 establishes and rules out potential explanations for my findings. Section 8 concludes, and a detailed appendix extends my empirical analysis on many fronts.

2 Stylized Facts

This section presents the paper's novel stylized facts. Many market downturns and upturns happen during recessions, but the central insight from Figure 1 is that these directional changes do not occur randomly. At the onset of economic contractions, it is more likely to observe market downturns, which are then followed by upturns. I present this stylized fact by introducing a feasible and straightforward strategy that trades on this pattern. The second stylized fact is a strong state-dependent autocorrelation in returns. I document this via simple OLS regressions. Section 3 presents a model of expected and realized returns that optimally accounts for these business cycle patterns.

The trading strategy switches from stocks (e.g., S&P 500) into cash (e.g., 1-month T-Bill return) during bad economic times and market downturns, while it stays fully invested in stocks in the remaining periods. Specifically, suppose at the end of the month, the U.S. industrial production growth (year-over-year) series and the realized market return are both below their 12-month moving average, indicating a contractionary period and a market downturn, respectively. In this case, the strategy goes 100% into cash. Otherwise, the strategy goes 100% into stocks. The series for the market return and the risk-free rate come from Kenneth R. French's website.⁶ Appendix A presents robustness with respect to the choice of the 12-month moving average and with respect to the sample period.

Column (1) of Table 1 reports performance measures for this simple trading strategy.

⁶https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

From January 1980 to December 2019, the strategy earned an annualized mean excess return of 7.24% with an annualized standard deviation of 12.64% by being fully invested in stocks 77% of the time. For comparison, a fixed allocation strategy that is invested in stocks the same unconditional amount of time, that is, a mix of 77% in equities and 23% in T-Bills, generates an annualized mean excess return of 6.33% and volatility of 11.82%, as shown in Column (4). These numbers imply that the simple strategy produces an overall Sharpe ratio increase of 6.92% over the 77/23 strategy. Moreover, the simple strategy generates an annualized CAPM alpha of 1.63% (t-statistic = 1.67). Panel B of Table 1 reports average maximum drawdowns in NBER expansions and recessions to help us understand in which phase of the business cycle the strategy loses money relative to the fixed allocation portfolio. In essence, the strategy benefits by reducing the maximum drawdowns during recessions. For instance, the average maximum drawdown during recessions is 16.49%, which is significantly lower than the corresponding 18.96% for the 77/23 strategy.

Figure 2 shows the cumulative monthly returns for investments in the market portfolio, the risk-free asset, the 77/23 portfolio, the main strategy that conditions on the growth and return signals, and a strategy that only conditions on the growth signal. All strategies invest \$1 in July 1927 and close the position in December 2019. The gains of the main strategy (assuming no transaction costs) would accumulate to around \$10,000 at the end of 2019 (see the dark blue line) versus about \$2000 for the fixed allocation 77/23 strategy (dotted black line) and \$6,000 for the buy-and-hold strategy (dotted gray line). The figure confirms the central message of Table 1. The strategy outperforms its benchmark by reducing risk-taking in low economic growth periods during market downturns. This result does not contradict the long empirical evidence on countercyclical expected returns (e.g., [Lettau and Ludvigson \(2010\)](#), [Lustig and Verdelhan \(2012\)](#)). For instance, a strategy that goes into cash during the entire duration of the contractionary period significantly underperforms by generating around \$443 at the end of the sample (see the dotted light blue line). However, as I will argue, there is a mismatch between macro risk and compensation for risk in the first months of contractionary periods. Therefore, these months are particularly bad times to invest in stocks, and an active strategy would like to cash out during these times and buy back in once the downward pressure on prices

Table 1
Performance of Trading Strategies

This table reports performance measures for four different strategies. The “100% stocks” denotes the buy-and-hold strategy that is fully invested in stocks. The “77/23 mix” denotes the portfolio that assigns a 77% weight to equities and 23% to T-Bills. The remaining two strategies switch from stocks into cash during bad times and market downturns, while they stay fully invested in stocks in the remaining periods. Market downturns are months in which returns are below their 12-month moving average. Bad times are periods when the industrial production growth series (“Ind. prod.’) or the common growth component (“Common growth”) are below their 12-month moving average. For each strategy, I report the Sharpe ratio increase relative to the Sharpe ratio obtained from a fixed allocation strategy that invests in stocks the same unconditional amount of time. I report t-statistics in brackets below the coefficient estimates and standard errors adjust for heteroskedasticity. I annualize all returns, expressed as a percentage per year, by multiplying monthly returns by 1200. The evaluation period is from January 1980 to December 2019.

	Ind. prod. (1)	Common growth (2)	100% stocks (3)	77/23 mix (4)
Time invested in stocks	77%	75%	100%	77%
Mean excess returns	7.24	7.87	8.15	6.33
Std. dev. returns	12.64	11.68	15.21	11.82
Sharpe ratio increase	6.92	25.81	0.01	-
α -Mkt	1.63 [1.67]	3.26 [2.72]	- -	- -
NBER economic state	Panel B: Average maximum drawdowns			
Total	7.84	6.47	11.21	8.51
Expansions	6.70	5.57	9.49	7.13
Recessions	16.49	14.02	24.23	18.96

dissipates. This is the first stylized fact.

I document two additional results. First, I can increase the strategy’s profit by improving the measure of economic activity. To do so, I estimate the growth signal from various macroeconomic indicators rather than just using the industrial production growth series. This result makes sense because business cycle fluctuations are about the comovement of a broad range of economic indicators. Column (2) of Table 1 shows results for a strategy that uses the common growth component estimated out of sample from nine real-time variables that measure different aspects of the overall U.S. macroeconomic

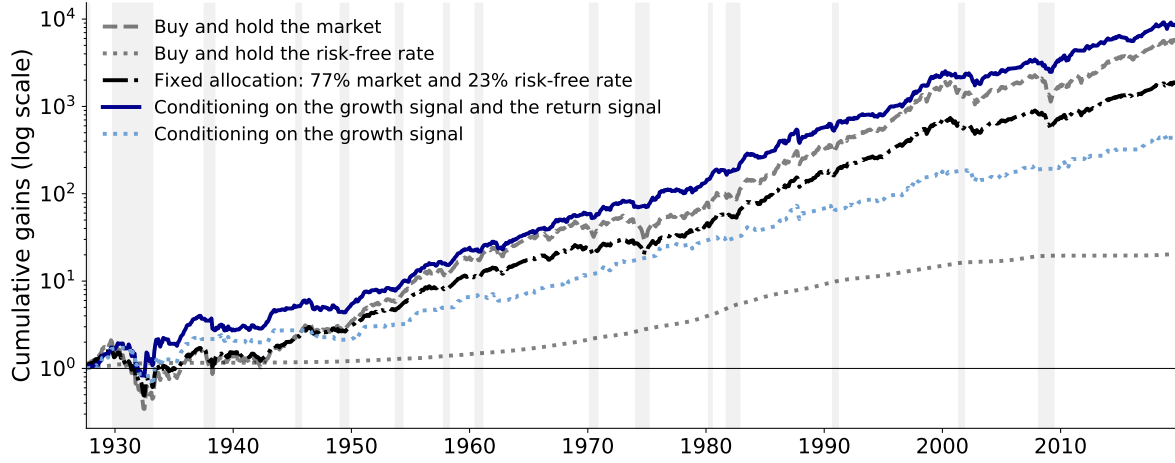


Fig. 2. **Cumulative gains.** The figure shows the cumulative monthly returns for one-dollar investments in the market portfolio, the risk-free asset, the 77/23 portfolio, the main strategy that conditions on the growth and return signals, and a strategy that only conditions on the growth signal. I describe the construction of the portfolios in the caption of Table 1. Light-shaded bars represent recessions, as defined by the National Bureau of Economic Research. The y-axis is on a log scale. The sample period is from July 1927 to December 2019.

activity. I introduce and estimate this variable in Section 3. Specifically, a bad economic time occurs when the common growth component is below its 12-month moving average. Under this broader measure of economic activity, the strategy is fully invested in stocks 75% of the time and increases Sharpe ratios by around 25% over its 75/25 fixed allocation benchmark. Moreover, it increases the annualized CAPM alpha to 3.26% by reducing the average maximum drawdowns during recessions to 14.02%.

Second, the autocorrelation in returns is business cycle dependent. To assess the state-dependent serial correlation, Table 2 presents regression results for the following specification:

$$r_{t+1}^e = a_0^j + (b_0^j + b_1^j \cdot I_{j,t}) r_t^e + \epsilon_{t+1}. \quad (1)$$

where r_{t+1}^e denotes the market excess return. The regressors are the lagged market excess return and an indicator variable that takes the value of one if we are in a contractionary period according to a growth signal j (e.g., when industrial production growth is low relative to its moving average) and zero otherwise, ϵ_{t+1} is an error term, and b_0^j and b_1^j are the parameters of interest.

As shown in Columns (1) through (3) of Table 2, conditioning on the state of the business cycle reveals that during bad economic times returns are strongly positively

Table 2
Regression Evidence: Serial Correlation in Returns

This table reports regression estimates for four different specifications. In all cases, the dependent variable is the monthly market excess return, r_{t+1}^e . The independent variables are a constant; the previous month excess market return r_t^e ; an indicator for bad economic times, $I_{\text{Ind.Prod.},t}$ ($I_{\text{Comm.Growth},t}$), which equals one if industrial production growth (the common growth component) is below its 12-month moving average; and the NBER recessionary indicator, $I_{\text{NBER},t}$. I report t-statistics in brackets below the coefficient estimates and standard errors adjust for heteroskedasticity. I multiplied the coefficient $\hat{a}_0$ by 1200. The sample period is from January 1980 to December 2019.

Coefficient	Variable	Estimated coefficients [t-statistics]			
		(1)	(2)	(3)	(4)
$\hat{a}_0$	1	7.43 [3.05]	7.24 [2.97]	8.57 [3.54]	7.52 [3.09]
$\hat{b}_0$	r_t^e	-0.01 [-0.03]	-0.01 [-0.21]	-0.03 [-0.63]	0.06 [1.38]
$\hat{b}_{\text{Ind.Prod.}}$	$I_{\text{Ind.Prod.},t} \cdot r_t$	0.12 [1.92]			
$\hat{b}_{\text{Comm.Growth}}$	$I_{\text{Comm.Growth},t} \cdot r_t$		0.10 [1.94]		
$\hat{b}_{\text{NBER}}$	$I_{\text{NBER},t} \cdot r_t$			0.39 [3.68]	
R_{adj}^2		0.011	0.012	0.028	0.00

autocorrelated. For instance, when the industrial production growth series is below its 12-month moving average, the point estimate is 0.11 ($= \hat{b}_0 + \hat{b}_{\text{Ind.Prod.}}$) with a t-statistic of 1.92 on the difference. Notably, the point estimate increases to 0.36 ($= \hat{b}_0 + \hat{b}_{\text{NBER}}$) when I use the ex post NBER recessionary index.⁷ These statistically significant positive coefficients during bad times are evidence of price momentum in the aggregate market and imply that a string of past negative (positive) returns forecast future negative (positive) returns. Conversely, the point estimates for the autocorrelation coefficient in good economic times (equal to $\hat{b}_0$) are negative, but not statistically significant. This mild negative autocorrelation in returns means that past bad returns forecast future good returns. In the next sections, I will show that, although the point estimate $\hat{b}_0$ is

⁷The NBER's Business Cycle Dating Committee often determines this dummy variable with a substantial lag, which makes it contemporaneously unavailable to investors. For instance, the NBER committee did not announce the June 2009 business cycle until September 20, 2010.

small and statistically indistinguishable from zero, exploiting mild reversals in returns during expansions is very important in a cumulative sense for forecasting future returns. Finally, as shown in Column (4), without conditioning on the business cycle, we would have missed this sign flipping pattern in the autocorrelation of returns.

3 Econometric Framework

The econometric model has two main ingredients. The first, described in Section 3.1, is a regime-switching model for realized stock returns, where I treat expected returns as a latent variable that follows a first-order autoregressive process. The model allows for both much more volatile expected return fluctuations during recessions and a negative correlation between return surprises and innovation in expected returns. As shown in Section 3.2, these two features allow the model to optimally account for the business cycle dependence in the autocorrelation of returns and to determine how the optimal filtering technique uses the noisy signals in lagged returns to form an expected returns estimate. The second ingredient, described in Section 3.3, is an econometric model that extracts a diffusion index from a large cross-section of mixed-frequency macro variables to identify the economic regimes. In Section 3.4, I model both ingredients jointly via a dynamic linear prediction pool and describe the estimation procedure.

3.1 *Excess Stock Returns*

Let r_{t+1}^e be the return on the aggregate stock market in excess of the risk-free rate,

$$r_{t+1}^e = \mu_t + \varepsilon_{t+1}^r, \quad (2)$$

where μ_t denotes the conditional expected excess return and ε_{t+1}^r is the unexpected excess return component. Excess stock returns, r_{t+1}^e , are observable, while I treat conditional expected excess returns as a latent variable that follows an AR(1) process,

$$\mu_{t+1} = \mu_0 + \rho (\mu_t - \mu_0) + \varepsilon_{t+1}^\mu, \quad (3)$$

where ε_{t+1}^μ is the shock to expected returns and the parameters μ_0 and ρ denote the unconditional mean and persistence of μ_t , respectively. Because I am not imposing any economic restrictions in equations (2) and (3), the process of expected returns does not need to be rational. As discussed below, it can incorporate a rational risk premium component and near-term forecastable cash flow changes that are not yet incorporated into prices. The state-space system is open-minded about the underlying sources that drive variations in expected returns.

I introduce information about the state of the business cycle via the distribution of the shocks ε_{t+1}^r and ε_{t+1}^μ . Specifically, the two shocks are conditionally Gaussian,

$$(\varepsilon_{t+1}^r, \varepsilon_{t+1}^\mu)' \sim i.i.d. \mathcal{N}(\underline{0}, \Sigma(S_t)) \quad \text{with} \quad \Sigma(S_t) = \begin{bmatrix} \sigma_r^2(S_t) & \sigma_{r\mu}^2(S_t) \\ \sigma_{\mu r}^2(S_t) & \sigma_\mu^2(S_t) \end{bmatrix}, \quad (4)$$

where S_t is latent and denotes the economic regime indicator that signals the state of the business cycle, that is, whether the economy is in a contractionary or expansionary period. S_t is determined by analyzing the comovement of a broad range of macroeconomic variables and will be specified shortly in Section 3.3. Finally, the parameter $\sigma_{r\mu}^2(S_t) = \rho_{\mu,r} \cdot \sigma_\mu(S_t) \cdot \sigma_r(S_t)$ measures the covariance between unexpected returns, ε_{t+1}^r , and shocks to expected returns, ε_{t+1}^μ , where I further assume that the correlation parameter, $\rho_{\mu,r}$, is stable across regimes.

Conditional Expected Returns. When expected returns are not observable, the econometrician's estimate for the equity premium implied by the model of returns described above is a weighted average of past realized excess returns and the probability of being in each economic regime.

Let $p(r_{t+1}^e | \Theta_j, r_{1:t}^e)$ denote the predictive density obtained from equations (2), (3), and (4) by conditioning on Θ_j and $r_{1:t}^e$, which denote the vector of model parameters associated with state $j \in \{Exp., Rec.\}$ and the observed excess returns up to time t (i.e., $\{r_1^e, \dots, r_t^e\}$), respectively. It follows that the aggregate predictive density of returns is given by

$$p(r_{t+1}^e | \Theta, r_{1:t}^e) = (1 - \hat{\pi}_{t|t}) \cdot p(r_{t+1}^e | \Theta_{Exp.}, r_{1:t}^e) + \hat{\pi}_{t|t} \cdot p(r_{t+1}^e | \Theta_{Rec.}, r_{1:t}^e), \quad (5)$$

where $\hat{\pi}_{t|t} = E(S_t = Rec. | \mathcal{F}_t)$ represents the probability of being in the recessionary regime at time t conditional on the information set available at time t .⁸ In essence, equation (5) is a Markov-switching mixture of individual predictive densities, in which the forecasting weights assigned to each individual model are time varying and correspond to the probability of being in a period of broad macroeconomic expansion, $1 - \hat{\pi}_{t|t}$, or contraction, $\hat{\pi}_{t|t}$.

The expected returns estimate is readily available from the aggregate predictive density in (5),

$$\hat{\mu}_{t+1|t} = E_t[r_{t+1}^e] = \int \int r_{t+1}^e p(r_{t+1}^e | \Theta, r_{1:t}^e) p(\Theta | r_{1:t}^e) d\Theta dr_{t+1}^e, \quad (6)$$

where I drop the Θ argument by integrating out the parameter vector from its posterior distribution, $p(\Theta | r_{1:t}^e)$. The filtered one-month ahead excess return forecast, $\hat{\mu}_{t+1|t}$, is the key estimate of this paper, and I use it for the portfolio choice in Section 5.

3.2 *Inspecting the Mechanism*

To improve the clarity of the discussion that follows, throughout this section I use the parameter estimates in Table 4 obtained from the in-sample estimation of the model, that is, using all data available from January 1965 to December 2019. I describe the estimation procedure and the dataset used in Section 4.

State-dependent autocorrelation in returns. The model imposes enough structure to identify a state-dependent autocorrelation in returns. Conditioning on a specific economic regime, S_t , equations (2) and (3) imply that the serial correlation in returns is given by

$$corr_{S_t}(r_{t+1}^e, r_t^e) = \rho \cdot \phi(S_t) + \rho_{\mu,r} \cdot \sqrt{\phi(S_t)(1 - \phi(S_t))(1 - \rho^2)}, \quad (7)$$

⁸Formally, $\hat{\pi}_{t|t}$ is given by $\hat{\pi}_{t|t} = \int \int \mathbb{1}_{S_t=1} p(S_t | \Theta_{bc}, \mathcal{F}_t) p(\Theta_{bc} | \mathcal{F}_t) d\Theta_{bc} dS_t$. Section 3.3 provides the details.

where I further define the parameter $\phi(S_t)$ as the fraction of variance in excess returns explained by μ_t (i.e., the conditional R -squared),

$$\phi(S_t) = \frac{Var_{S_t}(\mu_t)}{Var_{S_t}(r_t^e)}. \quad (8)$$

Depending on the parameter values, the model-implied autocorrelation can be positive or negative. The negative component of (7) arises if unexpected returns and shocks to expected returns are negatively correlated (i.e., $\rho_{\mu,r} < 0$). Empirically, I find that this is the case. The posterior median estimate for $\rho_{\mu,r}$ is equal to -0.95, which is consistent with implied estimates in the literature (e.g., [Campbell 1991](#); [Campbell and Ammer 1993](#)). An intuitive way to understand this negative correlation is by imposing the present-value relation to decompose unexpected returns, ε_{t+1}^r , into revisions in expectations about the current and future dividend growth, CFN_{t+1} , and revisions in expected future returns, DRN_{t+1} , (i.e., $\varepsilon_{t+1}^r = CFN_{t+1} - DRN_{t+1}$). Broadly, suppose expected dividend growth rates are not too volatile. In that case, unexpected returns are driven primarily by expected return shocks. Therefore, the correlation between these two shocks will be negative and close to minus one. [Pástor and Stambaugh \(2009\)](#) provide conditions for CFN_{t+1} and DRN_{t+1} under which $\rho_{\mu,r} < 0$.⁹

The positive component of (7) comes from the persistence and variability of expected returns. As shown in Table 4, the posterior median of ρ is 0.97, pointing towards reasonably persistent expected excess returns, consistent with [Fama and French \(1988\)](#), [Ferson, Sarkissian, and Simin \(2003b\)](#), [Pástor and Stambaugh \(2009\)](#), and [Binsbergen and Koijen \(2010\)](#). Moreover, the fraction of variance in excess returns explained by μ_t , as measured by $\phi(S_t)$, is state dependent and higher during recessions. The posterior median value for $\phi(S_t = \text{Exp.})$ during expansions is around 0.92%, suggesting that expected returns are fairly stable during these times. In contrast, expected returns are much more volatile during recessions, with $\phi(S_t = \text{Rec.})$ values of around 14%. This business cycle variation in $\phi(S_t)$ could be consistent with a variety of leading asset-pricing models in which variations in expected returns are usually linked to movements in state variables

⁹In essence, $\rho_{\mu,r}$ is expected to be negative if discount rate shocks are more important than cash flow shocks for explaining variations in unexpected returns. Alternatively, if cash flow and discount rate news are negatively correlated, it follows that $\rho_{\mu,r} < 0$.

linked to the economy (e.g., [Campbell and Cochrane 1999](#); [Bansal and Yaron 2004](#); [Gabaix 2012](#)). These state variables generally vary over time in a persistent fashion, giving rise to persistent expected returns. Moreover, for $\phi(S_t)$ to vary across economic regimes, we would need these state variables that determine risk premia to fluctuate differently across economic regimes and be more volatile during recessions.¹⁰

The central finding is that the relative strength of these two offsetting components in equation (7) varies with the business cycle. In expansions, the positive component coming from persistent and mildly variable expected returns is overwhelmed by the countervailing effect of return innovations that are highly negatively correlated with shocks to expected returns. In this state, a positive return shock induces a long string of mild negative returns that cumulatively bring prices back towards their original price. As shown in Panel A of Table 3, this generates a small negative return autocorrelation; that is, we have mean reversion in returns.¹¹ Conversely, in recessions, when expected returns are much more volatile, the positive component of (7) dominates, generating a strong positive return autocorrelation. Therefore, in recessions, a negative shock to returns sets off a string of negative returns, which add momentum to the original price decrease. Notably, as shown in Panel A of Table 3, the widths of the 90% credible intervals for the model-implied autocorrelations are reasonably tight, highlighting that the model imposes enough econometric structure to sharpen inference about the state-dependent autocorrelation in returns.

The OLS regression coefficients in Table 2 of Section 2 suggest a strikingly high amount of autocorrelation conditional on low economic growth periods and mild negative autocorrelation in good economic times. For instance, when the common growth series is

¹⁰Leading asset-pricing models, per se, do not generate different degrees of return predictability across economic regimes. However, simple extensions might generate variations in return predictability. For instance, [Gargano \(2013\)](#) modifies the long-run risk model of [Bansal and Yaron \(2004\)](#) by linking expected consumption volatility to left tail events in the cash flow processes. This extension successfully generates business cycle fluctuations in return predictability. Similarly, [Cujean and Hasler \(2017\)](#) propose an equilibrium model that generates higher stock return predictability in bad times. In their model, investors use different models to forecast the business cycle, which leads them to interpret the same news in different ways. Return predictability rises in bad times because this is when disagreement spikes, and in their model, variation in disagreement commands a positive risk premium.

¹¹I report population values. To do so, I sampled 10,000 draws of the model parameters from its posterior distribution. I then evaluated equation (7) at those values, which, for each regime, gave me 10,000 different values of the model-implied autocorrelation in returns. Finally, I computed the 5th, 50th, and 95th percentiles of this posterior distribution.

Table 3
Model-Implied Serial Correlation in Returns and Kalman Gains

This table reports the 5, 50, and 95 percentiles for the model-implied autocorrelation in returns and the Kalman gain. To compute these statistics, I sampled model parameters from their posterior distribution. The estimation sample ranges from January 1965 to December 2019.

	Panel A			Panel B		
	Serial correlation in returns			Kalman gain		
	5%	50%	95%	5%	50%	95%
Expansion	-1.40	-1.317	-1.23	-0.014	-0.013	-0.011
Recession	5.75	6.21	6.61	0.054	0.059	0.066

below its 12-month moving average, the point estimate is 0.09 ($= \hat{b}_0 + \hat{b}_{\text{Common.Growth}}$) with a t-statistic of 1.94 on the difference. Alternatively, when the series is above its moving average, the point estimate equals -0.01 ($= \hat{b}_0$). In Appendix A.2, I show that such high and low values are statistically indistinguishable from the model-implied regression coefficients based on simulations from its posterior distribution.¹² Significantly, as shown next, the estimate of expected returns improves upon the OLS regression in equation (1) because it not only uses the previously realized return but also considers the entire history of lagged returns.

Realized returns are informative about expected returns. When expected returns are not observable, realized returns are noisy signals about the past (persistent) expected returns. To show how lagged excess returns are informative about future excess returns, I consider the particular case in which the economy stays in one economic regime throughout. Once I condition on the economic regime S_t , the model is linear. Hence, I can use Kalman-filtering steps to track the Gaussian distribution of the equity premium

¹²I computed this predictive check as follows. First, I draw 10,000 parameters from the posterior distribution of the model described by equations (2) and (3) and of the model described in Section 3.3 that specifies the process for the business cycle. For each parameter draw, I simulate a trajectory for returns and the common growth component. Both of these trajectories have the same number of observations as in the actual sample (660 monthly observations). I then compute the model-implied regression coefficients by running the following specification on simulated data: $r_{t+1}^e = a_0 + (b_0 + b_1 \cdot I_{\text{Comm.Growth},t}) r_t^e + \epsilon_{t+1}$, where $I_{\text{Comm.Growth},t}$ takes a value of one whenever the simulated common growth component is low relative to its 12-month moving average and zero otherwise.

μ_t^j , where the j annotation indicates that I am conditioning on state j .

The Kalman filter operates in two main steps. The first consists of computing the expected value of μ_t^j , conditional on the information set at time $t - 1$,

$$\mu_{t|t-1}^j = E [\mu_t | \Theta_j, r_{1:t-1}^e, S_t = j] = \mu_0 + \rho (\mu_{t-1|t-1}^j - \mu_0). \quad (9)$$

The second step updates the previous estimate, $\mu_{t|t-1}^j$, by incorporating the new information that arrived at time t ,

$$\mu_{t|t}^j = E [\mu_t | \Theta_j, r_{1:t-1}^e, r_t^e, S_t = j] = \mu_{t|t-1}^j + \kappa_{j,t} (r_t^e - \mu_{t|t-1}^j), \quad (10)$$

where $\kappa_{j,t}$ represents the Kalman gain, which is a function of time and the parameters linked to state j . Hence, the new estimate of expected returns is a weighted average of the previous forecast and the new information, that is, the forecast error $(r_t^e - \mu_{t|t-1}^j)$. The weight given to this new information is precisely determined by the Kalman gain $\kappa_{j,t}$.

I write the Kalman gain for period one, for which we have an analytical expression, to build intuition on how the model parameters affect $\kappa_{j,1}$,¹³

$$\kappa_{j,1} = \frac{\frac{\sigma_{\mu,j}^2}{1 - \rho^2} + \sigma_{r\mu,j}^2}{\frac{\sigma_{\mu,j}^2}{1 - \rho^2} + \sigma_{r,j}^2 + 2\sigma_{r\mu,j}^2}. \quad (11)$$

This equation has a straightforward interpretation. When expected returns explain a substantial fraction of the variance in excess returns (i.e., $\frac{\sigma_{\mu,j}^2}{1 - \rho^2}$ is large relative to $\sigma_{r,j}^2$), a lower-than-expected realized excess return suggests that past expected returns are relatively low. If expected returns are persistent, then they will also be low going forward, which leads to positive updates about future returns in response to prior return surprises. That is, the Kalman gain is positive. Conversely, when fluctuations in expected returns do not explain much of the variations in realized excess returns (i.e., $\frac{\sigma_{\mu,j}^2}{1 - \rho^2}$ is low relative to $\sigma_{r,j}^2$), this positive effect is dominated by the negative correlation between return

¹³To compute this expression, I initialized the Kalman filter from the unconditional distribution for μ^j .

innovations and expected return news, $\sigma_{r\mu,j}^2$. In this case, the lower-than-expected excess return is more likely to be attributed to positive shocks to expected returns. Therefore, past (persistent) expected returns are relatively high, implying future high discount rates. That is, the Kalman gain is negative so that recent returns that were lower than expected lead to an increase in the forecasts for future returns. Panel B of Table 3 shows the posterior distribution of the Kalman gain in each state. Notably, under the posterior estimates, news about r_t^e is negatively weighted in expansionary periods but positively weighted in recessionary times.

To gain further insight into this mechanism, I further express $\mu_{t|t}^j$ in (10) as a weighted average of past excess returns rather than past return surprises. To do so, I substitute the Kalman gain at time t , $\kappa_{j,t}$, with the steady-state Kalman gain,¹⁴ κ_j , and replace the unconditional mean of the equity premium, μ_0 , with its sample mean. I then combine equations (9) and (10) and iterate backward:

$$\mu_{t|t}^j = \sum_{s=1}^t \delta_{s,t}^j \cdot r_s^e \quad \text{with} \quad \sum_{s=1}^t \delta_{s,t}^j = 1, \quad (12)$$

where the weights on past returns, $\delta_{s,t}^j$, are a function of the steady-state Kalman gain. Figure 3 plots these weights on lagged returns for two different cases. The first uses the steady-state Kalman gains in the expansionary regime. Under this case, returns exhibit time-series reversal (see Panel A of Table 3), the Kalman gain is negative (see Panel B of Table 3), and recent past returns are negatively weighted to predict future returns (see the dashed line in Figure 3). Because the weights must sum to one, more distant lags are positively weighted. The second represents the weights using the posterior median estimates in the recessionary regime. Here, returns exhibit time-series momentum, the Kalman gain is positive, and, hence, past returns are positively weighted to predict future returns. However, recent returns are more heavily weighted (see the black line in Figure 3).¹⁵

¹⁴The steady-state Kalman filter gain for regime j is given by $k_j = (P_j + \sigma_{r\mu,j}^2)(P_j + \sigma_{r,j}^2 + 2\sigma_{r\mu,j}^2)^{-1}$, where P_j solves the following Ricatti equation: $P_j = \rho^2 (P_j - (P_j + \sigma_{r,j}^2)(P_j + 2\sigma_{r\mu,j}^2 + \sigma_{r,j}^2)(P_j + \sigma_{r,j}^2)) + \sigma_{\mu,j}^2$. See [Simon \(2006\)](#) for a textbook treatment on the Kalman filter for when the shocks in the measurement equation and state-transition equation are correlated, and for convergence properties of the Ricatti equation.

¹⁵This case is akin to the time-series momentum strategy (e.g., [Moskowitz et al. \(2012\)](#); [Hurst, Ooi,](#)

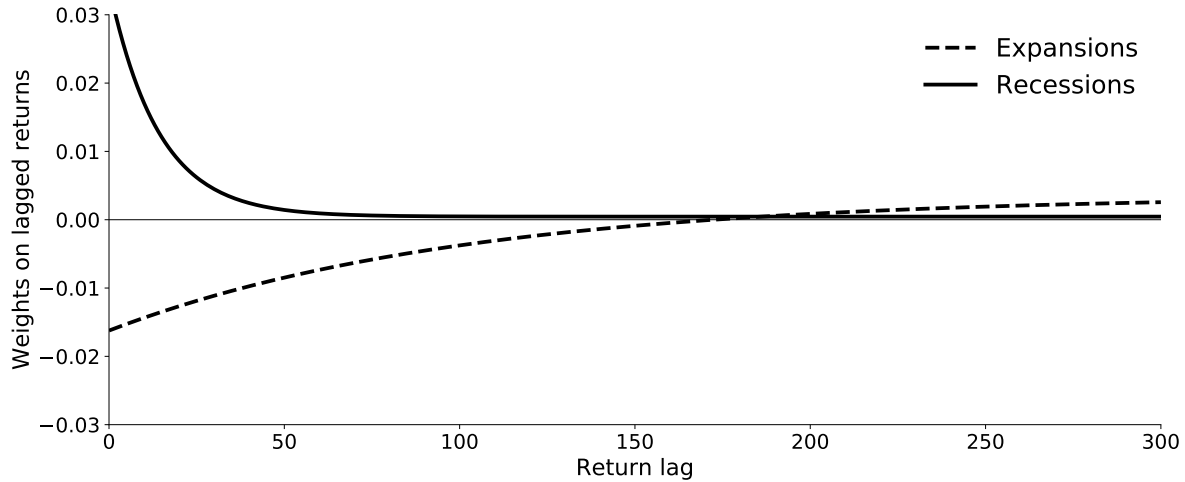


Fig. 3. **State-dependent weights on lagged returns.** This figure shows the weights on lagged excess returns when computing the measure of expected returns. I derive these weights by conditioning on each economic regime and iterating backward the prediction and updating steps in the Kalman filter algorithm. To do so, I substitute the Kalman gain at time t for the steady-state Kalman gain and use the sample mean as an estimate of the unconditional equity premium. To compute the steady-state Kalman gain, I used the posterior median estimates shown in Table 4.

Throughout this section, to build intuition into the model mechanism, I considered the case in which the economy remained in one single regime. However, the filtered measure of expected returns, $\hat{\mu}_{t+1|t}$, in equation (6) dynamically combines these predictive densities, which differ in the way they aggregate the information contained in lagged returns using time-varying weights. These dynamic model weights correspond to the probability of being in an expansion or recession. From an econometric perspective, the measure $\hat{\mu}_{t+1|t}$ accommodates parameter instability (e.g., [Paye and Timmermann, 2006](#); [Rapach and Wohar, 2006](#); [Spiegel, 2008](#)) and allows for large breaks in the model coefficients by tying them to business cycle turning points.

Alternative model specifications. In the benchmark specification model of returns described in equations (2), (3), and (4), I make two main assumptions. First, I allow for expected returns shocks and unexpected returns to be correlated as measured by the parameter $\rho_{\mu,r}$. Second, I allow for the fraction of variance in excess returns explained by μ_t to take different values in recessions and expansions, as measured by $\phi(S_t = Rec.)$ and $\phi(S_t = Exp.)$, respectively. In the sections that follow, I show that these two ingredients

and [Pedersen \(2017\)](#)), which equally weights lag returns over a short period of time to predict future returns. The main difference is that the forecasting weights used here are optimal in the sense that they are derived from optimal filtering techniques.

generate a good model of expected returns.

In that model, I also assume that $\rho_{\mu,r}$ and the persistence of expected returns, ρ , are stable across regimes. But, it could well be the case that $\rho_{\mu,r}$ and ρ are state dependent and vary with the business cycle.¹⁶ Moreover, as shown in equation (7), if ρ and $\rho_{\mu,r}$ increase during recessions and decrease during expansions, then these time-varying parameters could account for the state-dependent autocorrelation in returns. In Appendix A.2, I explore these alternative specifications. I find that $\rho_{\mu,r}$ is stable across regimes. That is, I cannot reject the null hypothesis that $\rho_{\mu,r}$ takes different values across the business cycle. With respect to the persistence of expected returns, I find that ρ is slightly lower during recessions. Based on predictive checks, I find that this specification generates an autocorrelation in returns that is too low relative to the data moments shown in Table 2. Finally, I perform a formal specification test and find that the benchmark model of returns is preferred over models in which ρ and $\rho_{\mu,r}$ depend on the state S_t .

3.3 State of the Business Cycle

Next, I describe how I identify the economic regime, S_t , which is an essential input in the model of stock returns.

Business cycle fluctuations are determined by analyzing the comovement of a broad range of macroeconomic indicators. Let $y_{i,t+1}$ denote the i -th variable at time $t + 1$ and let it evolve as

$$y_{i,t+1} = \gamma_i z_{t+1} + e_{i,t+1} \quad \text{for} \quad i = 1, \dots, N, \quad (13)$$

where z_{t+1} is the latent common growth component, whereas $e_{i,t+1}$ is the latent specific component. The parameter γ_i measures the exposure of the i -th variable to the state variable z_{t+1} . Moreover, I am assuming that there are a total of N macroeconomic

¹⁶Economically, if $\rho_{\mu,r}$ varies across regimes, it would imply that the contributions of discount rate shocks and cash flow shocks in explaining the variance of return news are state dependent. Broadly, if return news is driven primarily by discount rate shocks (cash flow shocks) in one state, then the correlation will be close to minus one (zero) during that state. In theory, $\rho_{\mu,r}$ can even be positive if, for instance, discount rate shocks and cash flow shocks have a positive correlation that is sufficiently high. Alternatively, if ρ is lower during recessions, it would imply that risk premia shocks die out more quickly during bad times relative to good times.

indicators from which we can identify the latent variables. Keep in mind that the y_i 's are observable but z and the e_i 's are not.

Based on the regime-switching model of [Hamilton \(1989\)](#) and [Chauvet \(1998\)](#), I assume that z_{t+1} is generated by an AR(1) process with Markov-switching deviations in the conditional mean, persistence, and volatility:

$$z_{t+1} = \mu_z(S_{t+1}) + \phi_z(S_{t+1})(z_t - \mu_z(S_t)) + \sigma_z(S_{t+1})\epsilon_{z,t+1}, \quad (14)$$

with $\epsilon_{z,t+1} \sim \text{i.i.d. } N(0, 1)$. S_t is the latent economic regime indicator used throughout Section 3. The regime-switching conditional mean, $\mu_z(S_{t+1})$, and volatility, $\sigma_z(S_{t+1})$, attempt to capture transitions from periods of broad economic growth and low volatility, called expansions, to periods of broad economic contractions and high volatility, called recessions.¹⁷ I further assume that transitions between regimes evolve according to a two-by-two transition matrix and that the specific component follows a simple AR(1) process

$$e_{i,t+1} = \mu_i + \psi_i e_{i,t} + \sigma_i \epsilon_{i,t+1}, \quad (15)$$

with $\epsilon_{i,t+1} \sim \text{i.i.d. } N(0, 1)$. The shocks $\epsilon_{z,t+1}$ and $\epsilon_{i,t+1}$ are independent of one another for all t and i .

Probability of a recession. Let $p(S_t|\Theta_{bc}, \mathcal{F}_t)$ denote the posterior distribution of the state S_t . This distribution conditions on the vector of model parameters, Θ_{bc} , associated with the process specified by equations (13), (14), and (15), and on the time t information set $\mathcal{F}_t$, which includes the macro and financial variables up to time t . Given $p(S_t|\Theta_{bc}, \mathcal{F}_t)$, we can compute the probability of being in a recession as follows,

¹⁷For normalization purposes, I define $\mu_z(S_{t+1}) = \mu_{z0} + \mu_{z1} \cdot \mathbb{1}_{\{S_{t+1}=Rec.\}}$, where $\mathbb{1}_{\{S_{t+1}=Rec.\}}$ is an indicator variable that equals one in the recessionary regime and zero otherwise. I further impose the following restriction: $\mu_{z1} < 0$. Hence, if $\mathbb{1}_{\{S_{t+1}=Rec.\}}$ equals zero, I say that the economy is in a period of broad economic growth at time $t+1$ (i.e., $\mu_{z0} > \mu_{z0} + \mu_{z1}$), whereas if $\mathbb{1}_{\{S_{t+1}=Rec.\}}$ equals one, I say that the economy is in a contractionary period in $t+1$. Similarly, I define $\sigma_z^2(S_{t+1}) = \sigma_{z,0}^2(1 + h_{\sigma_z} \cdot \mathbb{1}_{\{S_{t+1}=Rec.\}})$ and impose $h_{\sigma_z} > -1$ to guarantee that the conditional volatility remains positive in the recessionary period.

$$\hat{\pi}_{t|t} = E(S_t = Rec. | \mathcal{F}_t) = \int \int \mathbb{1}_{\{S_t = Rec.\}} p(S_t | \Theta_{bc}, \mathcal{F}_t) p(\Theta_{bc} | \mathcal{F}_t) d\Theta_{bc} dS_t, \quad (16)$$

where $\mathbb{1}_{\{S_t = Rec.\}}$ is an indicator variable that equals one if the economy is in an economic contraction at time t and zero otherwise. The estimate $\hat{\pi}_{t|t}$ represents the probability of being in the recessionary regime at time t conditional on the information set available at time t . I use $\hat{\pi}_{t|t}$ in the equation for expected returns in equation (6), and I plot it in Panel A of Figure 1.

Time aggregation. The empirical analysis that follows is tailored towards monthly data. However, I sharpen inference about $\hat{\pi}_{t|t}$ by considering the information contained in quarterly data. To link the monthly model-implied macro variables with the observed quarterly data, I use the following function:

$$\Delta y_{i,t+1}^{grt,o} = \sum_{j=1}^5 \frac{3 - |j - 3|}{3} \Delta y_{i,t+2-j}, \quad (17)$$

such that the monthly model variables $\Delta y_{i,t+2-j}$ time-aggregate to the observed quarterly growth rates $\Delta y_{i,t+1}^{grt,o}$, which are indicated by the o -subscript.¹⁸

Overall, the combination of data available at different frequencies, where each series starts at a different period; temporal aggregation; and accounting for missing observations because of the real-time nature of the data set in conjunction with the regime changes, leads to a high-dimensional state-space model.

3.4 Posterior Inference

To conduct posterior inference, I write the likelihood function as follows:

$$p(r_{1:T}^e | \Theta, \pi_{1:T}) = \prod_{t=2}^T \{(1 - \pi_t) p(r_t^e | \Theta, r_{1:t-1}^e, S_t = 0) + \pi_t p(r_t^e | \Theta, r_{1:t-1}^e, S_t = 1)\}. \quad (18)$$

¹⁸Older papers, [Hansen and Sargent \(1983\)](#) and [Heaton \(1995\)](#), and more recent ones, [Bansal, Kiku, and Yaron \(2016\)](#) and [Schorfheide, Song, and Yaron \(2018\)](#), account for time aggregation in their estimation in an asset-pricing setting.

Given the prior $p(\Theta)$ and the likelihood function in (18), Bayes' theorem provides the posterior aggregate density function

$$p(\Theta|r_{1:T}^e, \pi_{1:T}) \propto p(r_{1:T}^e|\Theta, \pi_{1:T})p(\Theta). \quad (19)$$

For posterior inference, I use a Bayesian Markov Chain Monte Carlo (MCMC) method. To focus the discussion on the forecasting results, I relegate details on the state-space models, the estimation method, regularity conditions, and identification of the model parameters to the appendix.

4 Data and Estimation Results

When computing the measure of expected returns in equation (6), it is critical to avoid any look-ahead biases arising from both the dataset used and the estimation procedure. I address this concern in two ways. First, I restrict the information set to those observations that were available to investors at each point in time. Second, I estimate the model out of sample in Section 5.

4.1 Data

I build a large dataset of real-time variables. Real-time data simulates the actual information set available to investors and accounts for the fact that macro variables are sometimes released with a delay or receive substantial revisions by the statistical agencies following publication.¹⁹

I selected nine real-time macroeconomic series that measure different parts of the economy and are available at the monthly or quarterly frequency. Specifically, I consider variables related to real output and income (e.g., industrial production, gross domestic product), employment and hours (e.g., initial jobless claims), consumer spending (e.g.,

¹⁹The fact that macro time series receive substantial revisions could potentially affect my estimates of the regimes and, if I did not take them into account, give me an unfair advantage relative to a real-time investor. For example, consumption growth rates change nearly 5 percentage points (quarterly, at annual rates) across different reporting vintages. I take the vintage data from the Philadelphia Fed's website: <https://www.philadelphiafed.org/surveys-and-data/real-time-data-research/survey-of-professional-forecasters>

Table 4
Prior and Posterior Estimates

This table reports the 5 and 95 percentiles of the model parameters' prior distributions along with the 5, 50, and 95 percentiles of their posterior distributions. The priors for the parameters are fairly agnostic. With respect to the assumed prior distribution, N, U, IG, denote the normal, uniform, and inverse gamma distribution, respectively. The estimation sample runs from January 1965 to December 2016.

	Prior distribution			Posterior distribution		
	Distr.	5%	95%	5%	50%	95%
μ_0	<i>N</i>	0.0034	0.0094	0.0053	0.0065	0.0076
ρ	<i>U</i>	0.80	0.99	0.958	0.970	0.982
$\rho_{\mu,r}$	<i>U</i>	-1.00	0.00	-0.984	-0.955	-0.926
$\phi(S_t = Rec.)$	<i>U</i>	0.00	0.20	0.093	0.149	0.181
$\phi(S_t = Exp.)$	<i>U</i>	0.00	0.20	0.006	0.009	0.015
$\sqrt{Var(r_t^e S_t = Rec.)}$	<i>IG</i>	0.0389	0.2890	0.043	0.060	0.072
$\sqrt{Var(r_t^e S_t = Exp.)}$	<i>IG</i>	0.0389	0.0389	0.0389	0.0389	0.0389

consumption expenditure), investment (e.g., private residential/nonresidential fixed investment), and inflation (e.g., the Consumer Price Index). All the series start at different periods, but all end in December 2019. The macro and financial indicators are the data counterpart of $y_{i,t}$ from Section 3.3, and I provide a detailed description of each in the Appendix B. Finally, the data for the excess market return, r^e , comes from Kenneth R. French's website.

4.2 Estimation Results

Table 4 provides percentiles of the prior and posterior distributions. I obtained these estimates in the in-sample estimation of the model using data between January 1965 and December 2019. Moreover, to facilitate an interpretation of the model parameters, I re-expressed the parameters in the covariance matrix, $\Sigma(S_t)$, in (4) in terms of the conditional R-squared, $\phi(S_t)$, in equation (8) and the conditional variance of realized excess returns $Var(r_t^e | S_t)$. The priors are fairly agnostic. For instance, the prior for $\rho_{\mu,r}$, the parameter that determines the correlation between unexpected returns and shocks to expected returns, is a uniform distribution on the interval $[-1, 0]$. For the

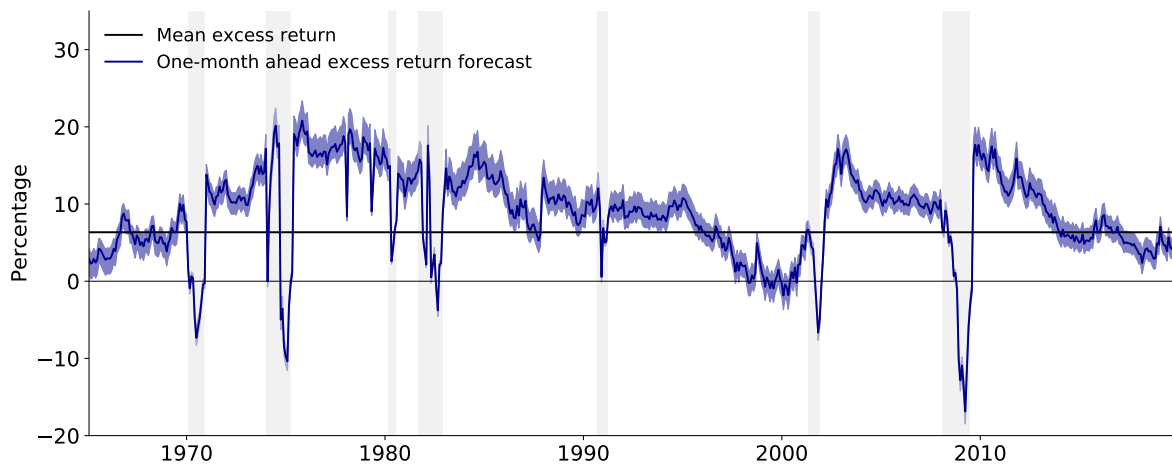


Fig. 4. **One-month ahead excess return forecast.** The figure shows the filtered series of expected returns ($\hat{\mu}_{t+1|t}$). The blue line represents the posterior median estimates, and the blue shaded areas denote the 90% credible intervals. The light gray shaded bars indicate NBER recession dates. The dark line graph denotes the mean equity premium computed using data from January 1965 to December 2019.

parameters $\phi(S_t = Exp.)$ and $\phi(S_t = Rec.)$, I use priors that are uniform on the interval $[0, 0.2]$, thereby allowing for expected returns to be equally volatile in the recessionary and expansionary regimes. Concerning the posterior distribution, the most important findings are that (1) discount rates are persistent, (2) return surprises and expected return news are strongly negatively correlated, and (3) discount rates are relatively stable in expansions and much more volatile during recessions.

Figure 4 plots the filtered one-month ahead excess return forecast, $\hat{\mu}_{t+1|t}$, as well as the mean excess return. The light gray bars indicate NBER recession dates. The figure shows that the filtered expected return series fluctuates around its unconditional mean. During recessions, the model updates forecasts of future excess returns in the same direction as recent realized returns. Because the market has usually recently crashed, the estimate of expected returns is negative. It remains low for several months following the onset of recessions before rising to higher-than-average levels once the economy starts to recover. Notably, negative expected returns are fairly rare and are concentrated in bad economic times. This pattern is my most striking finding. In expansions, the usual Campbell-Shiller logic prevails, and recent realized positive returns are associated with negative updates to expected returns. Prominently, the sequence of positive realized returns during the dot-com bubble in the late 1990s decreased the filtered expected return almost to values statistically indistinguishable from zero. That is, the reason for

the higher-than-expected realized returns during that period is that expected returns fell, causing an increase in prices that ultimately induces the positive returns.

5 Economic Relevance

This section shows that $\hat{\mu}_{t+1|t}$ is a good measure of conditional expected returns. I treat $\hat{\mu}_{t+1|t}$ as the conditional beta in a standard conditional CAPM framework. At the end of each month, I increase risk-taking when the estimate of expected returns is high and decrease it when $\hat{\mu}_{t+1|t}$ is low,

$$r_{t+1}^{e,\mu} = c \hat{\mu}_{t+1|t} r_{t+1}^e, \quad (20)$$

where r_{t+1}^e is the excess market return and the constant c controls for the average risk exposure of the strategy. To facilitate the interpretation, I follow [Moreira and Muir \(2017\)](#) and choose c such that the managed return, $r_{t+1}^{e,\mu}$, has the same unconditional volatility as the excess market return.

I perform a series of tests on $r_{t+1}^{e,\mu}$ and $\hat{\mu}_{t+1|t}$ to establish the economic relevance of my findings. Importantly, for all the results that follow, I estimate $\hat{\mu}_{t+1|t}$ on recursively increasing samples of real-time data. That is, at any given point in time, t , I only use data up to date t to estimate $\hat{\mu}_{t+1|t}$ as well as the economic regime S_t . Moreover, as described in Section 4, I am only using those observations that were publicly available at any point t . I start my out-of-sample computations in January 1980. Using data between January 1965 and December 1979 to estimate the model, I compute the expected return for January 1980. I then use data between January 1965 and January 1980 to estimate the model and compute the forecast for February 1980. I proceed recursively in this way up until December 2019.

5.1 Main Result

I focus on time-series regression of the form

$$r_{t+1}^{e,\mu} = \alpha + \beta r_{t+1}^e + \epsilon_{t+1}, \quad (21)$$

Table 5
Performance Evaluation: Alphas

In Panel A, I run times-series regressions of various trading strategies on the market excess return $r_{t+1}^{e,\mu} = \alpha + \beta r_{t+1}^e + \epsilon_{t+1}$. The strategies adjust the conditional beta based on an estimate of conditional expected returns. In Panel B, I include the time-series momentum factor of [Moskowitz et al. \(2012\)](#) (TMom) and the volatility-managed factor of [Moreira and Muir \(2017\)](#) (TVol) as additional controls in the regression. Standard errors adjust for heteroskedasticity and autocorrelation. I report the t-statistics in brackets. To facilitate an interpretation, I annualize all returns, expressed as a percentage per year, by multiplying monthly returns by 1200. All models are estimated out of sample on recursively increasing samples of data. The forecast evaluation period runs from January 1980 to December 2019.

Controls	Main (1)	Importance of the business cycle			
		NBER foresight (2)	Steady state (3)	Exp. $S_t = 0$ (4)	Rec. $S_t = 1$ (5)
Panel A: CAPM Alpha					
Mkt	5.46 [2.77]	5.89 [2.33]	1.38 [1.82]	0.82 [1.73]	3.91 [1.70]
Panel B: Multi-factor Alphas					
Mkt – TMom	4.56 [3.32]	4.86 [3.21]	1.82 [1.92]	1.58 [1.70]	1.98 [1.64]
Mkt – TVol	4.44 [3.19]	5.01 [2.57]	1.46 [1.97]	1.17 [1.64]	2.19 [1.65]
Mkt – TMom – TVol	4.02 [3.55]	4.51 [3.25]	1.71 [1.88]	1.58 [1.68]	1.26 [1.62]

where the parameter of interest is the unconditional alpha, which primarily measures the covariance between the one-month ahead excess return forecast, $\hat{\mu}_{t+1|t}$, and the unobservable market's conditional risk premium. Specifically, Equation 4 in [Lewellen and Nagel \(2006\)](#) decomposes (approximately) the unconditional alpha as

$$\alpha \approx \text{cov}(c\hat{\mu}_{t+1|t}, \gamma_t) - \frac{\gamma}{\sigma_M^2} \text{cov}(c\hat{\mu}_{t+1|t}, \sigma_t^2) \quad (22)$$

where γ_t and σ_t^2 denote the market's conditional risk premium and variance, while the corresponding unconditional moments are given by γ and σ_M^2 , respectively. Thus, if $\hat{\mu}_{t+1|t}$ covaries only with γ_t (or if γ_t is only weakly related to σ_t^2 , as results in [Moreira](#)

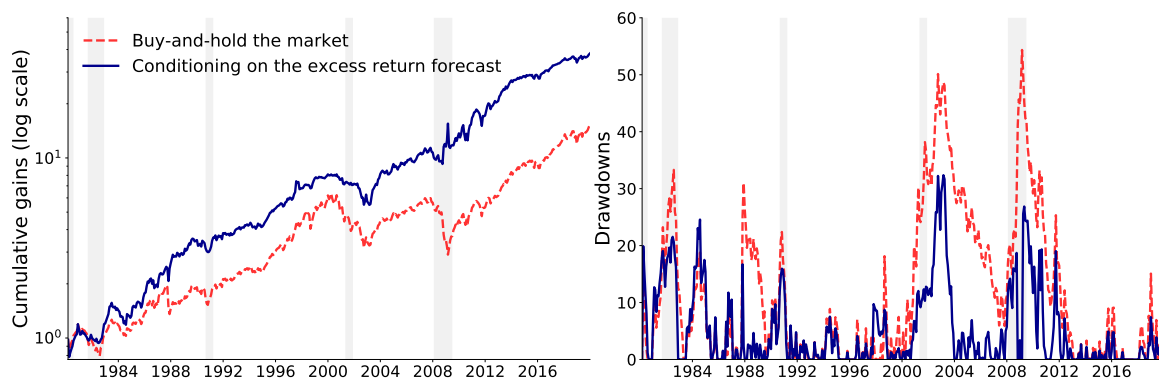


Fig. 5. **Cumulative returns and drawdowns.** The left panel plots cumulative monthly gains for the main specification (dark blue line) and the static buy-and-hold strategy (red dotted line). The y-axis is on a log scale. The right panel depicts the drawdown of each strategy. Both strategies have the same unconditional volatility. The sample runs from January 1980 to December 2019.

and Muir (2017) suggest), then $\alpha \approx \text{cov}(c\hat{\mu}_{t+1|t}, \gamma_t)$.²⁰

Column (1) of Table 5 reports results for the main specification. Panel A shows that the estimated market α is both economically large (5.46% annualized) and statistically significant (t-statistic = 2.77), thus suggesting that $\hat{\mu}_{t+1|t}$ is positively related with the unobservable market's conditional expected return.

To further quantify the economic relevance of my results, the left panel of Figure 5 depicts the cumulative excess return of the main strategy from 1980 to 2019. As the figure shows, the strategy (dark blue line) generates a relatively steady stream of positive returns that consistently outperform the fully invested buy-and-hold strategy (red dotted line). For instance, given a \$1 investment in January 1980, the strategy would generate a value of around \$38 in excess of the risk-free rate, which surpasses the 15 dollars generated by the strategy that buys and holds the market. Overall, the strategy increases Sharpe ratios over the buy-and-hold approach by around 28%. The right panel of Figure 5 plots the drawdown for both strategies. The main strategy's time under water is usually concentrated in recessionary periods but reverts shortly after. Likewise, the buy-and-hold approach incurs large market losses during recessions; however, it requires a long period to recoup these losses.

²⁰Indeed, the second term in equation (22) has only small effects on α , consistent with Lewellen and Nagel (2006). To see this, I estimate $\text{cov}(c\hat{\mu}_{t+1|t}, \sigma_t^2)$ using my measure of return forecast $c\hat{\mu}_{t+1|t}$ (scaled by c) and the previous month's realized variance computed from daily returns as a proxy for σ_t^2 . The estimate of $\text{cov}(c\hat{\mu}_{t+1|t}, \sigma_t^2)$ is -0.025%, and multiplying by $\frac{\gamma}{\sigma_M^2} = 0.0066/0.046^2 = 3.11$, gives a value of -0.07% monthly.

During recessions, past returns are positively weighted when estimating the current expected return. Therefore, it is important to control for the time-series momentum factor of Moskowitz et al. (2012). Panel B of Table 5 shows that the alpha drops to 4.56% from 5.46% once I control for the market and the time-series momentum factor (TMom), but it is still highly statistically significant with a t-statistic of 3.32. Moreover, the strategy also overlaps with the volatility-managed market factor (TVol) of Moreira and Muir (2017), since both take less risk in bad times (e.g., see Figure 4). Panel B shows that the alpha is equal to 4.44% (t-statistic = 3.19) when controlling for the market and TVol. Finally, controlling for TMom and TVol at the same time shows the alpha dropping by around 1.4 percentage points. Overall, these results indicate that the measure of expected returns, $\hat{\mu}_{t+1|t}$, is related to simple strategies that rely on the positive return autocorrelation and to strategies that avoid high volatility periods. Nevertheless, the alpha estimate of 4.0% (t-statistic = 3.55) is economically significant even after accounting for the TMom and TVol factors.

5.2 Robustness Checks

To assess the robustness of the main result, I perform an extensive battery of tests. I will briefly describe them here and provide the details in the appendix.

I document that the strategy generates similar alphas across different subsample periods and for a wide range of pricing factors. I find that the alphas are similar if I consider both conditional expected returns and time-varying volatility in the strategy's conditional beta. I also show that it produces fairly large alphas under tight leverage and short-selling constraints, even if I preclude the strategy from using them. Moreover, transaction costs would need to be incredibly high to make the strategy unprofitable. I show that the main strategy's performance is also robust to the regression-based timing measures of Treynor and Mazuy (1966) and Henriksson and Merton (1981). The strategy is also profitable if I use utility-based performance measures or the manipulation-proof measures of performance of Goetzmann et al. (2007). I further assess the economic relevance of the two key model ingredients (i.e., $\rho_{\mu,r} < 0$ and $\phi(S_t)$ to vary across economic states) by turning off each one of them separately and reestimating the model

out of sample. I find that the profitability of the strategy disappears once I impose these constraints. Moreover, I document similar profitability for characteristic-sorted portfolios, such as value, size, and momentum. I also extend the out-of-sample analysis and start in 1960 by using lagged information about the NBER recessionary index. I find similar profitability measures for this extended sample period.

Even though the return predictability results in Section 5.1 and the robustness checks discussed so far are out of sample, the sample period contains only five full business cycles. Thus, the general concern is that these results might be spurious or sample specific. To address this concern, in Appendix A.9 I extend the sample by looking at the cross-section of U.S. state portfolios and regional business cycles. This robustness check leads to consistent results.

Taken together, the positive estimates in unconditional alphas and the return predictability results in the appendix indicate that the econometric specification described in Section 3 provides a good measure of expected returns over monthly horizons.

6 Unveiling the Unconditional Alpha

6.1 *How Important Is the State of the Business cycle?*

This section shows that the trading strategy's alpha comes from timely identifying the business cycle state and exploiting the state-dependent autocorrelation in returns.

I start by showing results for the hypothetical scenario that assumes perfect NBER recessionary foresight.²¹ This exercise can be thought of as reflecting a situation in which investors could correctly foresee recessions with the ex post NBER accuracy. Column (2) of Table 5 shows that the ex post NBER economic state provides an increase in alphas relative to the primary strategy, producing, for example, a market alpha of 5.9%. This result is reassuring because it suggests that better estimates of business cycle turning points increase the strategy's performance.

The models under Columns (3), (4), and (5) in Table 5 do not condition on the state of the business cycle. In Column (3), I do not allow for time-varying recessionary

²¹To this end, I estimate the model again using the ex post NBER recessionary index for the state of the business cycle rather than the real-time recessionary probability estimates $\pi_{t|t}$.

probabilities in the expected returns measure. I do this by fixing $\pi_{t|t}$ to its unconditional value of $\pi_{ss} \approx 13\%$ for each period t in equation (6). Alternatively, the forecasts under Column (4) only consider the predictive density associated with the low variability of expected returns (i.e., $\phi(S_t = Exp.)$ in both states, mild reversals in returns, and negative forecasting weights). Finally, the forecasts under Column (5) consider the predictive density associated with the high ϕ (i.e., high $\phi(S_t = Rec.)$ in both states, momentum in returns, and positive forecasting weights).²² Notably, these three alternative models generate positive alphas of around 1.4%, 0.8%, and 3.9% respectively. However, these alphas are considerably smaller than the ones obtained under the main specification.

To help us understand where these alphas are coming from, Panel A of Table 6 reports the out-of-sample R -squared statistic, R_{OOS}^2 , computed over the whole sample period and the R_{OOS}^2 obtained by only using expansionary and recessionary periods.²³ The main specification delivers a strong predictor of excess market returns, with a monthly R_{OOS}^2 of around 1.7% (p -values below 0.01). To further analyze in which periods $\hat{\mu}_{t+1|t}$ outperforms its benchmark, in Figure A4 in the appendix, I follow [Goyal and Welch \(2008\)](#) and report the cumulative squared prediction errors of the benchmark model minus the cumulative squared prediction error of $\hat{\mu}_{t+1|t}$. The figure shows that the slope of the cumulative difference in squared prediction errors is mostly positive over the evaluation period. This result suggests that the predictive power of $\hat{\mu}_{t+1|t}$ is not concentrated in relatively short-lived periods.

Panel A of Table 6 also reports the R_{OOS}^2 obtained using only NBER recessionary and expansionary periods. Consistent with the literature, most of the predictability

²²To compute a measure of expected returns for these alternative models, I take the out-of-sample parameter estimates associated with each regime and filter (using those values) an estimate of expected returns.

²³The out-of-sample R -squared statistic is computed as follows: $R_{OOS}^2 = 1 - \frac{\sum_{t=m+1}^T (r_t^e - \hat{\mu}_{t+1|t})^2}{\sum_{t=m+1}^T (r_t^e - \hat{\mu}_{t+1|t}^{Null})^2}$, where $\hat{\mu}_{t+1|t}$ is the return forecast based on the model specification described in Section 3, while $\hat{\mu}_{t+1|t}^{Null}$ is the return forecast of a benchmark model. The literature on return predictability (e.g., see [Goyal and Welch \(2008\)](#)) commonly uses the prevailing mean forecast, defined as the average excess return computed up to month t , as a benchmark. However, given my Bayesian setting and the use of priors, the natural benchmark is the return forecast of a model that uses equation (2) and (3) but sets μ_t equal to a constant, that is, $r_{t+1}^e = \mu_0 + \epsilon_{t+1}^r$. This constant model serves as a natural benchmark since I use the same prior distribution for μ_0 , and, therefore, addresses the concern that some of the gains in OOS R^2 might be coming from ex post informative priors. I also computed the R_{OOS}^2 using the prevailing mean forecast as a benchmark and obtain very similar results. The reason is that I use “loose” priors throughout.

Table 6
Out-of-Sample R-squared and Drawdown

Panel A shows presents the out-of-sample R_{OOS}^2 across NBER economic states. I compute the out-of-sample R -squared statistic as follows: $R_{OOS}^2 = 1 - \sum_{t=m+1}^T (r_t^e - \hat{\mu}_{t+1|t})^2 / \sum_{t=m+1}^T (r_t^e - \hat{\mu}_{t+1|t}^{Null})^2$ where $\hat{\mu}_{t+1|t}$ is the return forecast based on the model specification described in Section 3, while $\hat{\mu}_{t+1|t}^{Null}$ is the return forecast of a benchmark model. As a benchmark, I use the return forecast of a model that uses equation (2) and (3) in the main text, but sets μ_t equal to a constant, that is, $r_{t+1}^e = \mu_0 + \epsilon_{t+1}^r$. I report p-values in parenthesis and R_{OOS}^2 as percentages. Panel B reports average maximum drawdowns across NBER economic states. The forecast evaluation period runs from January 1980 to December 2019.

Economic regime	100%- stocks (1)	Main (2)	Importance of the business cycle			
			NBER foresight (3)	Steady state (4)	Exp. $S_t = 0$ (5)	Rec. $S_t = 1$ (6)
Panel A: Out-of-sample R-squared R_{OOS}^2						
Total	–	1.671	1.683	0.656	0.493	-1.235
	–	(0.006)	(0.003)	(0.057)	(0.100)	(0.312)
Expansions	–	0.960	0.583	0.685	0.928	-3.361
	–	(0.021)	(0.033)	(0.058)	(0.038)	(0.825)
Recessions	–	3.769	4.930	0.570	-0.789	5.044
	–	(0.049)	(0.012)	(0.280)	(0.484)	(0.064)
Panel B: Average maximum drawdowns						
Total	11.21	5.097	4.695	6.278	5.778	9.383
Expansions	9.49	4.069	3.688	4.268	3.746	9.898
Recessions	24.23	12.862	12.297	21.465	21.120	12.042

comes from recessionary periods ($R_{OOS, Rec.}^2 \approx 4.0\%$).²⁴ In contrast to this literature, the striking degree of the strategy's predictability predominantly stems from being able to also predict market returns during expansions with an R_{OOS}^2 of around 1%.²⁵ Essentially, the main strategy allows for state-dependent time variation in the return forecast, thereby stabilizing the forecast and preventing overfitting in precisely those periods when expected returns do not move much.

²⁴See respectively, [Dangl and Halling \(2012\)](#), [Henkel, Martin, and Nardari \(2011\)](#), [Rapach, Strauss, and Zhou \(2010\)](#), and [Rapach, Zhou, et al. \(2013\)](#).

²⁵These are large statistical values for out-of-sample predictions, comparable in magnitude to the unconditional estimates reported in previous literature, such as the monthly R_{OOS}^2 of around 0.93% in [Kelly and Pruitt \(2013\)](#), who use a single factor extracted via partial least squares from 100 book-to-market ratios.

Alternative models predict well in some states but perform poorly in others. For instance, the model in Column (6) that assumes strong momentum in returns is able to predict well in recessions with monthly R_{OOS}^2 values around 5%, but performs poorly in expansions with negative statistics of -3% (a negative R_{OOS}^2 means that the simple recursive mean forecast produces lower forecasting errors). Conversely, the models that assume mild reversals in returns are able to predict well in expansions ($R_{OOS,Exp.}^2 \approx 0.9\%$), by sacrificing the predictability in recessions ($R_{OOS,Rec.}^2 \approx -0.8\%$).

Panel B of Table 6 reports the average maximum drawdown in expansions and recessions. The main takeaway is that exploiting mean reversion in returns during expansions reduces average drawdowns during these periods (e.g., see Column (5)). Conversely, a strategy that accounts for strong time-series momentum while in recessions decreases average drawdowns during these times (e.g., see Column (6)).

Overall, the lower average maximum drawdowns, together with the large R_{OOS}^2 statistics across economic states, suggest that a strong feature of the U.S. stock market is that it exhibits mild mean reversion in periods of broad economic growth but strong times-series momentum during contractions.

6.2 *Beta during Recessions*

This section shows that the strategy's unconditional alpha comes mainly from taking a negative risk exposure at the onset of a recession and by substantially increasing risk-taking shortly after.

Before running regressions, I first present this result graphically in Figure 6. I bucket the period- t excess return forecast (using only data up until time $t - 1$) into five bins (low to high). For each bin, I then report average values and 95 percent confidence intervals for four different measures. The top panel depicts results for the conditional risk exposure and the recession probabilities for month t . The bottom panel reports estimates for the excess returns realized over month t from a buy-and-hold strategy and the main trading strategy, respectively.

Figure 6 shows that when the excess return forecast is in the first bucket, average future excess returns are negative, and the probability of being in a recession is exceptionally

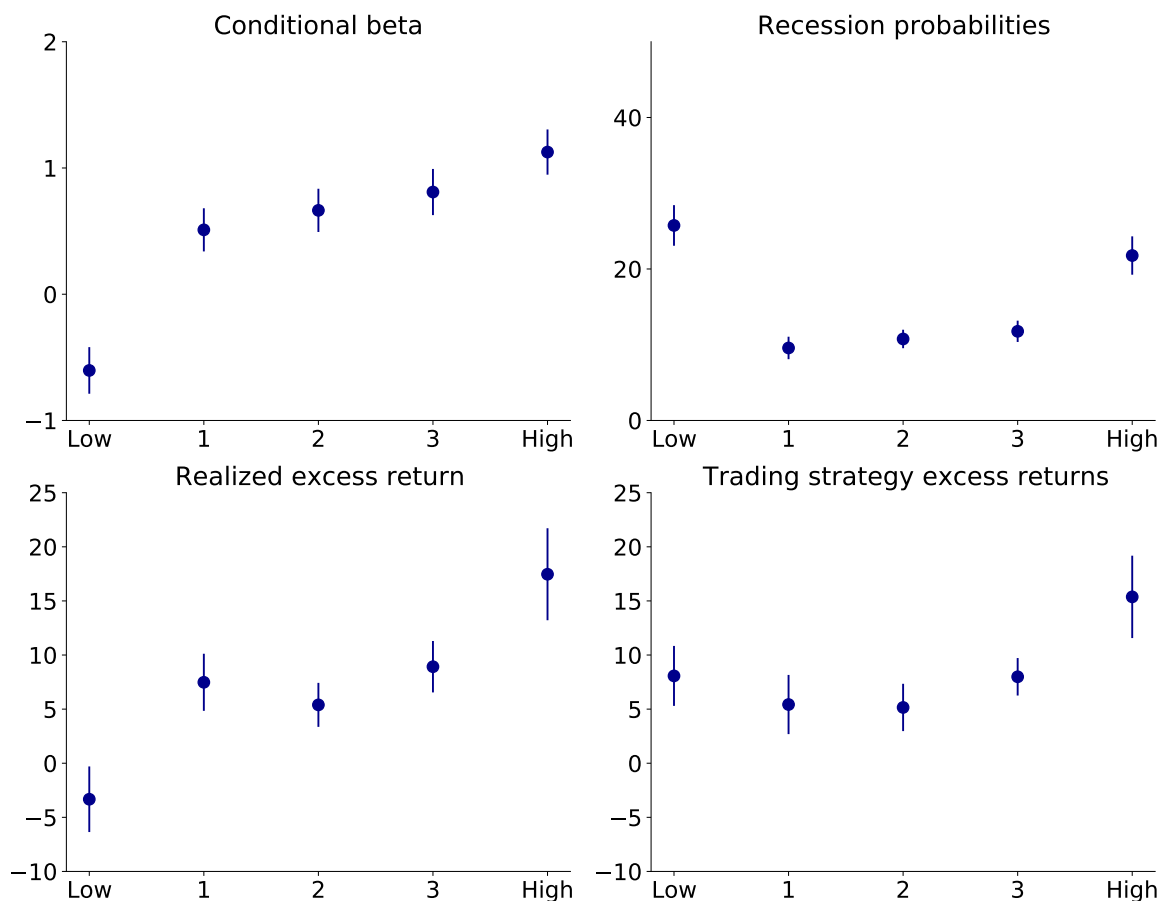


Fig. 6. **Sorts on the one-month ahead excess return forecasts.** This figure sorts the period- t excess return forecast for the aggregate market into five buckets. For each bucket, the figure reports average values and 95 percent confidence intervals for four different measures. The top panel depicts results for the conditional risk exposure and the recession probabilities for month t . The bottom panel reports estimates for the excess returns realized over month t from a buy-and-hold strategy and the main trading strategy, respectively.

high. The trading strategy generates a substantial part of its profits by avoiding these predictable negative returns. Note that, at such times, the conditional risk exposure is negative and the trading strategy's excess return is high. The figure further shows that when the excess return forecast is in the last bucket, the recession probability is also high and future excess returns are positive and large in magnitude. The strategy benefits from these higher-than-average returns by increasing risk-taking. Notably, in buckets 2 through 4 all four measures are fairly flat.

I next run the statistical analysis and estimate three monthly regressions. In all regressions, the dependent variable is $r_t^{e,\mu}$, the strategy's excess return in month t . The regressors are a combination of (a) the market excess return, r_t^e ; (b) the NBER

Table 7
Beta During Recessions

This table presents up- and down-market betas during recessions. The regressors are a combination of (a) the market excess return, r_t^e ; (b) the NBER recessionary indicator, $I_{\text{NBER},t}$; and (c) an up-market indicator variable, $I_{\text{up-market},t}$, that equals one if the cumulative excess market return in the past six months is positive and zero otherwise. Standard errors are adjusted for heteroskedasticity and autocorrelation. t -statistics are reported in brackets. To facilitate an interpretation, I annualize all returns, expressed as a percentage per year, by multiplying monthly returns by 1200.

Coef.	Variable	(1)	(2)	(3)
α_0	1	4.39 [2.95]	2.48 [2.22]	2.48 [2.22]
α_{rec}	$I_{\text{NBER},t}$	8.75 [2.07]	5.15 [2.05]	-4.40 [-0.85]
β_0	r_t^e	0.61 [4.77]	0.80 [8.64]	0.80 [8.64]
β_{rec}	$I_{\text{NBER},t} \cdot r_t^e$	—	-0.82 [-1.94]	-0.96 [-1.90]
$\beta_{rec,up}$	$I_{\text{NBER},t} \cdot I_{\text{up-market},t} \cdot r_t^e$	—	—	1.27 [1.94]
Obs.		480	480	480

recessionary indicator, $I_{\text{NBER},t}$; and (c) an up-market indicator variable, $I_{\text{up-market},t}$, that equals one if the cumulative excess market return in the past six months, including time t , is positive and zero otherwise.

Regression 1 in Table 7 fits an unconditional market model to $r_t^{e,\mu}$ and allows for a different constant in expansions and recessions:

$$r_{t+1}^{e,\mu} = \alpha_0 + \alpha_{rec} \cdot I_{\text{NBER},t} + \beta r_{t+1}^e + \epsilon_{t+1}. \quad (23)$$

Consistent with the previous sections' results, the estimated alphas in expansions ($4.4 = \alpha_0$) and recessions ($13.1 = \alpha_0 + \alpha_{rec}$) are large and statistically significant. Since the conditional CAPM holds in equation (20), these unconditional alphas are due to time-varying risk exposures. To explain the recessionary alpha, the next two regressions compute market beta differences in recessions.

Regression 2 in Table 7 fits a conditional CAPM with the NBER recessionary indicator,

$I_{\text{NBER},t}$, as an instrument:

$$r_t^{e,\mu} = \alpha_0 + \alpha_{rec} \cdot I_{\text{NBER},t} + (\beta_0 + \beta_{rec} \cdot I_{\text{NBER},t}) r_t^e + \epsilon_t. \quad (24)$$

This specification captures both alphas and market beta differences across economic states. A striking change is evident in the beta in recessionary periods. It is -0.82 lower (t -statistic = -1.94), suggesting that the strategy reduces risk-taking during recessions by completely cashing out. Indeed, the beta of the strategy, conditional on a recession, equals -0.02 ($=\beta_0 + \beta_{rec}$). This change in risk exposure explains almost half of the alpha differences in recessions, which shrinks from 8.75% (t -statistic = 2.07) to 5.15% (t -statistic = 2.05).

Regression 3 in Table 7 introduces an additional term to the previous specification to compute the strategy's risk exposure during a recession's up and down markets:

$$r_t^{e,\mu} = \alpha_0 + \alpha_{rec} \cdot I_{\text{NBER},t} + (\beta_0 + (\beta_{rec} + \beta_{rec,up} \cdot I_{\text{up-market},t}) \cdot I_{\text{NBER},t}) r_t^e + \epsilon_t. \quad (25)$$

The $\beta_{rec,up}$ of 1.27 (t -statistic = 1.94) in Column (3) shows that the strategy increases risk-taking once the market rebounds. In recessionary periods, the risk exposure of the strategy is -0.16 ($=\beta_0 + \beta_{rec}$) during down markets, but 1.11 ($=\beta_0 + \beta_{rec} + \beta_{rec,up}$) during up markets. This change in risk exposure during recessions explains the recessionary alpha. The point estimate for the alpha in recessions is -1.9% ($=\alpha_0 + \alpha_{rec}$) and not statistically significant.

The strategy overlaps with the volatility-managed portfolio of [Moreira and Muir \(2017\)](#) since both take less risk during recessions. However, during recessions, one main difference between these two strategies is when to get back in. Figure A3 in the appendix shows that the volatility strategy will slowly return to the market only when the volatility starts to decrease, not necessarily when the market rebounds. Therefore, volatility-timing strategies may stay out of the market for too long and miss subsequent higher-than-average excess returns.²⁶ In the appendix, I conduct formal spanning tests among these portfolios and find that their returns do not span each other.

²⁶In fact, the point estimate of the regression coefficient $\beta_{rec,up}$ in equation (25) is equal to 0.10 (t -statistic = 1.17) for the volatility-timing strategy.

6.3 *Beta during Expansions*

During expansions, the strategy's conditional risk exposure is related with common measures of expected returns documented in the literature. In the interest of brevity, I briefly describe the results here and present the estimation results in Appendix A.12.

During expansions, when survey measures of investors' expectations about future stock market returns (e.g., [Greenwood and Shleifer \(2014\)](#)), investor sentiment measures (e.g., [Baker and Wurgler \(2006\)](#)), or flows into equity-oriented mutual funds increase, the strategy takes a contrarian position and reduces risk-taking. In good times, these results are broadly consistent with studies in the behavioral literature suggesting that many investors hold extrapolative expectations about returns.²⁷ After a period of good stock market performance, one that generates high sentiment, many investors will become more bullish. Extrapolators will rebalance their portfolios into equity and, in doing so, will bid up current stock prices even further and reduce subsequent stock returns. Under this scenario, the strategy mildly decreases the portfolio weights on stocks. However, during recessions, this negative relation disappears, both economically and statistically.

During expansions, the strategy's profits also can be explained in terms of risk. When economic conditions deteriorate (but not enough to generate a recession) or when prices decrease more than dividends (representing an increase in risk premia), the strategy increases risk-taking. These results imply that, during expansions, the strategy has a strong countercyclical component. However, during recessions, this relation flips sign: an increase in the dividend-price ratio or a decrease in macro growth are positively related with the strategy's risk exposure.

7 Economic Mechanisms

What underlying economic mechanisms could explain my findings? I first discuss rational expectations models with countercyclical risk premia. Negative predictable returns in precisely bad times are puzzling from the perspective of these models. I then consider a

²⁷Survey evidence suggests that investors form beliefs about future returns by extrapolating past market performance. See, for instance, [Vissing-Jorgensen \(2003\)](#), [Amromin and Sharpe \(2009\)](#), [Bacchetta, Mertens, and Van Wincoop \(2009\)](#), and [Greenwood and Shleifer \(2014\)](#).

model in which investors do not have rational expectations about future cash flows. My findings appear to be consistent with a sluggish adjustment in stocks to recession news. I find support for this view in analysts' cash flow forecasts.

7.1 *Macrofinance Models with Countercyclical Risk Premia*

It is difficult to rationalize a negative equity premium at the onset of bad times. Rational asset-pricing models generally specify a stochastic discount factor which is a function of state variables linked to the economy, such as surplus consumption ratio (e.g., [Campbell and Cochrane \(1999\)](#)), conditional volatility of expected consumption growth (e.g., [Bansal, Kiku, and Yaron \(2012\)](#); [Bansal and Yaron \(2004\)](#)), time-varying probability of a large drop in consumption (e.g., [Gabaix \(2012\)](#); [Wachter \(2013\)](#)) or the health of the financial sector (e.g., the intermediary-based models of [He and Krishnamurthy \(2013\)](#); [Sannikov, Brunnermeier, et al. \(2012\)](#); [Moreira and Savov \(2017\)](#)). By imposing the main asset-pricing equation, these state variables determine variations in the equity premium. In these models, the early stages of a recession are precisely those periods when the compensation for taking risk is positive and high. Things are bad (e.g., consumption declines substantially relative to habit, real uncertainty spikes, the health of the financial sector deteriorates), and they could get worse.

In the context of these models, the unconditional alpha of the strategy would be either negative or close to zero, not positive and economically significant. As noted by [Lewellen and Nagel \(2006\)](#), when the conditional CAPM holds, the unconditional alpha depends on how the conditional risk exposure, β_t , covaries with the equity premium, γ_t , and with the conditional variance, σ_t^2 , that is, $\alpha \approx \text{cov}(\beta_t, \gamma_t) - \frac{\gamma}{\sigma_M^2} \text{cov}(\beta_t, \sigma_t^2)$ (e.g., see equation (22)). First, consider the case in which γ_t and σ_t^2 are not related and β_t covaries only with γ_t (i.e., $\alpha \approx \text{cov}(\beta_t, \gamma_t)$). A strategy that sells at the onset of recessions will tend to earn a negative unconditional alpha because it avoids taking risk (i.e., β_t is low) when things are scariest and compensation for risk is high (i.e., γ_t is high). Now consider the case in which γ_t and σ_t^2 are positively related, as many asset-pricing models imply.²⁸ In

²⁸For instance, in [Campbell and Cochrane \(1999\)](#), σ_t , γ_t , and γ_t/σ_t are decreasing functions of the surplus consumption ratio (see Figures 4, 5, and 6 in [Campbell and Cochrane \(1999\)](#)). Hence, γ_t is almost proportional to σ_t , but γ_t moves more than σ_t for any given movement in the surplus consumption

this case, the effect of time-varying volatility offsets the risk premium impact and shrinks the negative α towards zero. Hence, for each model, the net effect on alpha will depend on these two offsetting channels' relative strength. Consistent with this, [Moreira and Muir \(2017\)](#) simulate data from leading equilibrium asset-pricing models and show that the volatility-managed alpha, that is, the alpha from a strategy that reduces β_t when σ_t^2 is high, is either centered at zero (e.g., for long-run risk and intermediary-based models) or shows a weak negative deviation from zero alpha (e.g., for habits and time-varying rare-disasters models).

7.2 Present-Value Models

Next, I consider models that impose the economic restrictions implied by the present-value identity:

$$R_{t+1} = \left(\frac{P_{t+1} + D_{t+1}}{P_t} \right), \quad (26)$$

where R_{t+1} , P_{t+1} , and D_{t+1} represent the one-period return, price, and dividend for the aggregate stock market. Following [Campbell and Shiller \(1988\)](#), the log return can be decomposed approximately as

$$r_{t+1} \simeq \kappa_0 + \kappa_1 pd_{t+1} + \Delta d_{t+1} - pd_t, \quad (27)$$

where Δd_{t+1} denotes the log dividend growth rate and pd_t is the log price-dividend ratio. $\kappa_0 = \log(1 + \exp(\bar{pd})) - \kappa_1 \bar{pd}$ and $\kappa_1 = \exp(\bar{pd}) / (1 + \exp(\bar{pd}))$ are approximating constants that depend on the average level of the log price-dividend ratio $\bar{pd}$. By iterating forward this equation and imposing a no-bubble condition, $\lim_{T \rightarrow \infty} \kappa_1^T pd_{t+T} = 0$, we can write pd_t as follows:

$$pd_t = \frac{\kappa_0}{1 - \kappa_1} + \left[\sum_{j=1}^{\infty} \kappa_1^{j-1} \Delta d_{t+j} \right] - \left[\sum_{j=1}^{\infty} \kappa_1^{j-1} r_{t+j} \right]. \quad (28)$$

In the subsequent analysis, I assume that investors use this equation to price stocks.

ratio. In [Bansal and Yaron \(2004\)](#), γ_t and σ_t are also positively related. The conditional risk premium and volatility of the market portfolio are both increasing functions of expected consumption volatility, but volatility moves less than the risk premium since Sharpe ratios increase during periods of high economic uncertainty.

To do so, I further specify the laws of motion for Δd_{t+1} and r_{t+1} as

$$\begin{aligned}\Delta d_{t+1} &= g_t + \varepsilon_{t+1}^d, \\ g_{t+1} &= \gamma_0 + \gamma_1 (g_t - \gamma_0) + \varepsilon_{t+1}^g, \\ r_{t+1} &= \mu_t + \varepsilon_{t+1}^r, \\ \mu_{t+1} &= \delta_0 + \delta_1 (\mu_t - \delta_0) + \varepsilon_{t+1}^\mu,\end{aligned}\tag{29}$$

where the shocks are assumed to be $\varepsilon_{t+1}^i \sim N(0, \sigma_i^2)$, $i = \{d, g, r, \mu\}$. This specification is based on [Binsbergen and Koijen \(2010\)](#) and decomposes Δd_{t+1} (r_{t+1}) into a persistent component g_t (μ_t), and a transitory component ε_{t+1}^d (ε_{t+1}^r).

7.2.1 Rational Expectations

Under rational expectations, investors form expectations under the true measure for log returns and dividend growth, which gives rise to the following present-value relation:

$$pd_t = \frac{\kappa_0}{1 - \kappa_1} + \mathbb{E}_t \left[\sum_{j=1}^{\infty} \rho^{j-1} \Delta d_{t+j} \right] - \mathbb{E}_t \left[\sum_{j=1}^{\infty} \rho^{j-1} r_{t+j} \right], \tag{30}$$

where $\mathbb{E}_t [\cdot]$ represents the expectations computed under the actual statistical distribution as measured by an econometrician. Given the assumptions about the return and dividend growth processes in equation (29), [Binsbergen and Koijen \(2010\)](#) show that the present-value relation in equation (30) can be rewritten as

$$pd_t = A + B_1 (\mu_t - \delta_0) + B_2 (g_t - \gamma_0), \tag{31}$$

where $A = \frac{\kappa_0}{1-\rho} + \frac{\gamma_0 - \delta_0}{1-\kappa_1}$, $B_1 = -\frac{1}{1-\kappa_1\delta_1}$, and $B_2 = \frac{1}{1-\kappa_1\gamma_1}$ and all components are known at time t . The crucial implication of the present-value model is that the price-dividend ratio decreases with expected returns ($B_1 < 0$) and increases with expected growth rates ($B_2 > 0$).

Because rational investors and the econometrician form expectations under the true measure, I can substitute my expected returns estimate that uses data up to date t in the present-value relation in equation (31), and test whether $\hat{\mu}_{t+1|t}$ is consistent with the

Table 8
Implied Present-Value Model Parameters

The table reports the OLS regression results of the log price-dividend ratio on the one-month ahead excess return forecast $\hat{\mu}_{t+1|t}$ and an indicator variable that equals one if $\hat{\mu}_{t+1|t}$ is negative and is zero otherwise: $pd_t = a + (b_1 + b_1^{neg} \cdot I_{neg,t}) \hat{\mu}_{t+1|t} + \epsilon_t$. Section 3 provides the details on how $\hat{\mu}_{t+1|t}$ is constructed and estimated. Standard errors adjust for heteroskedasticity and autocorrelation. I report the t-statistics in brackets. The forecast evaluation period runs from January 1980 to December 2019.

Coef.	Variable	Estimated coefficients [t-statistics]	
		(1)	(2)
$\hat{b}_1$	$\hat{\mu}_{t+1 t}$	-25.94 [-2.55]	-60.02 [-6.93]
$\hat{b}_1^{neg}$	$I_{neg,t} \cdot \hat{\mu}_{t+1 t}$	— —	81.45 [5.08]
R-squared		0.06	0.18
Obs.		479	479

economic restrictions implied by the present-value model under rational expectations. To do so, I run the following OLS regression:

$$pd_t = a + b_1 \hat{\mu}_{t+1|t} + \epsilon_t, \quad (32)$$

The corresponding null hypothesis is $b_1 < 0$. Under the null, the log price-dividend ratio increases when the one-month ahead excess return forecast falls.²⁹ The results are summarized in Table 8. The estimate coefficient is negative: $\hat{b}_1 = -26$ with a t-statistic of -2.5 , consistent with results in [Binsbergen and Koijen \(2010\)](#). Based on annual data, Binsbergen and Koijen find a B_1 equal to -10.334 , which would translate into a B_1 estimate of approximately -27 for a monthly model.

Column (2) adds an additional variable to the regression that allows me to evaluate the extent to which $\hat{\mu}_{t+1|t}$ is consistent with the laws of gravity implied by the present-value

²⁹Due to the fact that the price-dividend ratio is also driven by expected dividend growth rates, the regression of pd_t on $\hat{\mu}_{t+1|t}$ in equation (32) suffers from an omitted variable bias problem. However, given that $B_2 > 0$, the bias on b_1 will be negative and in favor of the null hypothesis if risk premia are countercyclical, that is, if a rise in expected returns is associated with a decrease in future dividend growth. In this regard, [Lochstoer and Tetlock \(2020\)](#) find a negative correlation between cash flow news and discount rate news by exploiting the present-value equation at the firm level and then aggregating it out to the market portfolio level.

identity during times in which returns are predictably negative:

$$pd_t = a + (b_1 + b_1^{neg} \cdot I_{neg,t}) \hat{\mu}_{t+1|t} + \epsilon_t, \quad (33)$$

where $I_{neg,t}$ is an indicator that equals one if the econometrician's return forecast is negative and zero otherwise. When the expected return measure is negative, the regression coefficient has the wrong sign and equals 21 ($= \hat{b}_1 + \hat{b}_1^{neg}$) with a t-statistic of 5 on the difference. During these periods, I can reject the null hypothesis stated earlier. A decrease in the price-dividend ratio is associated with low expected returns instead of high expected returns. Conversely, in the rest of the periods, the usual Campbell-Shiller logic prevails, and high price-dividend ratios are associated with low expected returns.

Overall, under rational expectations, most models would suggest that the types of price movements common at the onset of recessions would be associated with higher, not lower, discount rate forecasts.

7.2.2 Subjective (and Sticky) Cash Flow Expectations

I also consider a model in which investors use the Campbell-Shiller identity in equation (28) to value stocks, but, similar to [Katz et al. \(2017\)](#), agents use subjective expectations, denoted by $\mathbb{E}_t^*[\cdot]$, to value stocks:

$$pd_t = \frac{\kappa_0}{1 - \kappa_1} + \mathbb{E}_t^* \left[\sum_{j=1}^{\infty} \rho^{j-1} \Delta d_{t+j} \right] - \mathbb{E}_t^* \left[\sum_{j=1}^{\infty} \rho^{j-1} r_{t+j} \right], \quad (34)$$

If subjective expectations are not rational, then there will be a wedge between the forecasts made under subjective beliefs, $\mathbb{E}_t^*[\cdot]$, and under the true measure of expectations, $\mathbb{E}_t[\cdot]$. The econometrician forecast, $\hat{\mu}_{t+1|t}$, will end up picking up this difference.

I break rational expectations by assuming that subjective beliefs about future cash flows are sluggish to react to new information. To simplify the analysis, I abstract from stickiness on the discount rate side. Drawing on the framework by [Mankiw and Reis \(2002\)](#), inattentive agents update their beliefs each period with probability $1 - \lambda$, and on

doing so, they acquire full information and have rational expectations:

$$\mathbb{E}_t^* [\Delta d_{t+j}] = (1 - \lambda) \mathbb{E}_t [\Delta d_{t+j}] + \lambda \mathbb{E}_{t-1}^* [\Delta d_{t+j}]. \quad (35)$$

The coefficient λ captures the degree of expectation rigidity. When λ is equal to zero, the model collapses to a perfectly rational expectations framework. A value of λ greater than zero, instead, suggests that information is slowly updated by (at least) some agents.

Given the expectation dynamics in (35), I can substitute the process for realized and expected dividend growth in (29) to obtain the subjective expectations about future cash flows:

$$\mathbb{E}_t^* [\Delta d_{t+j}] = (1 - \lambda) \sum_{k=0}^{\infty} \lambda^k \mathbb{E}_{t-k} [\Delta d_{t+j}] = \gamma_0 + (1 - \lambda) \sum_{k=0}^{\infty} \lambda^k \gamma_1^{j-1+k} (g_{t-k} + \gamma_0). \quad (36)$$

If I plug this equation into the Campbell-Shiller equation in (34) and assume $\mathbb{E}_t^* [\Delta r_{t+j}] = \mathbb{E}_t [\Delta r_{t+j}]$, it follows that

$$pd_t = A + B_1 (\mu_t - \delta_0) + (1 - \lambda) B_2 (g_t - \gamma_0) + (1 - \lambda) B_2 \sum_{j=1}^{\infty} (\lambda \gamma_1)^j (g_{t-j} - \gamma_0), \quad (37)$$

where, as before, $A = \frac{\kappa_0}{1-\rho} + \frac{\gamma_0 - \delta_0}{1-\kappa_1}$, $B_1 = -\frac{1}{1-\kappa_1\delta_1}$, and $B_2 = \frac{1}{1-\kappa_1\gamma_1}$. Moreover, given this expression for pd_t , I can easily derive the process for returns in (27) as a function of past shocks and expected returns:

$$r_{t+1} = \underbrace{\mu_t}_{\text{discount rates}} + \underbrace{\lambda B_2 (1 - \kappa_1 \lambda \gamma_1) \sum_{j=0}^{\infty} (\lambda \gamma_1)^j \varepsilon_{t-j}^g}_{\text{past cash flow shocks}} + \underbrace{\varepsilon_{t+1}^d + \kappa_1 B_1 \varepsilon_{t+1}^\mu + (1 - \lambda) \kappa_1 B_2 \varepsilon_{t+1}^g}_{\text{unexpected returns}}. \quad (38)$$

To interpret these formulas, note that if expectations are rational, that is $\lambda = 0$, the log price-dividend ratio takes the usual form in equation (31), and returns are given by $r_{t+1} = \mu_t + \varepsilon_{t+1}^d + \kappa_1 B_1 \varepsilon_{t+1}^\mu + \kappa_1 B_2 \varepsilon_{t+1}^g$. In this case, the conditional expectations of returns is given by μ_t . Negative realized returns may arise from positive discount rate shocks (ε_{t+1}^μ) as well as negative shocks to current (ε_{t+1}^d) and expected (ε_{t+1}^g) dividend growth. However, these shocks are not predictable and do not enter the conditional expectations of returns.

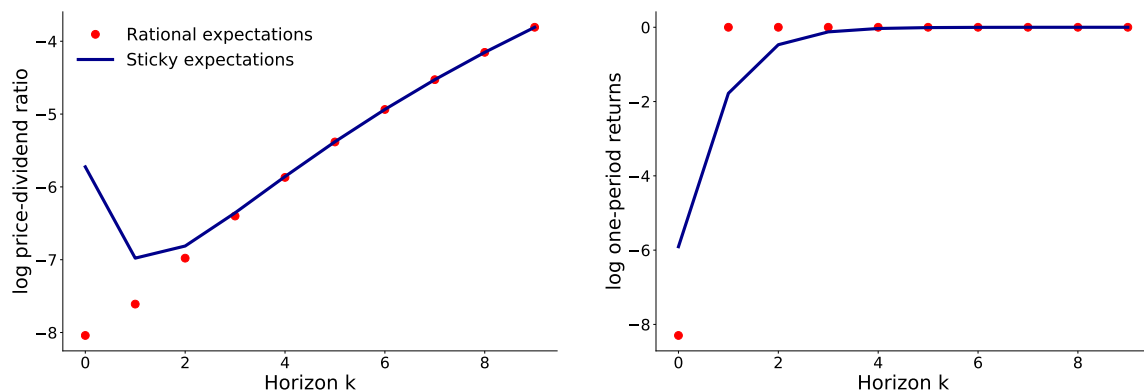


Fig. 7. **Impulse response to expected dividend growth shocks.** This figure plots the impulse response of the log price-dividend ratio (left panel) and the log one-period return (right panel) to expected dividend growth shocks ($\sigma_g \varepsilon_{t+1}$) of one standard deviation. I report results under rational expectations, that is $\lambda = 0$, and under sticky subjective expectations $\lambda > 0$. I use $\kappa_1 = 0.969$, $\gamma_1 = 0.91$, $\sigma_g = 0.0167$, and $\lambda = 0.17$.

If beliefs update sluggishly relative to the econometrician, that is $\lambda > 0$, then the log price-dividend ratio will depend on the infinite moving average of previous expected dividend growth rates (g_t). Importantly, shocks to g_t will affect the conditional expectations of returns, which now will be given by $\mu_t + \lambda B_2(1 - \kappa_1 \lambda \gamma_1) \sum_{j=0}^{\infty} (\lambda \gamma_1)^j \varepsilon_{t-j}^g$. Hence, we can have negative (and predictable to the econometrician) future price movements coming from the cash flow side, which offsets whatever is happening with the risk premia component (μ_t) embedded in current prices.³⁰

These price dynamics imply that overestimating future cash flows at the onset of recessions under the subjective measure imputes negative predictability of returns under the econometrician's one. If these mistakes are undone rather quickly, returns can be quite negative in the short run and mean revert shortly thereafter. To see this, Figure 7 plots the impulse response of the log price-dividend ratio $\left(\frac{\partial pd_{t+k}}{\partial \varepsilon_t^g}\right)$ and the log one-period return $\left(\frac{\partial r_{t+k}}{\partial \varepsilon_t^g}\right)$ to a one standard deviation surprise decrease in the expected dividend growth series. For the impulse responses, I take $\kappa_1 = 0.969$, $\gamma_1 = 0.354^{1/12}$, and $\sigma_g = 0.058/\sqrt{12}$ from Table II in [Binsbergen and Koijen \(2010\)](#) and use $\lambda = 0.17$ from Table III in [Bouchaud et al. \(2019\)](#). A negative shock to expected dividend growth immediately decreases the price-dividend ratio (left panel) and generates a negative return on stocks (right panel). Because subjective beliefs about lower cash flows are sluggish to react to this shock, expected dividends are too high relative to the rational

³⁰I thank the referees for this observation.

benchmark. Hence the price-dividend ratio and the realized return fall, but they do not fall enough (see the red dots). In the subsequent periods, agents sluggishly adjust their dividend growth forecast downwards to the rational level. In doing so, the price-dividend ratio slowly converges to the rational level, and returns on stocks continue to be negative. After four months, the shock is fully impounded into prices, and the price dynamics under subjective and rational expectations move in sync.

Under this subjective belief framework, my one-month ahead return forecast $\hat{\mu}_{t+1|t}$ equals $\mu_t + \lambda B_2(1 - \kappa_1 \lambda \gamma_1) \sum_{j=0}^{\infty} (\lambda \gamma_1)^j \varepsilon_{t-j}^g$. Hence, it incorporates two pieces. The first is the rational risk premia component, μ_t , that negatively affects price-dividend ratios (e.g., see the present-value restriction in (37)). The second, $\lambda B_2(1 - \kappa_1 \lambda \gamma_1) \sum_{j=0}^{\infty} (\lambda \gamma_1)^j \varepsilon_{t-j}^g$, is the near-term forecastable changes in cash flows that are not yet incorporated into prices. Rather than isolating these two pieces, $\hat{\mu}_{t+1|t}$ captures the sum of the two, which is the relevant object from a one-period market-timing perspective.

For long-term investors, the composition of expected returns might be relevant as cash flow shocks die out quicker than discount rate shocks since the latter seem to be more persistent. Consistent with this, [Moreira and Muir \(2019\)](#) find that determining the composition of shocks is important for setting the investment horizon in portfolio choice. Empirically, it is possible to disentangle these two expected return components. For instance, one can start with the state-space model of [Binsbergen and Koijen \(2010\)](#) described by equation (29) and impose the present-value restrictions under subjective expectations as in [De La O and Myers \(2021\)](#). By imposing sticky expectations, one gets the measurement equation for returns in (38). One could then use the diffusion index extracted in Section 3.3 to determine the economic regime in place and to sharpen inference about expected dividend growth and its shocks ε_t^g . These shocks, in turn, enter the measurement equation for returns. Finally, one can infer the parameter λ using forecast errors and forecast revisions as in [Bouchaud et al. \(2019\)](#). I leave these extensions of the model to future work.

7.2.3 *Analyst Forecast Errors and Revisions at the Onset of Recessions*

A model of subjective beliefs, where belief updating is sluggish relative to the econometrician, can explain the negative return predictability at the onset of recessions. Next, I provide suggestive evidence in favor of this explanation. Specifically, I show that downward revisions in analyst expectations about cash flows are exceptionally high, and forecasts are too optimistic when the equity premium forecast is negative.

To compute forecast errors and revisions, I proceed as follows.³¹ First, I construct cash flow subjective expectations using analyst-by-analyst earnings per share (EPS) forecasts from the Thomson Reuters unadjusted I/B/E/S Detail History File. I use analyst EPS forecasts rather than dividends per share forecasts because the former series starts in 1976, while the latter begins in 2003. I focus on the forecast made for the quarterly figures reported in the current and the next fiscal quarter (i.e., Forecast Period Indicator (FPI) variable equal to 6 or 7), and construct the median forecast using 30-day buckets around the quarterly fiscal period. To do so, I calculate for each firm the median of all forecasts submitted during each 30-day bucket. If an analyst makes more than one forecast for the same firm, the same fiscal quarter, and the same 30-day bucket, I consider only the last observation to ensure that I have the most up-to-date forecast. Next, I match these median forecasts with the actual EPS observation using the I/B/E/S unadjusted actuals file database. Consistent with prior studies, I use the CRSP cumulative adjustment factors to account for stock splits that transpired during the period between the EPS forecast and the earnings announcement. Finally, I define forecast errors as the difference between the actual EPS observation for fiscal quarter t and the corresponding median earnings forecast. Likewise, I define forecast revisions as the change in the median earnings forecast for fiscal quarter t between two consecutive 30-day buckets. I further normalize both of these differences by the stock price before the previous fiscal quarter announcement.

To test whether overestimates of future cash flows are particularly high when the

³¹I thank Freda Song Drechsler for posting the code that calculates earnings surprises. The code can be found at <https://www.fredasongdrechsler.com/full-python-code/pead>.

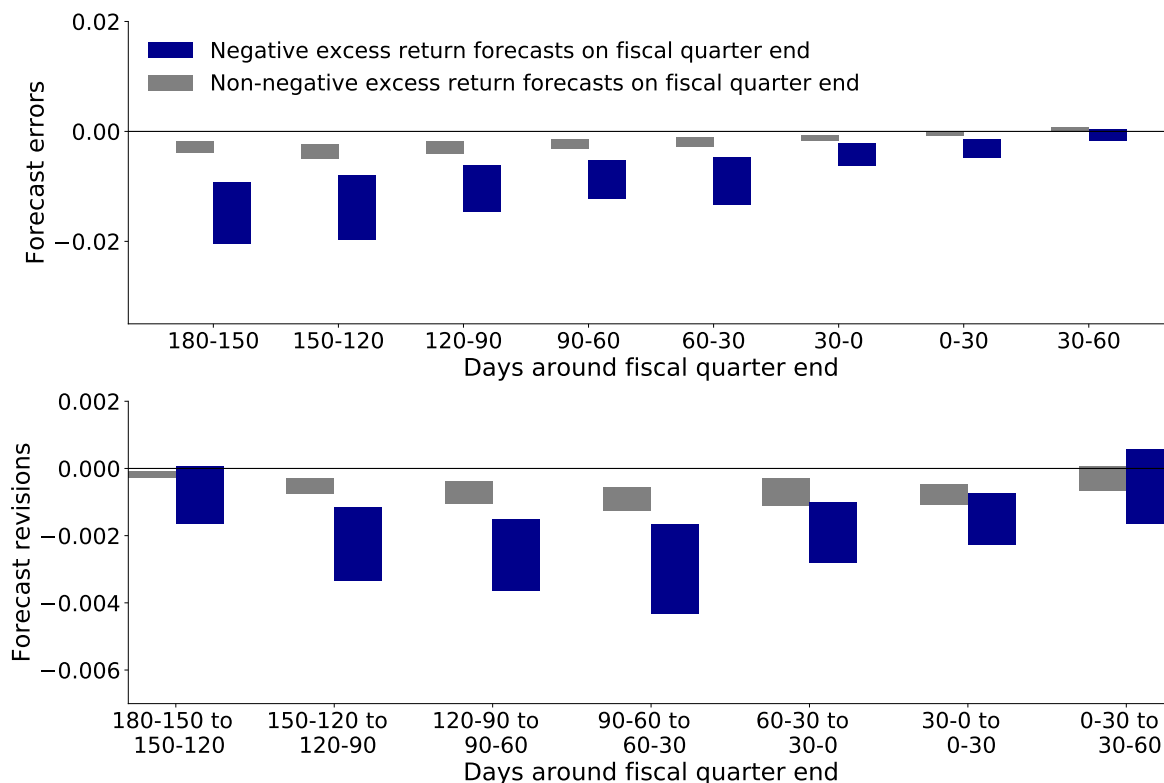


Fig. 8. **Analyst Forecast Errors and Revisions.** This figure plots forecast errors (top panel) and forecast revisions (bottom panel) for different horizons. I then calculate average forecast errors and revisions for fiscal quarter ends that contained a negative (blue boxes) or positive (gray boxes) equity premium forecast. The boxes denote 95 percent confidence intervals.

equity premium forecast is negative, I estimate:

$$FE_{f,t}(h) = a_h + a_h^{neg} \cdot I_{neg,t} + \epsilon_{f,t}, \quad (39)$$

where $FE_{f,t}(h)$ denotes the scaled forecast error for firm f at fiscal quarter t and for forecast horizon h ; $I_{neg,t}$ is an indicator that equals one when the equity premium forecast is negative in any month of fiscal quarter t and is zero otherwise; and $\epsilon_{f,t}$ is an error term. The estimated coefficients a_h and $a_h + a_h^{neg}$ are graphically summarized in the top panel of Figure 8, where the boxes denote 95 percent confidence intervals. Note that forecast errors rise with the forecast horizon and are negative on average, implying that the median analyst tends to overestimate earnings, consistent with results in [Bouchaud et al. \(2019\)](#). Importantly, overestimates in future cash flows are an order of magnitude higher when the equity premium forecast is negative. Notably, the error tends to be zero when the forecast is issued two months after the fiscal quarter end.

To test whether downward revisions of beliefs about cash flows are particularly high and adjust slowly when the equity premium forecast is negative, I run the following regression

$$FR_{f,t}(h, h') = a_{h,h'} + a_{h,h'}^{neg} \cdot I_{neg,t} + \epsilon_{f,t}, \quad (40)$$

where $FR_{f,t}(h, h')$ denotes the scaled forecast revision between forecast horizon h and h' for firm f at fiscal quarter t . The bottom panel of Figure 8 plots the estimated coefficients. The median analyst starts to adjust their expectations downwards six months before the fiscal quarter end (e.g., forecasts made between the 180–150 and 150–120 day buckets). In the subsequent months, the analyst continues to adjust sluggishly in the same direction. The rate of adjustment reaches a maximum in the first month of the current fiscal quarter end (e.g., between the 90–60 and 60–30 day buckets), and the rate decreases afterward. Hence, agents continue to revise downwards at lower speeds until the forecast revisions hover around zero, which happens in the second month after the fiscal quarter end. Crucially, these patterns are again an order of magnitude higher when the equity premium forecast is negative.

8 Conclusion

I have used data on a variety of economic indicators to identify business cycle turning points in real time. Using these estimates, I have documented a strong feature of the U.S. stock market: time-series reversals in periods of broad economic growth, but times-series momentum during contractions. I then propose a new measure of expected returns that utilizes the state-dependent autocorrelation in returns. This trading strategy earns large alphas, increases Sharpe ratios, and generates big utility gains for a mean-variance investor.

The positive alpha of the strategy is puzzling from the perspective of standard models with rational expectations and countercyclical risk premia. The strategy takes less risk at the onset of bad economic times, showing that returns are predictable negative, not high, during these periods. After cashing out, my strategy returns to the market during up markets in recessions. This result suggests that the average investors are late in acting

on the macroeconomic information that signals the onset of recessions. I then provide suggestive evidence in favor of this explanation by looking at analyst forecasts.

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Internet Appendix

The appendix consists of two main sections. In Section A, I present additional empirical results. In Section B, I provide the details of the forecasting model.

Appendix A Additional Empirical Results

Each section is self-contained and can be skipped without loss.

A.1 Robustness: Simple strategy

In this section, I provide several robustness checks to the simple strategy that trades on macro information. Specifically, in Table 1 in Section 2, given the data limitations, I restricted my analysis to the 1980:M1 through 2019:M12 sample period and used a 12-month moving average to compute the growth/valuation signals. Here, for each series, I consider the first observation in which we observe the series and consider various moving average lengths. To compute the Sharpe ratio increase, I compute the Sharpe ratio of the strategy and divide it by the Sharpe ratio of its fixed allocation benchmark computed over the same time period. Importantly, the main results do not change. First, the simple strategy generates Sharpe ratio increases over the fixed allocation benchmark and produces positive CAPM alphas. Second, the strategy benefits by reducing the maximum drawdowns during recessions. Third, macroeconomic variables play an important economic role in identifying economic contractions as measured by the Sharpe ratio increases. These results are robust with respect to moving average size. For instance, the simple strategy outperforms the buy-and-hold approach when I consider window sizes of length 6, 18, or 24 months instead of 12.

Table A1
Performance of Trading Strategies: Robustness

This table reports performance measures for six different strategies. The “100% stocks” denotes the buy-and-hold strategy, fully invested in stocks. The “76/24 mix” denotes the portfolio that assigns a portfolio weight of 76% to equities and 24% to T-Bills. The remaining four strategies switch from stocks into cash during bad times and market downturns, while staying fully invested in stocks in the remaining periods. For these strategies, market downturns are months in which returns are below their 12-month moving average. The “Ind. prod.” strategy defines a bad time whenever the industrial production growth series is below its 12-month moving average. The “Common growth” strategy defines a bad time when the common growth component of 24 real-time macroeconomic variables is below its 12-month moving average. See Section 3 for details on this variable’s construction. The “CAPE” strategy denotes bad times, or expensive periods, when the CAPE ratio is above its 12-month moving average. Finally, the “Slope” strategy denotes bad times when the yield curve slope is negative. For each strategy, I report the Sharpe ratio increase relative to that obtained from a fixed allocation strategy investing in stocks the same unconditional amount of time. I report t-statistics in brackets below the coefficient estimates and standard errors adjust for heteroskedasticity. I annualize all returns, expressed as a percentage per year, by multiplying monthly returns by 1200.

	Fixed allocation		Macro variables		Financial variables	
	100% stocks (1)	76/24 mix (2)	Ind. prod. (3)	Common growth (4)	CAPE (5)	Slope (6)
Time invested	1	0.76	0.76	0.75	0.73	0.93
Mean excess return	7.88	5.32	7.75	7.87	5.66	8.73
S.D.	18.47	12.47	15.26	11.68	16.15	14.28
Sharpe ratio incr.	–	–	18.99	31.80	-17.85	17.78
α -Mkt	–	–	2.38	3.26	-0.32	1.59
	–	–	[2.63]	[2.72]	[-0.38]	[2.22]
Average maximum drawdowns						
Total	17.13	10.72	8.69	6.47	14.52	9.65
Expansions	14.35	8.52	6.96	5.57	12.32	8.55
Recessions	30.44	21.27	16.96	14.02	25.02	19.00
CAPM alpha (α -Mkt) for different sizes of the moving average h						
$h = 6$ months			2.05	3.29	-0.53	1.45
			[2.26]	[2.76]	[-0.64]	[2.01]
$h = 12$ months			2.38	3.26	-0.32	1.59
			[2.63]	[2.72]	[-0.38]	[2.22]
$h = 18$ months			2.62	2.71	-0.77	1.39
			[2.88]	[2.29]	[-0.90]	[1.91]
$h = 24$ months			1.78	3.55	-0.39	1.50
			[1.97]	[2.92]	[-0.46]	[2.08]
Starting Date	1927-06	1927-06	1927-06	1980-12	1927-06	1976-06

A.2 Alternative Model Specifications

In the main article, I assume that the benchmark specification model of returns described in equations (2), (3), and (4) has a constant correlation between expected returns shocks and unexpected returns, $\rho_{\mu,r}$, and I allow for the fraction of variance in excess returns explained by μ_t to vary across economic regimes; that is, $\phi(S_t)$ could take different values in each state. As discussed in the main text, the estimation of the benchmark specification model points towards a fairly persistent predictable component in realized excess returns with a ρ estimate of 0.97. The estimate of $\rho_{\mu,r}$ is -0.95, which suggests that return surprises and expected return news are strongly negatively correlated. Finally, the estimation shows that discount rates are relatively stable in expansions and much more volatile during recessions.

To assess the robustness of this finding, I now modify the benchmark model in five different dimensions:

1. I set $\rho_{\mu,r} = 0$ and allow $\phi(S_t)$ to vary across states S_t .
2. $\rho_{\mu,r}$ is a free parameter to be estimated, and I impose $\phi(S_t = Exp.) = \phi(S_t = Rec.)$.
3. I set $\rho_{\mu,r} = 0$ and impose $\phi(S_t = Exp.) = \phi(S_t = Rec.)$.
4. I allow $\rho_{\mu,r}$ to vary across states S_t .
5. I allow the persistence parameter ρ to vary across states S_t .

The parameter estimates of these five alternative models are presented in Table A2. At the bottom, I also report the log marginal data density which provides the formal test in terms of model fit. Note that the difference in $\ln p(Y)$, denotes log posterior odds when the prior odds are equal to one.

There are several noteworthy aspects of these estimation results. First, based on $\ln p(Y)$, I conclude that the benchmark specification is preferred over these model alternatives. Second, when I set $\rho_{\mu,r} = 0$ or $\phi(S_t = Exp.) = \phi(S_t = Rec.)$, the log marginal data densities, $\ln p(Y)$, drops significantly, which implies that a strong feature in the data is $\rho_{\mu,r} < 0$ (see Specification 1) and that discount rates are fairly stable in expansions and much more volatile in recessions (see Specification 2).

Third, in Specification 4, I allow $\rho_{\mu,r}(S_t)$ to depend on the state S_t . Note that the 90% confidence intervals for $\rho_{\mu,r}(S_t = Exp.)$ and $\rho_{\mu,r}(S_t = Rec.)$ are remarkably similar. Using the posterior parameter estimates, I formally test the following null hypothesis:

$$H_0 : \rho_{\mu,r}(S_t = Exp.) = \rho_{\mu,r}(S_t = Rec.). \quad (A.1)$$

which I cannot reject at standard levels of significance. The 90% confidence interval on the difference equals $(-0.008, 0.017)$. Economically, this result suggests that the fraction of variance in return news explained by discount rate revisions does not vary across economic regimes.

Fourth, in Specification 5, I allow $\rho(S_t)$ to depend on the state S_t . The main result here are that the estimate of ρ is approximately 0.98 in expansions, while it is equal to 0.95 in recessions. I test whether the persistence coefficient of expected returns is higher in expansions via the following null hypothesis:

$$H_0 : \rho(S_t = Rec.) \leq \rho(S_t = Exp.), \quad (A.2)$$

which I cannot reject at standard levels of significance. The 90% confidence interval on the difference (i.e., $\rho(S_t = Exp.) - \rho(S_t = Rec.)$) equals $(0.027, 0.034)$. These results would suggest that risk premia shocks die out more quickly during recessions. For instance, based on the posterior median estimates, the half-life of a discount rate shock is 34 months in expansions and 13 months in recessions.³² However, we should be cautious with this analysis since this model is dominated, in terms of model fit, by the benchmark specifications because the log odds equals 10.

³²The half-life is defined as the number of months t required for the risk premia shock to be reduced to half, that is $\rho^t = 1/2$.

Model-implied autocorrelation and Kalman gains Next, I compute the model-implied autocorrelation of returns and the Kalman gains. In the main text, I showed that, conditioning on each economic regime, the autocorrelations of returns is given by

$$corr_{S_t}(r_{t+1}^e, r_t^e) = \rho(S_t) \cdot \phi(S_t) + \rho_{\mu,r}(S_t) \cdot \sqrt{\phi(S_t)(1 - \phi(S_t))(1 - \rho(S_t)^2)} \quad (A.3)$$

For each model, I evaluate (A.3) using parameter draws from their posterior distribution. Table A3 reports the 5th, 50th, and 95th percentiles from this computation.³³ Table A3 shows that the main model generates a strong positive autocorrelation in returns during recessions and mild negative autocorrelation in returns during expansions, with values of around 6.4% and -1.2% respectively. If I set $\rho_{\mu,r} = 0$, then the autocorrelation in returns during expansions turns positive and equals 1.2% (see Specification 1). Reverting back to $\rho_{\mu,r} < 0$ and setting $\phi(S_t = Exp.) = \phi(S_t = Rec.)$, produces a return autocorrelation in expansions and in recessions that is small and negative with values of around -1.2 (see Specification 2). The combination of $\rho_{\mu,r} = 0$ and $\phi(S_t = Exp.) = \phi(S_t = Rec.)$, implies a mild positive autocorrelation in returns throughout (see Specification 3). Allowing for $\rho_{\mu,r}$ to depend on S_t also produces a state-dependent pattern in the autocorrelation in returns. That is, strong positive autocorrelation in recessions and mild reversals during expansions (see Specification 4). This result is not surprising given that I could not reject the null hypothesis in (A.1). Finally, when ρ depends on S_t , the model-implied autocorrelation in returns is state dependent, but the serial correlation in returns during recessions drops substantially to 2.7% from 6.4% given that the persistence in expected returns drops during recessions (see Specification 5).

Table A3 also reports the 5th, 50th, and 95th percentiles for the steady-state Kalman gain using parameter draws from their posterior distribution. As explained in the main text, the Kalman gain determines the weight given to past returns when computing the estimate of expected returns. The Kalman gains across model specifications mimic the model-implied state-dependent autocorrelations in returns.

³³Specifically, I sampled 10,000 draws of the model parameters from their posterior distribution. I then evaluated equation (A.3) at those values, which gave me for each regime and each model 10,000 different values of the model-implied autocorrelation in returns. Finally, I then computed the 5th, 50th, and 95th percentiles.

Matching the state-dependent autocorrelation in returns. I now examine the extent to which these alternative model specifications account for the state-dependent autocorrelation in returns as implied by the OLS regression coefficients in Table 2 of Section 2 in the paper. To do so, for each model, I compute this predictive check as follows. First, I draw 10,000 parameters from the posterior distribution. For each parameter draw, I simulate trajectories for returns and the common growth component (described in Section 3.3). Both of these trajectories have the same number of observations as in the data (660 monthly observations). I then compute the model-implied regression coefficients by running the following specification:

$$r_{t+1}^e = a_0 + (b_0 + b_{Rec} \cdot I_{\text{Comm.Growth},t}) r_t^e + \epsilon_{t+1} \quad (\text{A.4})$$

where $I_{\text{Comm.Growth},t}$ takes a value of 1 whenever the simulated common growth component is low relative to its 12-month moving average and 0 otherwise.

Results are summarized in Table A4. I focus on the results from estimating the benchmark specification. All three OLS regression point estimates implied by the actual data are within the 5th and 95th percentile of the corresponding posterior predictive distribution. In particular, the benchmark model generates a sizeable return autocorrelation conditional on a recession, as implied by the regression coefficients, with a median value of 0.11 ($= \hat{b}_0 + \hat{b}_{Rec}$), which is remarkably close to point estimates from the data equal to 0.09. Moreover, consistent with the data, in good economic times, the autocorrelation in returns as implied by (A.4) is -0.01, but not statistically significant.

The five alternative model specifications are not able to capture this feature of the data. The only exception is Specification 4 since, in essence, the posterior estimates are very close to the benchmark specification (see Table A2).

I assess the economic relevance of these model ingredients in Appendix A.6.

Table A2
Alternative Model Specifications: Posterior Estimates

This table reports the 5, 50, and 95 percentiles of the posterior distribution for six different model specifications. The benchmark specification model of returns assumes a constant correlation between expected returns shocks and unexpected returns, $\rho_{\mu,r}$, and allows the fraction of variance in excess returns explained by expected returns, $\phi(S_t)$, to depend on the economic regime. In Specification 1, I fix $\rho_{\mu,r} = 0$ and allow $\phi(S_t)$ to vary across states S_t . In Specification 2, $\rho_{\mu,r}$ is a free parameter to be estimated, and I impose $\phi(S_t = Exp.) = \phi(S_t = Rec.) = \phi$. In Specification 3, I fix $\rho_{\mu,r} = 0$ and impose $\phi(S_t = Exp.) = \phi(S_t = Rec.) = \phi$. In Specification 4, I allow $\rho_{\mu,r}$ to vary across states S_t . In Specification 5, I allow the persistence parameter ρ to vary across states S_t . The estimation sample runs from January 1965 to December 2019.

	Benchmark specification				Specification 1 $\rho_{\mu,r} = 0$				Specification 2 $\phi(S_t) = \phi$		
	5%	50%	95%		5%	50%	95%		5%	50%	95%
μ_0	0.005	0.006	0.007	μ_0	0.005	0.006	0.007	μ_0	0.005	0.006	0.007
ρ	0.958	0.970	0.982	ρ	0.958	0.970	0.982	ρ	0.958	0.970	0.982
$\rho_{\mu,r}$	-0.984	-0.955	-0.926	$\rho_{\mu,r}$	0.000	0.000	0.000	$\rho_{\mu,r}$	-0.950	-0.924	-0.893
$\phi(S_t = Rec.)$	0.093	0.149	0.181	$\phi(S_t = Rec.)$	0.010	0.145	0.171	ϕ	0.010	0.012	0.014
$\phi(S_t = Exp.)$	0.008	0.009	0.011	$\phi(S_t = Exp.)$	0.012	0.013	0.014		–	–	–
$\sqrt{Var(r_t^e S_t = Rec.)}$	0.043	0.060	0.072	$\sqrt{Var(r_t^e S_t = Rec.)}$	0.045	0.058	0.069	$\sqrt{Var(r_t^e S_t = Rec.)}$	0.047	0.058	0.069
$\sqrt{Var(r_t^e S_t = Exp.)}$	0.039	0.039	0.039	$\sqrt{Var(r_t^e S_t = Exp.)}$	0.039	0.039	0.039	$\sqrt{Var(r_t^e S_t = Exp.)}$	0.039	0.039	0.039
$\ln p(Y)$		1475		$\ln p(Y)$		1286		$\ln p(Y)$		1473	
	Specification 3 $\rho_{\mu,r} = 0$ & $\phi(S_t) = \phi$				Specification 4 $\rho_{\mu,r}(S_t)$				Specification 5 $\rho(S_t)$		
	5%	50%	95%		5%	50%	95%		5%	50%	95%
μ_0	0.005	0.006	0.007	μ_0	0.005	0.006	0.007	μ_0	0.005	0.006	0.007
ρ	0.958	0.970	0.982	ρ	0.958	0.970	0.981	$\rho(S_t = Rec.)$	0.948	0.950	0.951
	–	–	–		–	–	–	$\rho(S_t = Exp.)$	0.979	0.980	0.982
$\rho_{\mu,r}$	0.000	0.000	0.000	$\rho_{\mu,r}(S_t = Rec.)$	-0.973	-0.949	-0.938	$\rho_{\mu,r}$	-0.976	-0.951	-0.916
	–	–	–	$\rho_{\mu,r}(S_t = Exp.)$	-0.979	-0.952	-0.921		–	–	–
ϕ	0.011	0.015	0.018	$\phi(S_t = Rec.)$	0.112	0.143	0.169	$\phi(S_t = Rec.)$	0.121	0.135	0.152
	–	–	–	$\phi(S_t = Exp.)$	0.005	0.007	0.011	$\phi(S_t = Exp.)$	0.006	0.007	0.014
$\sqrt{Var(r_t^e S_t = Rec.)}$	0.048	0.061	0.073	$\sqrt{Var(r_t^e S_t = Rec.)}$	0.049	0.063	0.075	$\sqrt{Var(r_t^e S_t = Rec.)}$	0.042	0.056	0.068
$\sqrt{Var(r_t^e S_t = Exp.)}$	0.039	0.039	0.039	$\sqrt{Var(r_t^e S_t = Exp.)}$	0.039	0.039	0.039	$\sqrt{Var(r_t^e S_t = Exp.)}$	0.039	0.039	0.039
$\ln p(Y)$		1283		$\ln p(Y)$		1468		$\ln p(Y)$		1465	

Table A3

Alternative Model Specifications: Model-implied Serial Correlation and Kalman Gains

This table reports the 5, 50, and 95 percentiles for the model-implied autocorrelation in returns and the Kalman gain for six different model specifications. To compute these statistics, I sample model parameters from their posterior distribution. The benchmark specification model of returns assumes a constant correlation between expected returns shocks and unexpected returns, $\rho_{\mu,r}$, and allows the fraction of variance in excess returns explained by expected returns, $\phi(S_t)$, to depend on the economic regime. In Specification 1, I fix $\rho_{\mu,r} = 0$ and allow $\phi(S_t)$ to vary across states S_t . In Specification 2, $\rho_{\mu,r}$ is a free parameter to be estimated, and I impose $\phi(S_t = Exp.) = \phi(S_t = Rec.) = \phi$. In Specification 3, I fix $\rho_{\mu,r} = 0$ and impose $\phi(S_t = Exp.) = \phi(S_t = Rec.) = \phi$. In Specification 4, I allow $\rho_{\mu,r}$ to vary across states S_t . In Specification 5, I allow the persistence parameter ρ to vary across states S_t . The estimation sample runs from January 1965 to December 2019.

	Benchmark specification			Specification 1 $\rho_{\mu,r} = 0$			Specification 2 $\phi(S_t) = \phi$		
	5%	50%	95%	5%	50%	95%	5%	50%	95%
Model-implied serial correlation in returns									
Expansion	-1.320	-1.190	-1.081	1.129	1.251	1.413	-1.391	-1.258	-1.145
Recession	5.919	6.457	6.872	13.262	14.035	14.637	-1.391	-1.258	-1.145
Kalman gains									
Expansion	-0.013	-0.012	-0.011	0.009	0.012	0.014	-0.014	-0.013	-0.011
Recession	0.058	0.064	0.068	0.133	0.140	0.146	-0.014	-0.013	-0.011
	Specification 3 $\rho_{\mu,r} = 0 \text{ \& } \phi(S_t) = \phi$			Specification 4 $\rho_{\mu,r}(S_t)$			Specification 5 $\rho(S_t)$		
	5%	50%	95%	5%	50%	95%	5%	50%	95%
Model-implied serial correlation in returns									
Expansion	1.154	1.446	1.737	-1.418	-1.280	-1.147	-0.963	-0.872	-0.790
Recession	1.154	1.446	1.737	4.481	5.373	6.031	2.384	2.721	3.264
Kalman gains									
Expansion	0.009	0.013	0.017	-0.014	-0.013	-0.012	-0.010	-0.009	-0.008
Recession	0.009	0.013	0.017	0.045	0.055	0.060	0.024	0.028	0.033

Table A4
Alternative Model Specifications: Regression Evidence on the Serial Correlation in Returns

This table reports the 5, 50, and 95 percentiles for the model-implied regression coefficients. Using model simulated data, I run the following specification: $r_{t+1}^e = a_0^j + (b_0^j + b_1^j \cdot I_{\text{Comm.Growth},t}) r_t^e + \epsilon_{t+1}$, where $I_{\text{Comm.Growth},t}$ takes a value of 1 whenever the simulated common growth component is low relative to its 12-month moving average and 0 otherwise. The simulated data has the same number of observations as in the actual sample. I consider six different model specifications. The benchmark specification model of returns assumes a constant correlation between expected returns shocks and unexpected returns, $\rho_{\mu,r}$, and allows the fraction of variance in excess returns explained by expected returns, $\phi(S_t)$, to depend on the economic regime. In Specification 1, I fix $\rho_{\mu,r} = 0$ and allow $\phi(S_t)$ to vary across states S_t . In Specification 2, $\rho_{\mu,r}$ is a free parameter to be estimated, and I impose $\phi(S_t = \text{Exp.}) = \phi(S_t = \text{Rec.}) = \phi$. In Specification 3, I fix $\rho_{\mu,r} = 0$ and impose $\phi(S_t = \text{Exp.}) = \phi(S_t = \text{Rec.}) = \phi$. In Specification 4, I allow $\rho_{\mu,r}$ to vary across states S_t . In Specification 5, I allow the persistence parameter ρ to vary across states S_t . The estimation sample runs from January 1965 to December 2019.

	OLS	Benchmark specification			Specification 1 $\rho_{\mu,r} = 0$			Specification 2 $\phi(S_t) = \phi$		
		5%	50%	95%	5%	50%	95%	5%	50%	95%
	Est.									
$\hat{a}_0$	7.24	2.23	7.40	12.72	-0.52	7.06	14.62	6.49	7.84	9.20
$\hat{b}_0$	-0.01	-0.08	-0.01	0.05	-0.01	0.07	0.14	-0.08	-0.02	0.05
$\hat{b}_{\text{Rec}}$	0.10	0.01	0.12	0.23	-0.06	0.05	0.16	-0.10	0.01	0.12
	OLS	Specification 3 $\rho_{\mu,r} = 0 \ \& \ \phi(S_t) = \phi$			Specification 4 $\rho_{\mu,r}(S_t)$			Specification 5 $\rho(S_t)$		
		5%	50%	95%	5%	50%	95%	5%	50%	95%
	Est.									
$\hat{a}_0$	7.24	3.34	7.56	11.88	2.41	7.45	12.49	2.53	7.64	12.71
$\hat{b}_0$	-0.01	-0.06	0.01	0.07	-0.09	-0.02	0.05	-0.07	-0.01	0.06
$\hat{b}_{\text{Rec}}$	0.10	-0.10	0.01	0.12	0.02	0.13	0.24	-0.06	0.05	0.16

Table A5
Multi-Factor Alphas and Subsample Analysis

This table reports the alphas from time-series regressions of the main strategy returns on different asset-pricing factors. With respect to the pricing factors, I consider the Fama-French three factors, the momentum factor (Mom), the time-series momentum factor (TMom), and the volatility-managed factor (TVol). I also show results for subsample regressions. Standard errors adjust for heteroskedasticity and autocorrelation. I report the t-statistics in brackets. To facilitate interpretation, all returns are annualized in percent per year by multiplying monthly returns by 1200. The estimation sample runs from January 1980 to December 2019.

	Mkt			Mkt		FF3	FF3
	Mkt	TMom	TVol	TMom	TVol	Mom	TMom
	(1)	(2)	(3)	(4)	(5)	(6)	TVol
							Mom
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
α : 1980-2019	5.38	4.46	4.33	3.91	6.12	4.94	4.67
	[2.76]	[3.41]	[3.19]	[3.69]	[2.50]	[2.08]	[2.95]
α : 1980-2000	4.63	4.57	3.88	3.82	4.43	4.46	3.61
	[2.82]	[2.66]	[2.15]	[1.99]	[2.15]	[2.32]	[1.90]
α : 2001-2019	6.95	3.05	5.98	3.11	6.93	6.25	3.40
	[2.49]	[2.63]	[3.37]	[2.56]	[2.59]	[2.28]	[3.77]

A.3 Considering Different Pricing Anomalies and Subsample Periods

Table A5 shows that the main specification has a positive alpha with respect to a wide range of pricing factors and across different subsample periods. I present results for several combinations of factors including the Fama-French three factors, the momentum factor, the time-series momentum factor, and the volatility-managed factor.

Reassuringly, for the whole 1980–2019 sample period, all alphas are statistically significant and economically large, ranging from 3.91% to 6.12%. One might worry that these results are entirely driven by the Great Recession that began in December 2007 and ended in June 2009. To address this concern, I also analyze the performance of the strategy across two subsamples ranging from 1980 to 2000 and from 2001 to 2019, respectively. In both subsamples, the alphas are economically large and statistically significant.

Table A6
Leverage, Short-Selling, and Transaction Costs

This table reports various performance measures for several alternative constraints on the main strategy and for different transaction costs. The alternative constraints consist of restricting the risk exposure to be above zero (i.e., no short-selling), below 1 (i.e., no leverage), or below 1.5 (50% standard margin requirement). With respect to the transaction costs, I consider 1, 10, and 20 basis points. Standard errors adjust for heteroskedasticity and autocorrelation. I report the t-statistics in brackets. To facilitate an interpretation, I annualize all returns, expressed as a percentage per year, by multiplying monthly returns by 1200. All models are estimated out of sample on recursively increasing samples of data. The forecast evaluation period runs from January 1980 to December 2019.

Panel A: Leverage and short-selling constraints									Panel B: Transaction costs			
weight		Distribution of Weights ω_t						Sharpe	α for different transaction costs			
limits	$ \Delta\omega $	5%	25%	50%	75%	95%	$E[R]$	increase	None	1bps	10bps	20bps
$[-, -]$	0.23	-0.31	0.56	0.93	1.15	1.52	10.31	28.32	5.38 [2.35]	5.41 [2.36]	5.15 [2.27]	4.87 [2.17]
$[0, -]$	0.20	0.00	0.56	0.93	1.15	1.52	9.13	28.28	3.19 [2.77]	3.21 [2.81]	3.00 [2.64]	2.76 [2.44]
$[0, 1.5]$	0.18	0.00	0.56	0.93	1.15	1.50	9.09	30.28	3.22 [2.93]	3.23 [2.95]	3.04 [2.79]	2.82 [2.61]
$[0, 1.0]$	0.11	0.00	0.56	0.93	1.00	1.00	7.78	31.73	2.63 [3.17]	2.64 [3.18]	2.52 [3.05]	2.38 [2.90]

A.4 Leverage, Short-selling, and Transaction Costs

I also show that the strategy produces fairly large benefits even with tight leverage and short-selling constraints. Moreover, transaction costs would need to be incredibly high to make the strategy unprofitable.

Panel (a) of Table A6 shows the distribution of the portfolio weights under different scenarios of leverage and short-selling constraints together with various performance measures. The first row is the benchmark strategy, which does not impose any constraints. Notably, the benchmark strategy uses modest levels of short-selling and leverage: the 5% and 95% quintiles of the weight distribution are -0.31 and 1.52, respectively.³⁴ These

³⁴The 1% and 99% quintiles are -1.45 and 1.98, respectively.

numbers suggest that the performance of the main strategy is not due to unrealistically high levels of short-selling or leverage. To see this, in the second row, I preclude the strategy from going short, while in the third and fourth row, I add a leverage constraint by imposing a cap on the upper bound of the portfolio weights of 1.5 (50% standard margin requirement) and 1.0 (no leverage), respectively. Several aspects of these restrictions are noteworthy; here, I simply mention a few. First, these constraints substantially reduce trading activity as measured by the mean absolute difference of the monthly weights, $|\Delta\omega|$. However, the overall performance of the strategies does not decrease by much. For instance, even under the extreme case of no short-selling and leverage, the strategy earns a significant alpha of 2.63% and increases Sharpe ratios relative to the buy-and-hold approach by 31%.

Panel (b) of Table A6 shows that the performance of the strategy survives realistic levels of transaction costs. I report the estimated CAPM alpha under various trading costs: 1, 10, and 20 basis points. To put these numbers into perspective, [Fleming, Kirby, and Ostdiek \(2003\)](#) estimate a cost of around 1 basis point of the traded value in the futures market. Overall, the results of Panel (b) suggest that transaction costs would need to be incredibly high to eliminate the market-timing gains.

A.5 Additional Market-Timing Tests

In this section, I present five additional timing-related performance measures. Column (1) of Table A7 presents the results. The first measure attempts to capture the trade-off between risk and returns via a utility function. Following [Fleming, Kirby, and Ostdiek \(2001\)](#), I equate the realized quadratic utility obtained under the dynamic portfolio strategy with the one obtained under the static buy-and-hold strategy,

$$\sum_{t=0}^{T-1} \left[(r_{t+1}^{e,\mu} - \Delta) - \frac{\gamma}{2(1+\gamma)} (r_{t+1}^{e,\mu} - \Delta)^2 \right] = \sum_{t=0}^{T-1} \left[(r_{t+1}^e) - \frac{\gamma}{2(1+\gamma)} (r_{t+1}^e)^2 \right] \quad (\text{A.5})$$

Table A7
Additional Market-Timing Tests

This table presents five market-timing tests for the different specifications considered and described in the main text. Panel A shows a utility-based performance measure (Performance fee), the manipulation-proof measures of performance (MPPM) of [Goetzmann et al. \(2007\)](#), and the correlation between the portfolio weights at time t and the $t + 1$ realized excess returns ($Corr(\hat{\mu}_{t+1|t}, r_{m,t+1}^e)$). Panel B and C show results for the timing measures of [Treyner and Mazuy \(1966\)](#) and [Henriksson and Merton \(1981\)](#). Column (1) presents results for the benchmark specification model of returns, which assumes a constant correlation between expected returns shocks and unexpected returns, $\rho_{\mu,r}$, and allows the fraction of variance in excess returns explained by expected returns, $\phi(S_t)$, to depend on the economic regime. In Specification 2, I fix $\rho_{\mu,r} = 0$ and allow $\phi(S_t)$ to vary across states S_t . In Specification 3, I fix $\rho_{\mu,r} = 0$ and impose $\phi(S_t = Exp.) = \phi(S_t = Rec.) = \phi$. In Specification 4, $\rho_{\mu,r}$ is a free parameter to be estimated, and I impose $\phi(S_t = Exp.) = \phi(S_t = Rec.) = \phi$. I report the t-statistics in brackets. All models are estimated out of sample on recursively increasing samples of data. The forecast evaluation period runs from January 1980 to December 2019.

	Model Constraints			
	Main	$\rho_{\mu,r} = 0$	$\rho_{\mu,r} = 0$ $\phi(S_t) = \phi$	$\phi(S_t) = \phi$
	(1)	(2)	(3)	(4)
Panel A: Performance measures				
Performance fee (bps)	18.95	-9.91	-0.60	-3.49
MPPM	7.79	4.56	4.78	4.10
$Corr(\hat{\mu}_{t+1 t}, r_{m,t+1}^e)$	0.13	0.04	0.06	-0.05
Panel B: Results for the timing measures of Treyner and Mazuy (1966)				
Regression: $r_{p,t+1} = \gamma_0 + \gamma_1 r_{m,t+1}^e + \gamma_2 (r_{m,t+1}^e)^2 + \epsilon_{t+1}$				
γ_0	-0.01 [-0.00]	3.88 [1.74]	-2.47 [-2.34]	2.02 [2.68]
γ_1	0.65 [4.79]	0.47 [2.31]	0.91 [8.31]	0.94 [13.08]
γ_2	2.16 [1.77]	-0.33 [-0.28]	1.30 [2.97]	-0.82 [-2.03]
Panel C: Results for the timing measures of Henriksson and Merton (1981)				
Regression: $r_{p,t+1} = \gamma_0 + \gamma_1 r_{m,t+1}^e + \gamma_2 max(-r_{m,t+1}^e, 0) + \epsilon_{t+1}$				
γ_0	-5.44 [-1.77]	2.36 [0.68]	-4.46 [-2.45]	2.78 [1.95]
γ_1	0.88 [8.39]	0.49 [2.82]	1.02 [8.22]	0.88 [12.30]
γ_2	0.52 [2.37]	0.03 [0.14]	0.25 2[.28]	-0.14 [-1.45]

and solve for Δ for a given level of risk aversion γ .³⁵ The parameter Δ can be interpreted as the monthly maximum performance fee that an investor would be willing to pay to have access to the dynamic strategy. As shown in Panel A, the performance fee for the main strategy is substantial. For instance, an investor with a quadratic utility and a risk aversion coefficient of 3 is willing to pay a monthly fee of around 19 basis points to have access to the $\hat{\mu}_{t+1|t}$ estimate.

The second market-timing test is the manipulation-proof performance measure (hereafter MPPM) of [Goetzmann et al. \(2007\)](#). Traditional measures (e.g., alphas, appraisal ratios, Sharpe ratios) are vulnerable because they can be manipulated.³⁶ The MPPM statistic is robust to manipulations and can be written as:

$$MPPM = \frac{1}{(1 - \rho)\Delta t} \ln \left(\frac{1}{T} \sum_{t=1}^T \left[\frac{1 + r_{p,t}}{1 + r_{f,t}} \right]^{1-\rho} \right), \quad (\text{A.6})$$

where $r_{f,t}$ denotes the risk-free rate and $r_{p,t}$ denotes the per-period (not annualized) portfolio return. I select the coefficient ρ (≈ 2.7) such that holding the market is optimal for an uninformed manager.³⁷ Panel A shows that the MPPM statistic for the main specification is equal to 7.8% and can be interpreted as the annualized continuously compounded excess return certainty equivalent of the portfolio.

The third market-timing test is straightforward. A successful market-timing investor would increase their exposure to the risky asset when future returns are high and will decrease it when future returns are low. Hence, the correlation between the forecasting weights $\hat{\mu}_{t+1|t}$ and the time $t + 1$ realized returns, $r_{m,t+1}^e$, should be positive. Indeed, the last row of Panel A shows a positive correlation of 13% for the main specification.

The fourth measure is based on the quadratic regression of [Treyner and Mazuy \(1966\)](#):

$$r_{p,t+1} = \gamma_0 + \gamma_1 r_{m,t+1}^e + \gamma_2 (r_{m,t+1}^e)^2 + \epsilon_{t+1}, \quad (\text{A.7})$$

³⁵More generally, the quadratic utility function can be viewed as a second-order approximation to the investor's utility function. I assume a risk aversion coefficient of 3 to compute the performance fee. However, I verified that the outperformance is robust to different perturbations to γ .

³⁶Portfolio managers can use dynamic strategies or derivatives to alter the distribution of returns to artificially enhance any particular performance measure or deliberately generate measurement errors to their benefit. For details and examples, see [Goetzmann et al. \(2007\)](#).

³⁷Specifically, I set $\rho = \frac{\ln[E(1+\hat{r}_m)] - \ln(1+\hat{r}_f)}{\text{Var}(1+\hat{r}_m)}$ where $\hat{r}_m$ and $\hat{r}_f$ are the unconditional mean of the market and risk-free rate.

where the coefficient γ_2 measures the return timing performance. As shown in Panel B of Table A7, the γ_2 coefficient associated with the main predictor is significantly positive. As expected, given that the quadratic regression term $(r_{m,t+1}^e)^2$ captures timing abilities, the abnormal return γ_0 for the main specification is small and insignificant.³⁸

The fifth measure is due to [Henriksson and Merton \(1981\)](#) and replaces the quadratic term $(r_{m,t+1}^e)^2$ with $\max(-r_{m,t+1}^e, 0)$:

$$r_{p,t+1} = \gamma_0 + \gamma_1 r_{m,t+1}^e + \gamma_2 \max(-r_{m,t+1}^e, 0) + \epsilon_{t+1} \quad (\text{A.8})$$

Similarly, a positive γ_2 suggests market-timing ability. Relative to the measure of [Treynor and Mazuy \(1966\)](#), the emphasis of this market-timing measure is on how well it predicts during market downturns. Panel C of Table A7 shows the results for this specification. The positive γ_2 under the main specification provides further supporting evidence of the market-timing abilities during downturns.

³⁸After controlling for $(r_{m,t+1}^e)^2$, a positive γ_0 would suggest superior stock selection, which is not considered in this paper.

Table A8
Economic Importance of Model Ingredients

This table presents results of the trading strategy under various forecasting model constraints. Column (1) presents results for the benchmark specification model of returns, which assumes a constant correlation between expected returns shocks and unexpected returns, $\rho_{\mu,r}$, and allows the fraction of variance in excess returns explained by expected returns, $\phi(S_t)$, to depend on the economic regime. In Specification 2, I fix $\rho_{\mu,r} = 0$ and allow $\phi(S_t)$ to vary across states S_t . In Specification 3, I fix $\rho_{\mu,r} = 0$ and impose $\phi(S_t = Exp.) = \phi(S_t = Rec.) = \phi$. In Specification 4, $\rho_{\mu,r}$ is a free parameter to be estimated, and I impose $\phi(S_t = Exp.) = \phi(S_t = Rec.) = \phi$. Standard errors adjust for heteroskedasticity and autocorrelation. I report the t-statistics in brackets. All models are estimated out of sample on recursively increasing samples of data. The forecast evaluation period runs from January 1980 to December 2019.

	Main (1)	Model Constraints		
		$\rho_{\mu,r} = 0$ $\rho_{\mu,r} = 0$ (2)	$\rho_{\mu,r} = 0$ $\phi(S_t) = \phi$ (3)	$\phi(S_t) = \phi$ (4)
α - Mkt	5.46 [2.77]	3.05 [1.58]	0.82 [1.03]	-0.05 [-0.07]
Sharpe ratio increase $\times 100\%$	28.23	-14.87	-0.96	-5.27
R^2_{OOS}	1.671	-0.126	0.493	-0.232
R^2_{OOS} (p-values)	(0.026)	(0.304)	(0.100)	(0.697)

A.6 Importance of the Two Key Model Ingredients.

To assess the economic relevance of the two key assumed model ingredients, I turn off each one of them separately and re-estimate the model. Table A8 presents the results. Column (1) presents results for the main specification, which assumes a constant correlation between expected returns shocks and unexpected returns, $\rho_{\mu,r}$, and allows the fraction of variance in excess returns explained by expected returns, $\phi(S_t)$, to depend on the economic regime. In Column (2), I fix $\rho_{\mu,r} = 0$ and allow $\phi(S_t)$ to vary across states S_t . In Column (3), I fix $\rho_{\mu,r} = 0$ and impose $\phi(S_t = Exp.) = \phi(S_t = Rec.) = \phi$. In Column (4), $\rho_{\mu,r}$ is a free parameter to be estimated, and I impose $\phi(S_t = Exp.) = \phi(S_t = Rec.) = \phi$.

The main takeaway is that the profitability of the strategy disappears when I impose the model constraints. The market alphas are small and statistically insignificant. The Sharpe ratios are lower than the Sharpe ratios of the buy-and-hold strategy. The R^2_{OOS} statistics are negative. These results stress the economic importance of the two key model

Table A9
Characteristic-Sorted Portfolio Returns

This table presents results of the economic and statistical significance of the Characteristic-sorted portfolio excess returns. I consider only the results for the main specification estimated out of sample using an expanding window. Standard errors adjust for heteroskedasticity and autocorrelation. I report the t-statistics in brackets. To facilitate interpretation, all returns are annualized in percent per year by multiplying monthly returns by 1200. The forecast evaluation period runs from January 1980 to December 2016.

	Growth to Value			Big to Small			Losers to Winners		
	(1)	(3)	(5)	(1)	(3)	(5)	(1)	(3)	(5)
Panel (a): Market -timing $\omega_t \propto E_t[r_{t+1}]$									
alpha - FF3 Mom	5.67	5.39	4.91	6.28	4.08	2.36	8.25	4.29	4.41
	[2.69]	[2.97]	[2.27]	[3.21]	[2.68]	[1.46]	[1.99]	[2.53]	[2.51]
Original Sharpe	0.53	0.64	0.59	0.59	0.53	0.42	0.13	0.55	0.64
New Sharpe	0.68	0.83	0.75	0.79	0.66	0.50	0.21	0.67	0.83
$R_{OOS}^2(\%)$	1.38**	2.09***	2.20***	1.88***	2.12***	1.89***	3.29***	2.17***	1.04*

ingredients and complement the results in Appendix A.2, where I assessed the statistical significance by computing the log marginal data density, $\ln p(Y)$, for each specification.

A.7 Characteristic-Sorted Portfolio Returns

Table A9 shows the results when, instead of timing the excess market return, I time characteristic-sorted portfolio returns in excess of the risk-free rate. Specifically, I consider the first, third, and fifth quintile on stock book-to-market ratios, market capitalization, and momentum portfolios from Kenneth R. French's website.³⁹ I show results only for the main specification. As expected, the strategy works equally well for these returns. The strategy produces positive alphas with respect to FF3 and the Momentum factors and it increases Sharpe ratios. Finally, the last row also reports out-of-sample R^2 above 1% per month.

³⁹https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html

Table A10
Performance Evaluation: Extended Period

This table shows results by introducing stale information pertaining to the economic conditions to the forecasting strategy. The columns correspond to the induced lag in months in the NBER recession dummy. Standard errors adjust for heteroskedasticity and autocorrelation. I report the t-statistics in brackets. The sample runs from January 1960 to December 2016. The training sample runs from January 1930 to December 1959.

	Lag in the forecasting weights relative to the NBER Recessionary Index							
	0	1	2	3	...	6	9	12
Time Period 1960 - 2016								
alpha - Mkt	6.10 [3.20]	5.16 [2.73]	5.35 [2.83]	5.30 [2.81]		4.38 [2.36]	1.48 [0.85]	0.79 [0.51]
Original Sharpe	0.33	0.33	0.33	0.33		0.33	0.33	0.33
New Sharpe	0.52	0.46	0.48	0.47		0.43	0.27	0.27
$R^2_{OOS}(\%)$	1.94***	1.48***	1.60***	1.59***		1.11**	-0.41	-0.52

A.8 Extended Sample Period

Above, I report out-of-sample results based on a 1980:01 to 2019:12 sample split. In this section, I demonstrate the robustness of the out-of-sample forecasts to an earlier sample period. Ideally, I should predict recursively the economic states using real-time data as I do in the main text. However, most of the macro data available to use start in 1960. Thus, the earliest that I can start the out-of-sample forecasts using this approach is from 1980 onwards (considering that I need approximately 20 years of data as a training sample).

To deal with this issue, I make use of the ex post NBER recession indicator. This allows me to extend the forecasting sample period as early as 1960, which uses around 30 years of data as the training sample (from 1930:01 to 1959:12). With this in mind, I modify the aggregate predictive density in equation (5) in the main text as follows:

$$p(r_{t+1}^e | r_{1:t}^e, \Theta, \pi_t) = (1 - I_{rec,t-h}) p(r_{t+1}^e | r_{1:t}^e, \Theta_{S_t=0}, S_t = 0) + I_{rec,t-h} p(r_{t+1}^e | r_{1:t}^e, \Theta_{S_t=1}, S_t = 1) \quad (\text{A.9})$$

where $I_{rec,t-h}$ is the NBER recession indicator, which takes a value of one if period $t - h$ is recessionary and a value of zero if it is expansionary. Importantly, I introduce stale information about the economic state by lagging the recessionary indicator by h periods

at the time of doing the forecasts and form portfolios as before. This exercise simulates the scenario of an investor who needs h periods to identify that they are in a recession. For instance, when $h = 0$, the investor perfectly foresees recessions and expansions, while $h = 3$ means that they need around three months of data to identify business cycle turning points.

Table A10 shows the results. I draw two important conclusions from these results. First, the strategy is still profitable even under stale information about the cycle. However, the outperformance of the strategies vanishes after the six-month lag. Note that for values of h greater than six, the alphas are insignificant and small. Similarly, the R^2_{00S} statistics are negative. Second, there is strong evidence that the strategy works well for the extended sample period.

A.9 Stocks and Business Cycles in U.S. States

In this section I extend the sample by looking at the cross-section of U.S. state portfolios and regional business cycles. The motivation for this exercise comes, on the one hand, from finance literature underscoring the significance of geographic location in financial decisions. For example, [Pirinsky and Wang \(2006\)](#) finds that stock returns of firms headquartered in the same location exhibit substantial comovement, whereas results in [Korniotis and Kumar \(2013\)](#) suggest that changes in local macroeconomic conditions influence returns on local stocks. On the other hand, the macroeconomic literature shows that state-level business cycle phases are not in sync with national phases.⁴⁰ For instance, [Owyang et al. \(2005\)](#) and [Hamilton and Owyang \(2012\)](#) find that state recessions differ significantly in the timing of business cycle turning points, with some states entering or recovering from recessions before others. Therefore, the cross-section of U.S. states provides additional data to assess the robustness of my main result. This robustness check leads to consistent results.⁴¹

⁴⁰References on geographic segmentation in financial markets include: [Coval and Moskowitz \(1999\)](#), [Pirinsky and Wang \(2006\)](#), [Hong, Kubik, and Stein \(2008\)](#), and [Korniotis and Kumar \(2013\)](#). References on papers that characterize regional business cycles include: [Crone \(2005\)](#), [Owyang, Piger, and Wall \(2005\)](#), [Owyang, Piger, Wall, and Wheeler \(2008\)](#), [Hamilton and Owyang \(2012\)](#).

⁴¹[Hong, Torous, and Valkanov \(2007\)](#) show that industry portfolios are tightly linked to economic indicators. Specifically, they find that industry returns forecast various indicators of economic activity. Hence, alternatively, I could have used industry portfolios and constructed industry-level business cycles.

A.9.1 State-level Database and Estimation

To construct value-weighted state portfolios, I obtain stock returns from the Center of Research in Security Prices (CRSP). I restrict the sample to NYSE, Amex, or NASDAQ firms with common equity (i.e., CRSP share code 10 and 11). To form state portfolios, I use the location of the firms' headquarters, consistent with Coval and Moskowitz (1999), Pirinsky and Wang (2006), Hong et al. (2008), and Korniotis and Kumar (2013). I obtain the location data from Compustat. I then cross-check, on an annual basis, the historic records of firms' location from the Compact Disclosure database. This allows me to identify those firms that changed the state of their headquarters in any given year. Finally, I exclude states with less than 10 firms.

To construct the state-level business cycle phases, I use six state-level macroeconomic indicators that are available at monthly or quarterly frequencies. The monthly variables are nonagricultural payroll employment (first available observation: 1990.01), unemployment rate (1976.01), initial jobless claims (1986.02). The quarterly variables are real wage and salary disbursements (1998.03), personal income (1948.03), and a state-level house price index (1975.03). In estimating the economic regimes, I incorporate the series when they become available. As described in Section 3.3, the state-space model is flexible enough to handle mixed-frequency data and to account for missing data coming from the different publication lags.

A.9.2 State-level Results

For each state, I fit the model of expected returns described in Section 3. Given the data availability, I start the out-of-sample computations in January 1990 and end in December 2019. I use data from January 1980 to December 1989 to initialize the model parameters.

State-level Regime Switching. There is substantial state-specific variation in the timing and duration of economic contractions. To highlight the difference in the timing, Figure A1 depicts the diffusion of the 2007–2009 national recession across the 48 contiguous

However, one of the significant hurdles in constructing business cycle phases in U.S. industries is data unavailability. Most of the industry macro data is at the annual frequency.

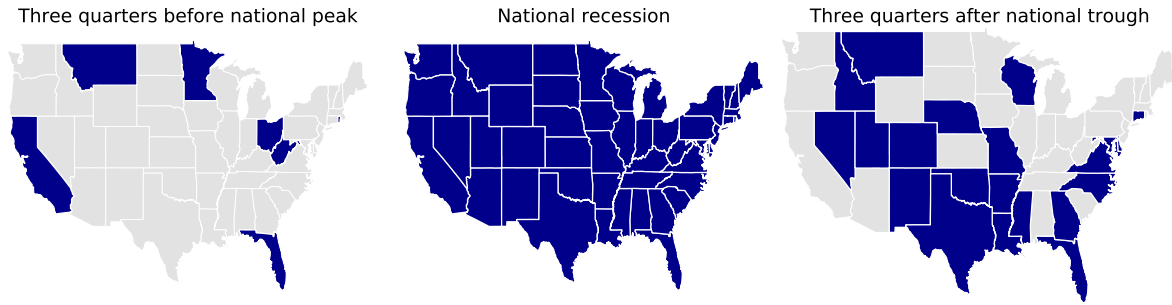


Fig. A1. **National Recession 2007–2009** This figure depicts the diffusion of the 2007–2009 national recession across the 48 contiguous states. A U.S. state is in recession whenever its recession probability is above 50 percent.

ous states.⁴² The left panel shows that six states were already in a recession even three quarters before the nation as a whole went into one (see the dark blue shaded areas). These include states hit particularly hard by the collapse of the housing market (e.g., California and Florida) or by the significant contraction observed in the manufacturing sector (e.g., Ohio). At the national peak, all 48 contiguous states were in a recession, as shown in the middle panel. Finally, the right panel shows that three quarters after the national trough, many states remained in recessionary conditions.⁴³

There are also significant cross-sectional differences in the expected duration of state-level expansions and recessions. Table A11 reports the expected duration of expansions and recessions. The average expected duration of a recession (expansion) is 12 months (51 months) with a standard deviation of 6 months (19 months). The table also reports the probability of remaining in each regime. The average probability of remaining in recessions and expansions is 97% and 90%, respectively. These numbers suggest that these economic regimes are persistent. Table A11 also reports the percentage of time that state-level business cycles are in sync with the national cycle. Following [Harding and Pagan \(2002\)](#), I compute the degree of concordance as

$$C_{i,US} = \frac{1}{T} \sum_{t=1}^T [S_{it}S_{US,t} + (1 - S_{it})(1 - S_{US,t})], \quad (\text{A.10})$$

where S_{it} and $S_{US,t}$ denote the corresponding economic states. The average state was in

⁴²I say that a U.S. state is in recession whenever its recession probability is above 50 percent.

⁴³Similar results can be found for the other four national recessions in my sample. The dates for these recessions are 1980.01–1980.07, 1981.07–1982.11, 1990.07–1991.03, and 2001.03–2001.11.

the same regime as the country 80% of the time. The standard deviation is equal to 5%.

Table A11
State-Level Expansions and Recessions

This table reports the expected duration and the probability of remaining in each regime for each state. Following [Harding and Pagan \(2002\)](#), the degree of concordance is given by $C_{i,US} = \frac{1}{T} \sum_{t=1}^T [S_{it}S_{US,t} + (1 - S_{it})(1 - S_{US,t})]$, where S_{it} and $S_{US,t}$ denote the corresponding economic states. The null column measures the concordance between state i and the U.S. under the null hypothesis that the business cycles of both entities are uncorrelated. The estimation sample runs from January 1980 to December 2019.

State	Expansions		Recessions		Concordance		State	Expansions		Recessions		Concordance	
	Prob. Remaining	Expected Duration	Prob. Remaining	Expected Duration		Null		Prob. Remaining	Expected Duration	Prob. Remaining	Expected Duration		Null
United States	0.98	42.79	0.84	6.36	1.00	0.83	—	—	—	—	—	—	—
Alabama	0.98	41.20	0.91	10.84	0.79	0.74	Montana	0.99	133.29	0.97	34.29	0.79	0.71
Alaska	0.75	4.04	0.97	36.62	0.14	0.14	Nebraska	0.97	35.30	0.89	9.11	0.82	0.73
Arizona	0.98	48.69	0.89	9.40	0.85	0.77	Nevada	0.98	53.73	0.88	8.18	0.91	0.84
Arkansas	0.98	57.92	0.92	12.43	0.86	0.78	New Hampshire	0.99	70.64	0.91	11.23	0.87	0.81
California	0.98	56.09	0.93	14.30	0.85	0.75	New Jersey	0.98	57.00	0.91	11.67	0.87	0.79
Colorado	0.98	57.87	0.93	13.92	0.83	0.77	New Mexico	0.99	91.07	0.92	12.20	0.92	0.83
Connecticut	0.98	41.20	0.93	15.17	0.77	0.68	New York	0.98	62.82	0.92	13.28	0.86	0.78
Delaware	0.96	23.61	0.87	7.41	0.81	0.76	North Carolina	0.98	59.07	0.93	14.40	0.90	0.80
Florida	0.99	79.88	0.97	29.49	0.79	0.70	North Dakota	0.97	30.38	0.84	6.24	0.86	0.80
Georgia	0.98	55.51	0.93	14.80	0.87	0.77	Ohio	0.97	32.89	0.84	6.08	0.88	0.81
Hawaii	0.97	39.43	0.95	19.33	0.71	0.65	Oklahoma	0.97	36.34	0.86	7.38	0.85	0.77
Idaho	0.98	60.38	0.89	8.87	0.90	0.83	Oregon	0.98	46.61	0.90	9.77	0.90	0.79
Illinois	0.98	44.80	0.91	11.23	0.82	0.75	Pennsylvania	0.98	46.40	0.90	10.09	0.84	0.74
Indiana	0.98	40.76	0.92	11.77	0.83	0.74	Rhode Island	0.98	41.58	0.91	11.72	0.79	0.73
Iowa	0.98	50.26	0.90	10.15	0.88	0.80	South Carolina	0.99	68.94	0.88	8.43	0.92	0.83
Kansas	0.98	62.23	0.89	9.23	0.91	0.83	South Dakota	0.97	30.65	0.91	11.12	0.76	0.69
Kentucky	0.98	43.67	0.92	12.16	0.81	0.74	Tennessee	0.98	55.69	0.90	9.95	0.89	0.82
Louisiana	0.99	69.24	0.93	14.21	0.86	0.82	Texas	0.99	73.70	0.93	15.32	0.83	0.79
Maine	0.98	40.06	0.92	12.58	0.82	0.75	Utah	0.99	94.18	0.92	13.05	0.78	0.76
Maryland	0.98	48.25	0.89	8.90	0.84	0.79	Vermont	0.98	46.07	0.88	8.01	0.89	0.81
Massachusetts	0.98	52.76	0.95	19.61	0.76	0.69	Virginia	0.98	57.80	0.93	14.49	0.88	0.79
Michigan	0.97	37.34	0.89	8.91	0.85	0.77	Washington	0.98	44.39	0.89	8.80	0.87	0.78
Minnesota	0.98	43.23	0.91	11.30	0.85	0.77	West Virginia	0.96	25.74	0.89	9.11	0.79	0.70
Mississippi	0.98	46.17	0.91	11.04	0.87	0.78	Wisconsin	0.98	56.45	0.88	8.63	0.92	0.83
Missouri	0.98	44.77	0.90	10.06	0.83	0.77	Wyoming	0.98	64.53	0.84	6.43	0.90	0.86

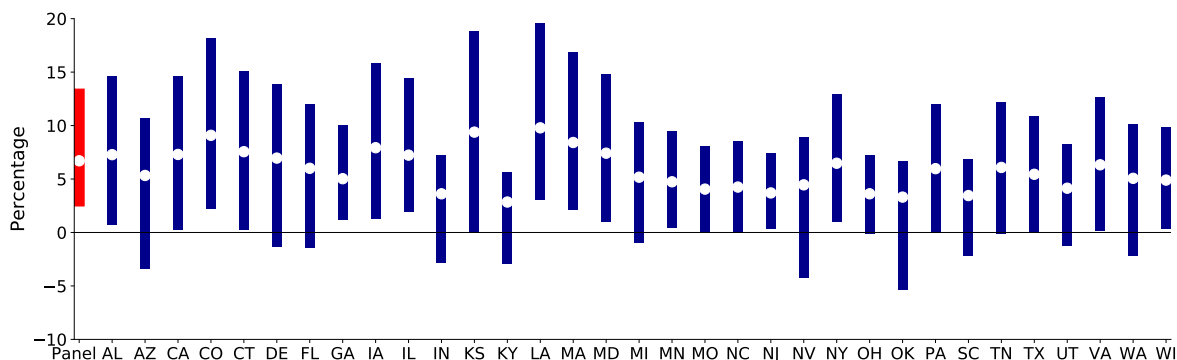


Fig. A2. **State-level Unconditional Alphas.** For each U.S. state, I run time-series regression of the managed excess state return on the nonmanaged excess state return $r_{j,t+1}^{e,\mu} = \alpha_j + \beta_j r_{j,t+1}^e + \epsilon_{j,t+1}$. The white dots denote OLS point estimates, while the blue shaded bars denote 95 percent confidence intervals. The red shaded bar denotes estimates from a pooled OLS regression that double cluster the standard errors at the month-state level. To facilitate an interpretation, I annualize all returns, expressed as a percentage per year, by multiplying monthly returns by 1200. All models are estimated out of sample on recursively increasing samples of data. I exclude states with less than 10 firms. The forecast evaluation period runs from January 1990 to December 2019.

State-level Unconditional Alphas. I take the individual states as the unit of analysis and construct regional-based trading strategies by adjusting the conditional beta, $c_j \hat{\mu}_{j,t+1|t}$, as follows

$$r_{j,t+1}^{e,\mu} = c_j \hat{\mu}_{j,t+1|t} r_{j,t+1}^e. \quad (\text{A.11})$$

The first subscript, j , is the index for the U.S. states, $\hat{\mu}_{j,t+1|t}$ denotes the state j excess return forecast for time $t + 1$ using data up to date t , and $r_{j,t+1}^e$ is the excess return for state j . Similarly, c_j is a normalization constant equalizing the volatility of $r_{j,t+1}^{e,\mu}$ and $r_{j,t+1}^e$.

Figure A2 plots the unconditional alphas for each U.S. state in my sample. The white dots denote OLS point estimates, while the blue shaded bars denote 95 percent confidence intervals. Consistent with the main finding, all point estimates are positive and statistically significant in most cases. This finding is also robust to including the state-level time-series momentum factor and the volatility-managed factor as controls. Finally, the red shaded bar denotes the 95 percent confidence interval on the point estimate from a pooled OLS regression. To determine this, I stack all states and dates and run a pooled panel regression and double cluster the standard errors at the month-state level. The panel regression specification utilizes both time-series and cross-sectional variations in state-managed excess returns and state excess return, increasing my statistical tests'

Table A12
State-level Portfolio Sorts

This table reports the performance estimates of a trading strategy that uses the cross-section of state-level expected excess return measures. Each month, I sort U.S. states into five portfolios based on the predicted state-level expected excess return measure $\hat{\mu}_{j,t+1|t}$. The first (fifth) portfolio is the equally-weighted portfolio of the 20% of state portfolios with the lowest (highest) expected return measure. The long-short portfolio is the portfolio that captures the difference in the returns of the fifth and the first portfolio. To facilitate an interpretation, I annualize all returns, expressed as a percentage per year, by multiplying monthly returns by 1200. All models are estimated out of sample on recursively increasing samples of data. I exclude states with less than 10 firms. The forecast evaluation period runs from January 1990 to December 2019.

Return statistics	Portfolios					long - short
	1	2	3	4	5	
Mean excess return	6.36 [1.46]	8.41 [2.32]	8.88 [2.88]	9.35 [2.66]	11.30 [2.45]	4.95 [1.91]
α - Mkt	-2.79 [-1.86]	-1.02 [-0.62]	0.63 [0.48]	1.72 [0.72]	2.70 [1.95]	5.50 [1.82]

power. The pooled annualized alpha is equal to 6.7%, with a t-statistic of 2.6.

To what extent the state-level unconditional alphas in Figure A2 come from comovement with the national expected return factor versus state-specific factors? To answer this question, each month, I sort U.S. states into five buckets based on the predicted state-level expected excess return measure $\hat{\mu}_{j,t+1|t}$. For each bucket, I then compute equally-weighted portfolios based on the state portfolios. Finally, to difference out common national factors, I take the difference in returns between the 5th and 1st portfolio. Table A12 reports the results. The long-short return is economically large and statistically significant. The long portfolio earns a annualized excess return of 11.3%, whereas the short portfolio gains 6.36%. The difference in returns equals 4.95% (t-statistics = 1.9). The mean excess returns of the other three portfolios lie between the mean excess return of the short and long portfolios, respectively. Similar results hold true when I compute CAPM alphas.

Overall, this result suggests that the state-level analysis is not entirely driven by common national variations and provides incremental information for robustness.

Table A13
Performance Evaluation: Timing Volatility

I run time-series regressions of various trading strategies on the market excess return $r_{t+1}^{e,\mu} = \alpha + \beta r_{t+1}^e + \epsilon_{t+1}$. The strategies adjust the conditional beta based on the estimate of conditional expected excess returns, $\hat{\mu}_{t+1|t}$, described in the main text and an estimate of conditional volatility σ_t^2 . For σ_t^2 I use the previous month's realized variance. Standard errors adjust for heteroskedasticity and autocorrelation. I report the t-statistics in brackets. To facilitate an interpretation, I annualize all returns, expressed as a percentage per year, by multiplying monthly returns by 1200. The forecast evaluation period runs from January 1980 to December 2019.

	Conditional beta		
	$c \hat{\mu}_{t+1 t}$ ERF (1)	$c \frac{1}{\sigma_t^2}$ Vol (2)	$c \frac{\hat{\mu}_{t+1 t}}{\sigma_t^2}$ ERF + Vol (3)
alpha-mkt	5.46 [2.21]	3.51 [1.71]	5.30 [2.71]

A.10 Timing Volatility

In this section, I combine the information in time-varying expected returns and volatility. Specifically, I modify the conditional beta in equation (20) by scaling the expected return by the inverse of the conditional variance. The managed portfolio return is then

$$r_{t+1}^{e,\mu-\sigma} = c \frac{\hat{\mu}_{t+1|t}}{\sigma_t^2} \cdot r_{t+1}^e, \quad (\text{A.12})$$

where again I set c such that $r_{t+1}^{e,\mu-\sigma}$ has the same unconditional volatility as r_{t+1}^e . The main motivation for the use of these weights comes from the optimal dynamic strategy of a mean-variance investor who levers their position up or down on the risky asset over time to maximize the unconditional Sharpe ratio of the portfolio (see [Daniel and Moskowitz \(2016\)](#)). To compute the weights, I use the main specification for $\hat{\mu}_{t+1|t}$. For σ_t^2 , I follow [Moreira and Muir \(2017\)](#) and use the previous month's realized variance, which is computed from the daily data. Specifically, I compute σ_t^2 as follows: $\sigma_t^2 = \sum_{d=1}^{T_t} (r_{d,t}^e - \frac{\sum_{d=1}^{T_t} r_{d,t}^e}{T_t})^2$, where t denotes the month and T_d the total number of days in month t .

Table A13 reports the results. Column (2) shows that a strategy that adjusts the conditional beta using the inverse of the conditional volatility (i.e., $\frac{c}{\sigma_t^2}$) generates an alpha equal to 3.51% (t-statistic = 1.71). Column (3) shows that if I consider both expected returns and volatility, the alpha equals 5.30% (t-statistic = 2.71)

A.10.1 Sorts on the One-Month Ahead Excess Return Forecasts

For the aggregate market and U.S. state portfolios, I bucket the period- t excess return forecast (using only data up until time $t - 1$) into ten bins (low to high). For each bin, I then report average values and 95% confidence intervals for the average conditional risk exposure and the trading strategy's excess return over the subsequent month. The top, middle, and bottom panel use as conditional beta $c \hat{\mu}_{t+1|t}$, $c \frac{1}{\sigma_t^2}$, and $c \frac{\hat{\mu}_{t+1|t}}{\sigma_t^2}$, respectively.

Figure A3 shows that all strategies take substantially less risk when the excess return forecast (ERF) is in the first decile. Note that, at such times, the conditional risk exposure is negative or close to zero, and the trading strategies' excess returns are positive or zero, but not negative. The figure further shows that when the excess return forecast is in the last decile, the volatility-timing strategies have relatively low risk exposure. At such times, volatility is high, and, as shown in Figure 6 in the main text, the probability of being in a recession is also high. Therefore, volatility-timing strategies stay out of the market for too long and miss subsequent higher-than-average excess returns.

A.10.2 Spanning Test

In this section, I conduct formal spanning tests among these three different portfolios. Specifically, I consider the strategies that adjust risk exposure based on the following three conditional beta estimates: $c \hat{\mu}_{t+1|t}$ (ERF), $c \frac{1}{\sigma_t^2}$ (Vol), and $c \frac{\hat{\mu}_{t+1|t}}{\sigma_t^2}$ (ERF+Vol), respectively. I regress these portfolio returns on each other. The annualized alphas from these regressions are reported in Table A14. For ease of comparison, the three strategies have the same volatility.

Panel A of Table A14 reports results from regressing the Vol and ERF+Vol portfolio's returns on the ERF return. The intercepts are positive and significant in most cases, with t-statistics ranging from 1.46 to 3.57. These results indicate that the ERF strategy does

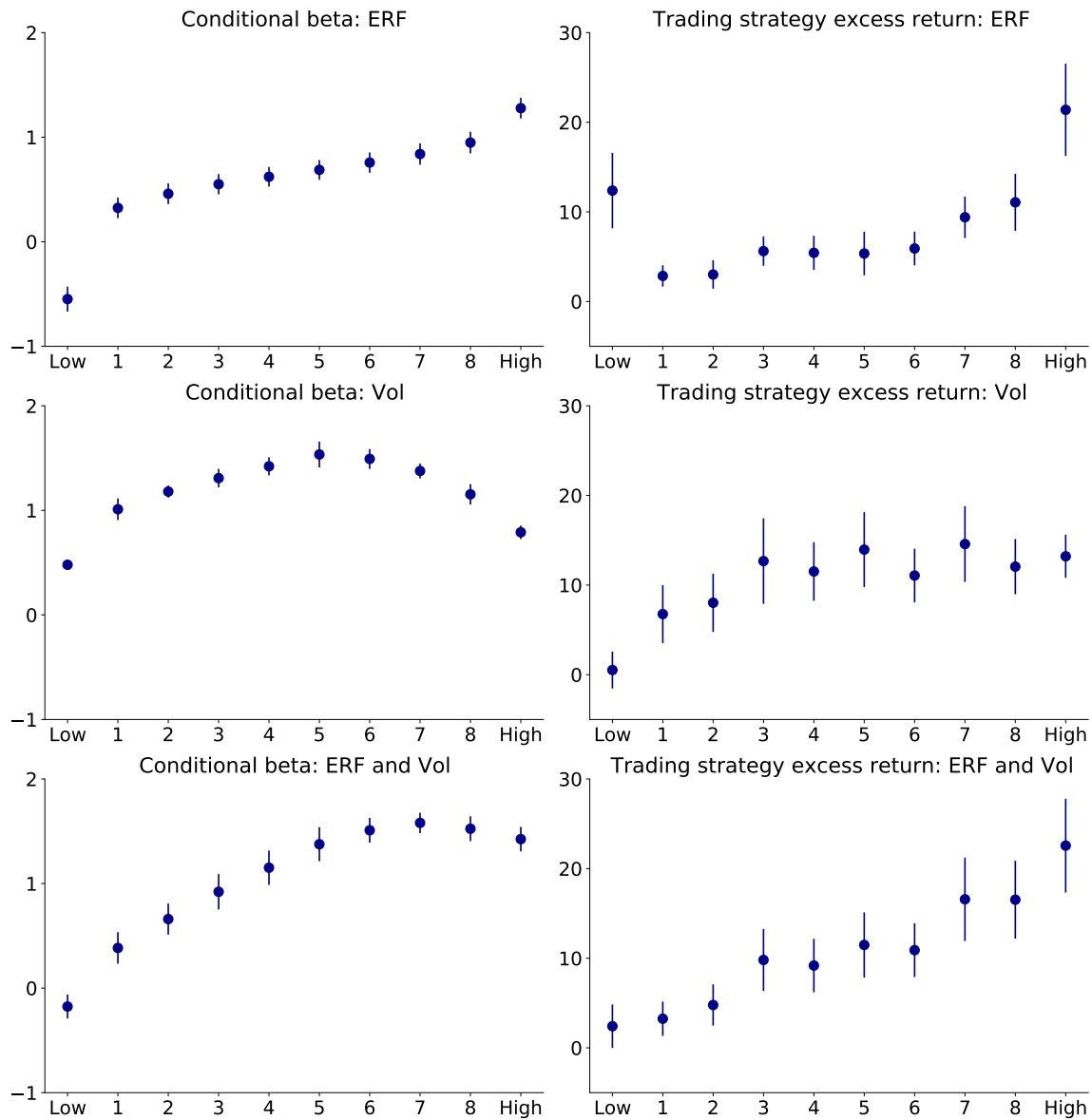


Fig. A3. **Sorts on the one-month ahead excess return forecasts.** This figure sorts the period- t excess return forecast (ERF) for the aggregate market and U.S. state portfolios into ten buckets. For each decile, the figure reports average values and 95 percent confidence intervals for the average conditional risk exposure and the trading strategies' excess returns over the subsequent month. The top, middle, and bottom panel use as conditional beta $c \hat{\mu}_{t+1|t}$, $c \frac{1}{\sigma_t^2}$, and $c \frac{\hat{\mu}_{t+1|t}}{\sigma_t^2}$, respectively. These estimates use data up to time $t - 1$.

not capture the Vol and ERF+Vol strategies. Panel B of Table A14 regresses the returns of the ERF and ERF+Vol strategies on the volatility-managed factor, Vol. All the alphas are positive and highly statistically significant, suggesting that the ERF and ERF+Vol strategies are not spanned by the volatility-managed factor. Finally, Panel C examines whether the ERF and Vol portfolios are spanned by the ERF+Vol portfolio. Columns

Table A14
Spanning Tests

This table presents the results of spanning tests. I consider three different strategies that adjust risk exposure based on the following three conditional beta estimates: $c \hat{\mu}_{t+1|t}$ (ERF), $c \frac{1}{\sigma_t^2}$ (Vol), and $c \frac{\hat{\mu}_{t+1|t}}{\sigma_t^2}$ (ERF+Vol), respectively. Panel A regresses Vol and ERF+Vol on ERF. Panel B regresses ERF and ERF+Vol on Vol. Panel C regresses ERF and Vol on ERF+Vol. For ease of comparison, the three strategies have the same volatility. I report annualized alphas from these regressions.

				Cross-sectional evidence: U.S. States		
	ERF (1)	Vol (2)	ERF + Vol (3)	ERF (4)	Vol (5)	ERF + Vol (6)
Panel A: Independent variable = returns to the strategy that uses $c \hat{\mu}_{t+1 t}$ (ERF)						
α	–	3.30	3.39	–	7.94	6.43
	–	[1.46]	[1.91]	–	[3.31]	[3.57]
R-squared	–	0.33	0.48	–	0.10	0.28
Observations	–	479	479	–	11847	11847
Panel B: Independent variable = returns to the strategy that uses $c \frac{1}{\sigma_t^2}$ (Vol)						
α	5.05	–	2.10	4.90	–	2.61
	[2.99]	–	[2.17]	[2.25]	–	[2.76]
R-squared	0.33	–	0.83	0.10	–	0.60
Observations	479	–	479	11847	–	11847
Panel C: Independent variable = returns to the strategy that uses $c \frac{\hat{\mu}_{t+1 t}}{\sigma_t^2}$ (ERF + Vol)						
α	3.07	-0.40	–	2.51	2.16	–
	[2.34]	[-0.39]	–	[1.92]	[1.43]	–
R-squared	0.48	0.83	–	0.28	0.60	–
Observations	479	479	–	11847	11847	–

(1) and (4) indicate that the ERF strategy is not spanned by the ERF + Vol portfolio. However, the alphas for the Vol strategy are small and statistically indistinguishable from zero. This result suggests that ERF + Vol portfolio spans the Vol portfolio.

A.11 Difference in Cumulative Squared Prediction Error

Figure A4 reports the cumulative squared prediction errors of the benchmark model minus the cumulative squared prediction error of $\hat{\mu}_{t+1|t}$ (see Section 3 for details on how to compute $\hat{\mu}_{t+1|t}$). I take as a benchmark the return forecast of a model that uses equation

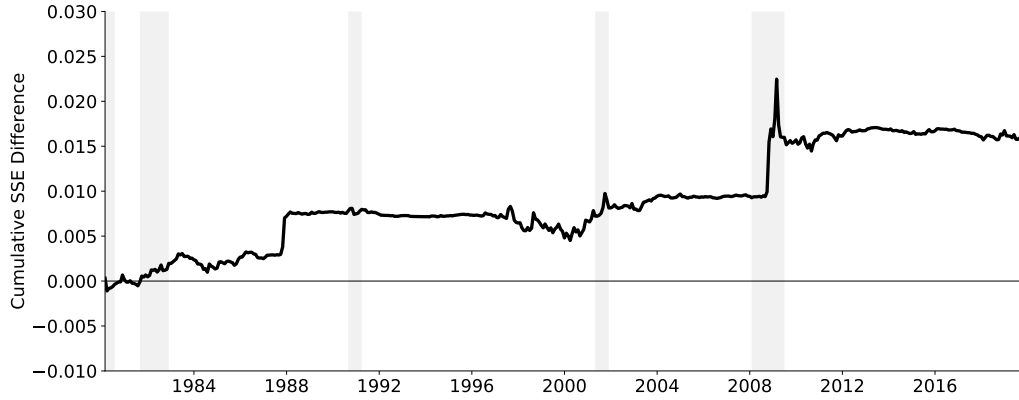


Fig. A4. **Difference in Cumulative Squared Prediction error.** This figure shows the cumulative squared prediction errors of the benchmark model minus the cumulative squared prediction error of main econometric framework described in Section 3 in the main text. The benchmark model is the return forecast of a model that uses equation (2) and (3) in the main text but sets μ_t equal to a constant, that is, $r_{t+1}^e = \mu_0 + \epsilon_{t+1}^r$. The forecast evaluation period runs from January 1980 to December 2019.

(2) and (3) in the main text but sets μ_t equal to a constant, that is, $r_{t+1}^e = \mu_0 + \epsilon_{t+1}^r$. The figure shows that the slope of the cumulative difference in squared prediction errors is mostly positive over the evaluation period. This result suggests that the predictive power of $\hat{\mu}_{t+1|t}$ is not concentrated in relatively short-lived periods or predictability pockets.

A.12 Beta During Expansions

This section analyzes the behavior of expected returns across the business cycle. I run contemporaneous time-series regressions of the strategy's risk exposure (i.e., $c \hat{\mu}_{t+1|t}$) onto five variables related to future stock market returns as previously documented in the literature.

The first variable that I consider is investors' expectations of future stock market returns, obtained from the American Association of Individual Investors (AAII) survey. Each week since 1987, the AAI has asked its members a simple question: "Do you feel the direction of the stock market over the next six months will be up (bullish), no change (neutral) or down (bearish)?" My measure of expectations is the monthly average of the percentage of "bullish" investors minus the percentage of "bearish" investors (i.e., %Bull - %Bear). [Greenwood and Shleifer \(2014\)](#) also rely on the AAI survey data. My second variable is the investor sentiment measure of [Baker and Wurgler \(2006\)](#). My

third variable is the monthly investor inflows into equity-oriented mutual funds from the Investment Company Institute. In each period, I scale the inflows by the aggregate capitalization of the U.S. stock market. The fourth variable is the year-on-year change in the Coincident Index, which measures the current state of the business cycle. The fifth and last variable is the dividend-price ratio, which is a typical measure of risk premia in the literature. I standardize these variables to ease the interpretation of the regression coefficients.

Table A15 presents the results. As shown in Columns (1), (2), and (3), during NBER expansions, when the survey measure of return expectations, sentiment, or flows into the equity markets increase, the strategy takes a contrarian position and reduces risk-taking.⁴⁴ In good times, these results are broadly consistent with studies in the behavioral literature suggesting that many investors hold extrapolative expectations about returns. After a period of good stock market performance, one that generates high sentiment, many investors will become more bullish. Therefore, *extrapolators* will rebalance their portfolios into equity, and, doing so, will bid up current stock prices even further. At this point, the stock market is arguably overvalued, and, hence, subsequent price changes are low, on average. In essence, the strategy exploits this mispricing by mildly decreasing the portfolio weights on stocks. For instance, see from Figure 3 that during expansions the strategy negatively weights lag returns to predict future returns. This is by and large consistent with the equilibrium model of Barberis, Greenwood, Jin, and Shleifer (2015) in which rational traders counteract the overvaluation caused by extrapolators. Therefore, my expected return estimates can be conceptualized as reflecting the return expectations of rational traders in their model. In contrast, extrapolation of returns does not explain my strategy's profits during recessions. Note that during these bad economic times the strategy follows the survey and sentiment measures as well as the flows into equity markets. Although, for the former two series, the point estimates are close to zero.

During expansions, the strategy's profits also can be explained in terms of risk. Columns (4) and (5) show that during expansions, when economic conditions deteriorate

⁴⁴The negative relation between the estimate of expected returns and the survey measures of return expectations in expansions is consistent with the findings of Greenwood and Shleifer (2014), who show that six investor surveys (including the AAII survey) are negatively correlated with traditional predictors of returns, such as the dividend-price ratio.

(but not enough to generate a recession) or when prices decrease more than dividends (representing an increase in risk premia), the strategy increases risk-taking. These results imply that during expansions, the strategy has a strong countercyclical component. Hence, profits also can be viewed as compensation for risk bearing.⁴⁵

While I cannot determine whether the strategy's profits during expansions are due to market inefficiency or as a result of rational variation in expected returns, my profits during recessions conflict with standard risk-based explanations. Specifically, during recessionary periods, an increase in the dividend-price ratio or a decrease in macro growth would suggest high expected returns and thus a time to buy stocks. Yet my strategy rebalanced the portfolio weights out of equity and into Treasuries.

⁴⁵For example, in a standard representative-agent asset-pricing model, risk premia vary over the business cycle and are relatively high in bad times when the economy is slowing down. For habit models, see [Campbell and Cochrane \(1999\)](#). For long-run risk models, see [Bansal and Yaron \(2004\)](#) and [Bansal et al. \(2012\)](#). For models with rare disasters, see [Rietz \(1988\)](#), [Barro \(2006\)](#), [Gabaix \(2012\)](#), and [Wachter \(2013\)](#).

Table A15
Regression of Risk Exposure on Contemporaneous Variables

I run time-series regressions of the strategy's risk exposure on various contemporaneous variables $c \hat{\mu}_{t+1|t} = (\omega_0 + \omega_{rec} \times I_{rec,t}) + (\beta_0 + \beta_{rec} \times I_{rec,t}) x_t + \epsilon_t$, where $I_{rec,t}$ denotes the NBER recessionary index and x_t denotes one of the contemporaneous variables considered. The data are monthly on the sample period 1987 to 2016 given that the %Bull - %Bear series from the American Association of Individual Investors survey starts in 1987. The sentiment measure of Baker and Wurgler (2006) ends in 2015:M9. Standard errors are adjusted for heteroskedasticity and autocorrelation. t -statistics are reported in brackets.

Contemporaneous variables	Dependent variable: $c \hat{\mu}_{t+1 t}$					
	(1)	(2)	(3)	(4)	(5)	(6)
%Bull - %Bear	-0.32 [-2.57]					-0.14 [-1.47]
%Bull - %Bear $\times I_{rec}$	0.36 [1.89]					-0.08 [-0.27]
Sentiment		-0.31 [-4.21]				-0.14 [-2.18]
Sentiment $\times I_{rec}$		0.33 [2.44]				-0.09 [-0.46]
Flow into equity markets			-0.17 [-2.51]			-0.07 [-1.32]
Flow into equity markets $\times I_{rec}$			0.75 [7.25]			0.67 [6.80]
Coincident indicator YoY				-0.33 [-4.51]		-0.25 [-3.62]
Coincident indicator YoY $\times I_{rec}$				0.56 [2.49]		0.58 [3.84]
Dividend-price ratio					0.24 [2.82]	0.15 [1.54]
Dividend-price ratio $\times I_{rec}$					-0.34 [-1.41]	-0.53 [-2.47]
ω_0	0.79 [9.84]	0.77 [8.65]	0.77 [8.41]	0.81 [10.32]	0.77 [8.80]	0.84 [14.06]
$\omega_{rec} \times I_{rec}$	-0.38 [-2.15]	-0.41 [-1.38]	0.03 [0.15]	-0.14 [-0.47]	-0.36 [-1.66]	0.37 [0.79]
R^2 -adjusted	0.19	0.17	0.20	0.22	0.14	0.54

Appendix B Forecasting Model: Data, Details, and Proofs

The dataset consists of macroeconomic variables measured at the monthly and quarterly frequency. All the series start at different periods, but all end in December of 2019

Table A16
Data Series Underlying the Aggregate Economic Activity Index.

The first column reports the macro series used to compute the monthly measure of economic activity. The second column denotes the source. The abbreviations are as follows: BLS = U.S. Bureau of Labor Statistics, BEA = U.S. Bureau of Economic Analysis, ETA = U.S. Employment and Training Administration. The third column reports the frequency. The fourth column gives the first observation available. All series end in December 2019.

Data series	Source	Frequency	First obs.
(1)	(2)	(3)	(4)
Industrial production index	BEA	Monthly	1962:M10
Nonfarm payroll employment	BLS	Monthly	1964:M11
Real personal income excluding current transfer receipts	BEA	Monthly	1959:M2
Aggregate weekly hours worked	BLS	Monthly	1971:M8
Initial jobless claims	ETA	Monthly	1967:M2
GDP	BEA	Quarterly	1965:Q3
Personal consumption expenditures	BEA	Quarterly	1965:Q4
Private residential fixed investment	BEA	Quarterly	1965:Q4
Private nonresidential fixed investment	BEA	Quarterly	1965:Q4

Table A17
Data Series Underlying the U.S. State Economic Activity Index

The first column reports the macro series used to compute the monthly measure of economic activity. The second column denotes the source. The abbreviations are as follows: BLS = U.S. Bureau of Labor Statistics, BEA = U.S. Bureau of Economic Analysis, ETA = U.S. Employment and Training Administration., FHFA = U.S. Federal Housing Finance Agency. The third column reports the frequency. The fourth column gives the first observation available. All series end in December 2019.

Data series	Source	Frequency	First obs.
(1)	(2)	(3)	(4)
Nonagricultural payroll employment	BLS	Monthly	1990:M1.
Unemployment rate	BLS	Monthly	1976:M1
Initial jobless claims	ETA	Monthly	1986:M2
Real wage and salary disbursements	BEA	Quarterly	1998:Q3
Personal income	BEA	Quarterly	1948:Q3
House price index	FHFA	Quarterly	1975:Q3

B.1 State-Space Representations

In this section, I describe the state-space representation of the forecasting model.

B.1.1 State-space Representation of the Forecasting Model Weights

Below, I describe the state-space representation of the forecasting model weights. The representation consists of a measurement equation:

$$y_{t+1} = A_{t+1} (H_0 + H_1 x_{t+1} + \nu_{t+1}) \quad \text{with} \quad \nu_{t+1} \sim N(0, R); \quad (\text{B.1})$$

where A_{t+1} is a selection matrix that accounts for changes in the vector of observables y_{t+1} . The state-transition equation is:

$$x_{t+1} = F_0(S_t, S_{t+1}) + F_1(S_{t+1})x_t + \omega_{t+1} \quad \text{with} \quad \omega_{t+1} \sim N(0, Q(S_{t+1})) \quad (\text{B.2})$$

I will describe each in turn.

Measurement equation. The measurement equation provides the link between the vector of observables y_{t+1} and the vector of state variables, x_{t+1} , this link is determined by the assumed processes in equation (13):

$$y_{i,t+1} = \gamma_i z_{t+1} + e_{i,t+1} \quad \text{for} \quad i = 1, \dots, N_{\text{Macro}}. \quad (\text{B.3})$$

To facilitate the model estimation, I further specify $y_{i,t+1}^* = y_{i,t+1} - \psi_i y_{i,t}$. Hence, equation (B.3) simplifies to

$$y_{i,t+1}^* = \gamma_i z_{t+1} - \gamma_i \psi_i z_t + \sigma_i \epsilon_{i,t+1} \quad \text{for} \quad i = 1, \dots, N_{\text{Macro}}$$

Conditional on the parameters, the vector of observables is given by $y_{t+1} = [y_{1,t+1}^*, y_{2,t+1}^{*,qrt}]'$. For the sake of exposition, I only consider two observables (i.e., $N_{\text{Macro}} = 2$), and I further assume that the first variable is available at the monthly frequency (e.g., industrial production growth), while the second variable is available only at the quarterly frequency (e.g., GDP growth). The link between the quarterly time-aggregated series with the model-implied monthly frequency is given by the following function described in the main text:

$$y_{2,t+1}^{*,qrt} = \sum_{j=1}^5 \frac{3 - |j - 3|}{3} y_{2,t+2-j} \quad (\text{B.4})$$

such that the monthly model variables $y_{2,t+2-j}$ time-aggregate to the observed quarterly growth rates $y_{2,t+1}^{*,qrt}$ (to simplify the notation, I further omitted the o -subscript used for the observable variables in the main text).

The selection matrix A_{t+1} and the vector of observables y_{t+1} are time dependent. Specifically, we have:

- If $t + 1$ is the last month of the quarter:

$$y_{t+1} = \begin{bmatrix} y_{1,t+1}^* \\ y_{2,t+1}^{*,qrt} \end{bmatrix}_{2 \times 1} \quad A_{t+1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}_{2 \times 2}$$

- If $t + 1$ is the first or second month of the quarter:

$$y_{t+1} = \left[y_{1,t+1}^* \right]_{1 \times 1} \quad A_{t+1} = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}_{2 \times 2}$$

To define the matrices associated with the measurement equation (i.e., H_0 , H_1 , and R), I first specify the state vector x_{t+1} as

$$x_{t+1} = \left[z_{t+1} \quad z_t \quad z_{t-1} \quad z_{t-2} \quad z_{t-3} \quad z_{t-4} \quad \sigma_1 \epsilon_{1,t+1} \quad \sigma_2 \epsilon_{2,t} \quad \sigma_2 \epsilon_{2,t-1} \quad \sigma_2 \epsilon_{2,t-2} \quad \sigma_2 \epsilon_{2,t-3} \quad \sigma_2 \epsilon_{2,t-4} \right]'_{12 \times 1}$$

Given the state vector x_{t+1} , it follows that

$$H_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}_{2 \times 1} \quad R = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}_{2 \times 2}$$

$$H_1 = \begin{bmatrix} \gamma_1 & -\gamma_1 \psi_1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ \frac{1}{3} \gamma_2 & \frac{2}{3} \gamma_2 - \frac{1}{3} \gamma_2 \psi_2 & \frac{3}{3} \gamma_2 - \frac{2}{3} \gamma_2 \psi_2 & \frac{2}{3} \gamma_2 - \frac{3}{3} \gamma_2 \psi_2 & \frac{1}{3} \gamma_2 - \frac{2}{3} \gamma_2 \psi_2 & -\frac{1}{3} \gamma_2 \psi_2 & 0 & \frac{1}{3} & \frac{2}{3} & \frac{3}{3} & \frac{2}{3} & \frac{1}{3} \end{bmatrix}_{2 \times 12}$$

State-transition equation. The state-transition equation provides the law of motion of the state variables in the model, which evolve according to

$$z_{t+1} = \mu_z(S_{t+1}) + \phi_z(S_{t+1})(z_t - \mu_z(S_t)) + \sigma_z(S_{t+1})\epsilon_{z,t+1}, \quad (\text{B.5})$$

Given x_{t+1} , it follows that

$$F_0(S_t, S_{t+1}) = \begin{bmatrix} \mu_z(S_{t+1}) - \phi_z(S_{t+1})\mu_z(S_t) \\ O_{11 \times 1} \end{bmatrix}_{12 \times 1} \quad F_1(S_{t+1}) = \left[\begin{array}{cccccc|cccccc} \phi_z(S_{t+1}) & 0 & 0 & 0 & 0 & 0 & & & & & \\ 1 & 0 & 0 & 0 & 0 & 0 & & & & & \\ 0 & 1 & 0 & 0 & 0 & 0 & & & & & \\ 0 & 0 & 1 & 0 & 0 & 0 & & & & & \\ 0 & 0 & 0 & 1 & 0 & 0 & & & & & \\ 0 & 0 & 0 & 0 & 1 & 0 & & & & & \\ \hline & & & & & & O_{6 \times 6} & & & & \\ & & & & & & & O_{6 \times 6} & & & \end{array} \right]_{12 \times 12}$$

And $Q(S_{t+1})$ is a sparse 12×12 matrix with the following elements in the ij -th position:

$$Q^{[1,1]}(S_{t+1}) = \sigma_{z,S_{t+1}}^2, \quad Q^{[7,7]}(S_{t+1}) = \sigma_1^2, \quad Q^{[8,8]}(S_{t+1}) = \sigma_2^2.$$

B.1.2 State-Space Representation of the Process for the Excess Market Return

Below, I describe the state-space representation of the process for the excess market return. In the main text, I defined:

$$\begin{aligned} r_{t+1}^e &= \mu_t + \varepsilon_{t+1}^r, \\ \mu_{t+1} &= \mu_0 + \rho(\mu_t - \mu_0) + \varepsilon_{t+1}^\mu, \end{aligned} \tag{B.6}$$

with $(\varepsilon_{t+1}^r, \varepsilon_{t+1}^\mu)' \sim iidN(\underline{0}, \Sigma(S_t))$, with

$$\Sigma(S_t) = \begin{bmatrix} \sigma_r^2(S_t) & \sigma_{r\mu}^2(S_t) \\ \sigma_{\mu r}^2(S_t) & \sigma_\mu^2(S_t) \end{bmatrix}.$$

To write the state-space model, I will use the same notation as in equations (B.1) and (B.2).

Measurement equation. For each time period $t + 1$, the vector of observables and the selection matrix are $y_{t+1} = r_{t+1}^e$ and $A_{t+1} = 1$, respectively. I write the rest of the matrices as follows:

$$x_{t+1} = \begin{bmatrix} \mu_{t+1} & \mu_t & \sigma_r(S_t)\varepsilon_{t+1}^r \end{bmatrix}'_{3 \times 1}, \quad H_0 = \begin{bmatrix} 0 \end{bmatrix}_{1 \times 1}, \quad H_1 = \begin{bmatrix} 0 & 1 & 1 \end{bmatrix}_{1 \times 3}, \quad R = \begin{bmatrix} 0 \end{bmatrix}_{1 \times 1}$$

State-transition equation. Given the state vector x_{t+1} , the matrices associated with the state-transition equation are given by

$$F_0 = \begin{bmatrix} \mu_0(1 - \rho) \\ 0 \\ 0 \end{bmatrix}_{3 \times 1}, \quad F_1 = \begin{bmatrix} \rho & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}_{3 \times 3}, \quad Q(S_t) = \begin{bmatrix} \sigma_r^2(S_t) & 0 & \sigma_{r\mu}^2(S_t) \\ 0 & 0 & 0 \\ \sigma_{r\mu}^2(S_t) & 0 & \sigma_\mu^2(S_t) \end{bmatrix}_{3 \times 3}.$$

B.2 Posterior Inference

B.3 Posterior Inference: Forecasting Model Weights

To estimate the model associated with the forecasting weights described in Section B.1.1, I use the Bayesian Gibbs sampling approach described in [Kim and Nelson \(1998\)](#).⁴⁶ The assumed prior distribution is quite diffuse, and it is similar to the one used in [Kim et al. \(1999\)](#). The estimation's main output is the posterior distribution of the economic states S_t . The posterior mean of S_t represents the real-time recessionary probabilities, π_t . Importantly, these probabilities correspond to the dynamic forecasting weights of the predictive model described in Section 3.

B.4 Posterior Inference: Process for Excess Returns

The state-space representation described in Section B.1.2 is linear conditional on the economic states S_t (estimated in the previous stage, Section B.3). Therefore, for posterior inference of the aggregate predictive model, I can use a standard Metropolis-within-Gibbs sampler.

Algorithm. Let Θ denote the parameter vector and $\mu_{1:T}$, be the sequence of hidden states. The Metropolis-within-Gibbs algorithm involves iteratively sampling from two conditional posterior distributions. For $s = 1, \dots, N_{sim}$, the main steps of the sampler are:

1. Initialize $\Theta^{(0)}$ at some value.
2. Draw $\mu_{1:T}^{(s)}$ conditional on $\Theta^{(s-1)}$ and the vector of observables $y_{1:T}$. I obtain draws of $\mu_{1:T}^{(s)}$ by applying the simulation smoother of [Durbin and Koopman \(2012\)](#).
3. Draw $\Theta^{(s)}$ conditional on $\mu_{1:T}^{(s)}$ and the vector of observables $y_{1:T}$. I am using a Random-Walk Metropolis-Hastings algorithm to generate draws of the model parameters, Θ , from their posterior distribution in (19). Furthermore, I assume that the proposal distribution is $N(\Theta^{(s-1)}, c \Omega)$ for $s = 1, \dots, N_{sim}$.

⁴⁶For a textbook description of the Gibbs sampler, see Chapter 10 of [Kim, Nelson, et al. \(1999\)](#).

For the covariance matrix of the proposal distribution, Ω , I follow [Schorfheide \(2005\)](#) and use the negative inverse of the posterior density Hessian evaluated at the posterior mode $\tilde{\Theta}$. I target an acceptance rate of approximately 30 percent by adequately modifying the scaling factor c . I initialize the algorithm by selecting $\Theta^{(0)}$ to be the posterior mode $\tilde{\Theta}$ (estimated using standard maximization algorithms) and generated 25,000 draws (i.e., $N_{sim} = 25,000$) from the posterior distribution. Finally, I set the burn-in period at 15,000 draws.

B.5 Identification

In this section, I establish the identification results of the predictive model. A simple way for checking identification is to rewrite the state-space model into an ARMA(1,1) process, where we know that the unknown model parameters can be consistently estimated provided that the roots of the MA component are normalized to lie on or outside the unit circle, and are different from the roots of the AR component, assuming that these roots also lie outside the unit circle (see [Griliches, Engle, Intriligator, and McFadden \(1983\)](#) for instance). In this section, I assume that the ARMA(1,1) model is identified, that is, a change in any parameter associated with the ARMA(1,1) would imply a distinct probability distribution for the aggregate market return $\{r_t^e\}_{t=1}^T$. I then compare the parameters of the ARMA(1,1) model with the parameters implied by the state-space representation.

ARMA(1,1) representation. To rewrite the model into an ARMA(1,1) process consider the following state-space model

$$\begin{aligned} r_{t+1}^e &= \mu_{t+1} + \varepsilon_{t+1}^r, \\ \mu_{t+1} &= \rho(S_t)\mu_t + \varepsilon_{t+1}^\mu, \\ (\varepsilon_{t+1}^r, \varepsilon_{t+1}^\mu)' &\sim iidN(\underline{0}, \Sigma(S_t)) \quad \text{with} \quad \Sigma(S_t) = \begin{bmatrix} \sigma_r^2(S_t) & \sigma_{r\mu}^2(S_t) \\ \sigma_{\mu r}^2(S_t) & \sigma_\mu^2(S_t) \end{bmatrix} \end{aligned} \tag{B.1}$$

where to simplify the subsequent notation I set the constant terms to zero, which can be identified from the conditional mean of r_{t+1}^e . I also slightly modify the timing in

the measurement equation in order to simplify the notation of the ARMA(1,1) process. However, this change in the timing does not affect the identification issues of the model that this section addresses.

To reformulate the state-space into an ARMA(1,1) model, we can combine the measurement equation with the state-transition equation as follows:

$$\begin{aligned} r_{t+1}^e &= \mu_{t+1} + \varepsilon_{t+1}^r, \\ &= (\rho(S_t)\mu_t + \varepsilon_{t+1}^\mu) + \varepsilon_{t+1}^r \\ &= \rho(S_t)r_t^e - \rho(S_t)\varepsilon_t^r + \varepsilon_{t+1}^\mu + \varepsilon_{t+1}^r \end{aligned}$$

Since conditional on being in regime S_t , ε_{t+1}^μ and ε_{t+1}^r are normally distributed, their sum, $\eta_{t+1} = \varepsilon_{t+1}^\mu + \varepsilon_{t+1}^r$, follows a normal distribution with zero mean and variance $\sigma_\eta^2(S_t) = \sigma_r^2(S_t) + \sigma_\mu^2(S_t) + 2\sigma_{r\mu}^2(S_t)$. Hence, if we let $\eta_{t+1} \sim N(0, \sigma_\eta^2(S_t))$ and rescale the variance, it follows that

$$\begin{aligned} r_{t+1}^e &= \rho(S_t)r_t^e - \rho(S_t)\frac{\sigma_r(S_t)}{\sqrt{\sigma_r^2(S_t) + \sigma_\mu^2(S_t) + 2\sigma_{r\mu}^2(S_t)}}\eta_t + \eta_{t+1} \\ &= \gamma(S_t)r_t^e + \psi(S_t)\eta_t + \eta_{t+1} \end{aligned} \tag{B.2}$$

which is a Markov-switching ARMA(1,1) process with AR coefficient $\gamma(S_t)$, and MA and variance parameters equal to $\psi(S_t)$ and $\sigma_\eta^2(S_t)$, respectively. Hence, the ARMA(1,1) has six free parameters. I impose two parameter restrictions to achieve identification. The first restriction fixes the steady-state variance of r^e conditional on the expansionary regime. The second sets the correlation parameter between the unexpected return and expected return shocks to be equal across regimes. Under these identifying restrictions, the number of model parameters equals the number of free parameters of the ARMA(1,1) representation and, therefore, the model is identified.